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Master Thesis

A Cosmic Cannibal Seen Through the Eyes of a New X-Ray Telescope:

Probing the Ionized Disk Atmosphere of 4U 1916–053

by

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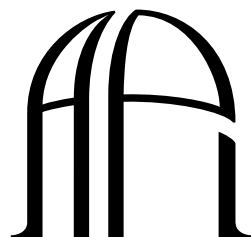
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Abstract

Absorption dips in high-inclination low-mass X-ray binaries provide repeated sight lines through material associated with the accretion disc. These dips therefore offer a way to investigate the structure and dynamics of gas around accreting compact objects. We investigate the ultracompact neutron-star binary 4U 1916–053 using simultaneous 2024 April observations with XMM-*Newton* (105.2 ks), XRISM (244.4 ks), and NuSTAR (47.1 ks). The EPIC-pn light curve changes from relatively isolated dips to broader attenuation and closely separated multiple dips. We divide the observation into three time-selected morphology intervals (sectors 1–3), defined by the changing light-curve morphology. For the persistent (non-dip) and dip selections in each interval, we jointly fit the RGS and *Resolve* spectra with a common continuum, a three-zone photoionised absorber, and a fixed scattered-continuum contribution. These six persistent and dip spectral data sets provide high-resolution absorption-line diagnostics in the RGS soft band and the *Resolve* Fe-K band. NuSTAR is used as an auxiliary continuum constraint.

The most significant spectroscopic result is the fitted positive line-centroid velocity shift of 1507–1736 km s^{−1} for absorption lines from the intermediate-ionisation absorber in all six data sets. Within the adopted model, this zone is consistently redshifted and has a lower column density than the highly ionised zone, which has the largest column but smaller, selection-dependent shifts. The lower-ionisation zone has a higher column density and lower ionisation in each dip selection than in the corresponding persistent selection, although the significance of these differences varies between parameters and sectors. Because the three-zone absorber decomposition may suffer from degeneracy, these zone-by-zone trends remain model dependent. We investigate several scenarios for this recurrent large redshift, including a static disc-rim bulge, a failed wind, and pure gravitational redshift. We conclude that a post-impact downstream flow offers the most plausible interpretation, consistent with a multiphase disc atmosphere disturbed by stream impact.

The same EPIC-pn observation contains three Type-I bursts, with bursts 2 and 3 occurring close to transitions between the time-selected morphology intervals. Because burst irradiation can temporarily perturb the persistent emission, we use these events as a secondary test of whether it could contribute to the subsequent evolution of the dip morphology. The burst fits indicate a short-lived continuum response, but they do not directly measure an absorber response or establish that the bursts caused the hours-scale morphology evolution.

Keywords: neutron stars; X-ray binaries; accretion discs; photoionised absorption; high-resolution X-ray spectroscopy; type-I bursts.

Chapter 1

Introduction

1.1 X-ray Astronomy

X-ray observations probe hot and energetic environments that are often inaccessible at other wavelengths. Their emission commonly comes from 10^6 – 10^8 K plasmas, accretion onto compact objects, or non-thermal particles in shocks and magnetised outflows. Continuum emission, absorption lines, and edges together constrain the column density, ionisation state, and velocity structure of the accretion flow and its surrounding material.

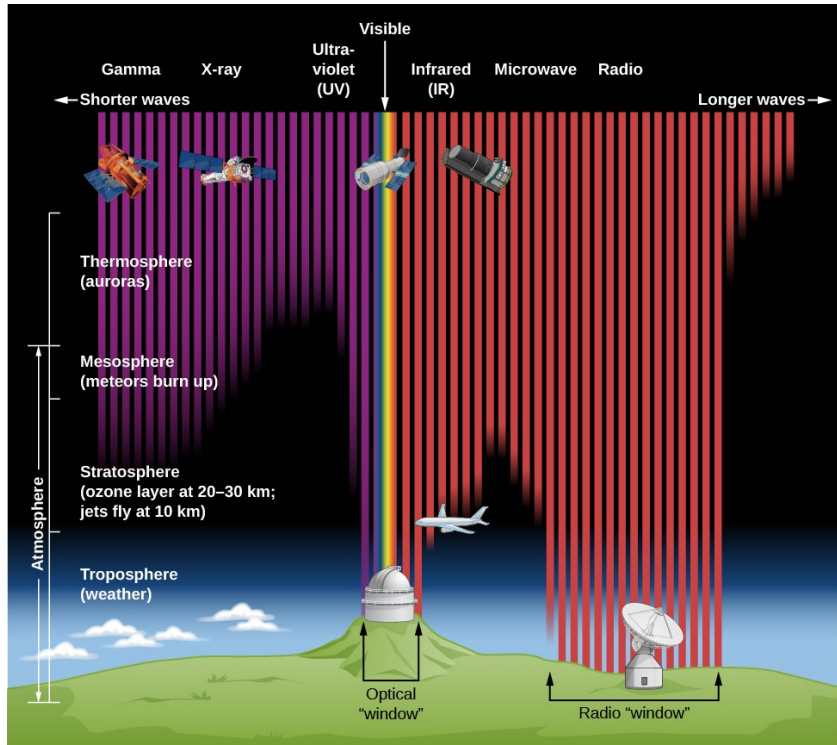


FIG. 1.1— Atmospheric transmission across the electromagnetic spectrum. Earth's atmosphere is opaque to X-rays and gamma rays. Adapted from STScI/JHU/NASA via NASA Imagine [NASA Imagine the Universe \(2013\)](#).

Earth's atmosphere is effectively opaque to X-rays (Fig. 1.1), so X-ray observations must be made from space. Rocket experiments in the early 1960s discovered Scorpius X-1, the first bright extrasolar X-ray source, and the diffuse X-ray background ([Giacconi et al. 1962](#)).

Uhuru, the first satellite dedicated to celestial X-ray astronomy, surveyed the sky and established major source classes, including X-ray binaries, supernova remnants, Seyfert galaxies, and clusters of galaxies ([Forman et al. 1978](#)). Later missions developed complementary capabilities: *Chandra* provides high-spatial-resolution X-ray imaging, whereas *XMM-Newton* combines a large effective area with high-resolution grating spectroscopy ([Weisskopf et al. 2000](#); [Jansen et al. 2001](#)). Combining time-resolved observations with high-resolution spectra allows source variability to be related to changes in the emitting and absorbing gas. Resolved lines and edges then constrain the ionisation

and velocity structure of gas in the accretion environment. The XRISM/*Resolve* microcalorimeter extends high-resolution spectroscopy into the Fe-K band (Tashiro 2022; XRISM Science Team 2020).

1.2 X-ray binaries as laboratories for disc accretion

X-ray binaries are among the brightest X-ray point sources in the sky and provide laboratories for studying accretion under strong gravity (Lewin et al. 1995). They reveal a wide range of physical processes and events, including mass transfer, disc accretion, ionised outflows, and, in neutron-star systems, thermonuclear X-ray bursts. Understanding the physics and efficiency of disc accretion is important for determining how mass and angular momentum are transferred onto compact objects, shaping the growth and spin evolution of neutron stars and black holes (Frank et al. 2002). Accretion-driven neutron-star growth can produce massive neutron stars whose masses constrain the dense-matter equation of state, while black-hole accretion physics helps frame how massive black holes grew rapidly in the early Universe (Özel & Freire 2016; Inayoshi et al. 2020). XRB evolution also provides pathways to compact binaries that may later merge through gravitational-wave emission (Tauris & van den Heuvel 2006). This thesis focuses on disc-associated plasma in a high-inclination neutron-star XRB.

An X-ray binary (XRB) contains a neutron star (NS) or black hole (BH) in orbit around a non-degenerate companion (Lewin et al. 1995; Verbunt 1993). Matter transferred from the donor releases gravitational potential energy as it falls towards the compact object. The resulting 10^7 – 10^8 K plasma radiates from the inner accretion disc, the neutron-star boundary layer, or a hot corona (see e.g., Shakura & Sunyaev 1973; Frank et al. 2002).

XRBs are commonly classified by donor mass as high-mass X-ray binaries (HMXBs) or low-mass X-ray binaries (LMXBs) (Lewin et al. 1995; Tauris & van den Heuvel 2006). HMXBs usually contain an O- or B-type donor with $M \gtrsim 10 M_\odot$. In Be/XRBs, the compact object often accretes from the donor’s circumstellar decretion disc, whereas supergiant HMXBs are typically wind-fed (Tauris & van den Heuvel 2006; Brown et al. 2019). In an LMXB, a donor with $M \lesssim 1 M_\odot$ fills its Roche lobe and transfers gas through the inner Lagrangian point (L_1) (Frank et al. 2002). The gas has enough angular momentum to form a rotationally supported disc around the compact object (Frank et al. 2002). The absence of a strong donor wind makes LMXBs particularly useful for studying plasma associated with the accretion disc.

1.2.1 Accretion-Disc Plasmas in LMXBs

Recent catalogues list 349 Galactic LMXBs (Avakyan et al. 2023). Most contain neutron stars; only ~ 60 are confirmed or candidate black-hole binaries (Corral-Santana et al. 2016). Of these black-hole systems, ~ 20 have dynamical mass measurements above $3 M_\odot$, while the rest are classified from their spectral or timing properties. A neutron

star also has a solid surface, which permits thermonuclear Type I X-ray bursts. LMXB orbital periods range from tens of minutes in ultracompact binaries to days in evolved systems (Bahramian & Degenaar 2023).

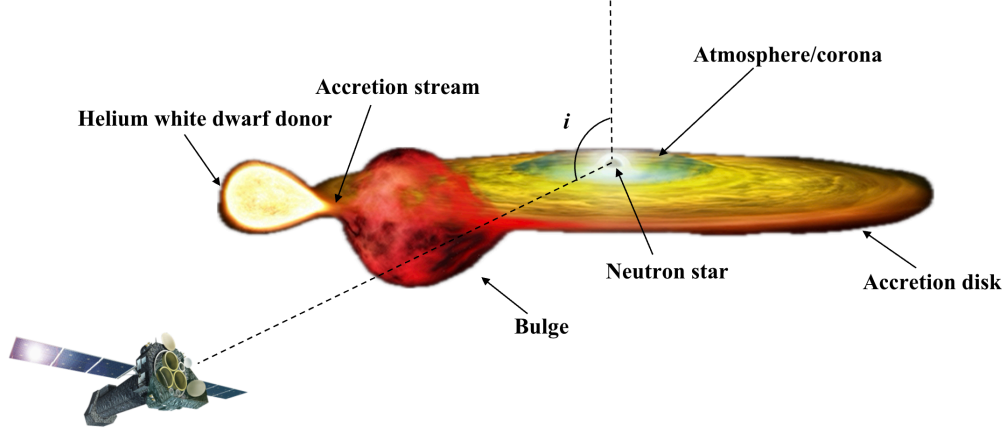


FIG. 1.2— Schematic geometry of a high-inclination ultracompact neutron-star LMXB with a helium white dwarf donor, showing Roche-lobe overflow, the accretion disc and outer bulge, and the inclination angle i . The XMM-*Newton* spacecraft icon marks the observer’s line of sight; its orientation is illustrative. Adapted from NASA/CXC/M. Weiss.

In the geometry shown in Fig. 1.2, the Roche-lobe overflow stream strikes the outer disc rim and can form a vertically extended “bulge” above the orbital plane (White & Swank 1982). The broad-band continuum of a neutron-star LMXB usually contains thermal emission from the accretion disc and neutron-star surface or boundary layer, together with a harder Comptonised component (Bahramian & Degenaar 2023). Absorption features are measured against this continuum. An example persistent-emission continuum from this thesis is shown in Chapter 4 (Fig. 4.2). X-rays from the central source photoionise gas in the disc atmosphere and any outflow, producing absorption lines and edges (Díaz Trigo & Boirin 2015). At high inclination, the line of sight passes close to the disc rim and can intercept dense outer-disc gas. This produces dips, short reductions in X-ray flux that recur near the orbital period (Frank et al. 1987; Díaz Trigo et al. 2006).

Disc-associated plasma can produce narrow absorption lines from highly ionised species, especially Fe XXV and Fe XXVI in the Fe K band (Ponti et al. 2015; Díaz Trigo & Boirin 2015). Figure 1.3 shows an example (Boirin et al. 2004). These lines are found mainly in high-inclination systems. The fitted absorber column density and ionisation also change during dips. These changes support an origin in the binary system rather than the interstellar medium (Díaz Trigo et al. 2006; Ponti et al. 2015).

Photoionisation models use line depths and ionic ratios to constrain the absorber column density and ionisation. The ionisation parameter is $\xi = L/(nr^2)$, where L is the ionising luminosity, n is the gas density, and r is the distance from the ionising source.

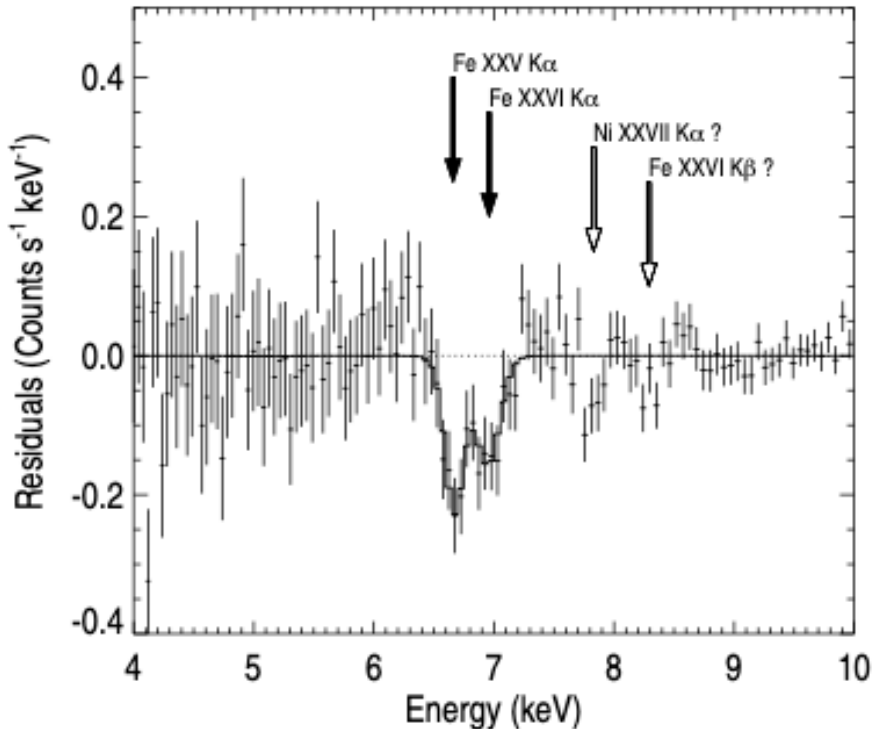


FIG. 1.3— XMM-Newton/EPIC-pn Fe K-band residuals from the persistent-emission spectrum of 4U 1916–053, showing narrow absorption lines from Fe XXV and Fe XXVI. Adapted from [Boirin et al. \(2004\)](#).

Photoionisation studies often find $\log \xi \gtrsim 3$, with ξ expressed in erg cm s^{-1} ([Díaz Trigo & Boirin 2015](#)).

Line-centroid shifts constrain motion along the line of sight. A blueshift indicates motion towards the observer and may trace an outflow, while an unshifted line is consistent with a bound disc atmosphere. For small shifts, the line-of-sight speed can be estimated as $v \simeq c \Delta\lambda/\lambda_0$, or equivalently $v \simeq c \Delta E/E_0$, where λ_0 and E_0 are the rest wavelength and energy. Together, the ionisation and velocity measurements help distinguish a bound atmosphere from an outflow, as discussed in Section 1.2.3.

1.2.2 High-inclination dipping systems

Dipping has been reported in about 20% of LMXBs ([Galloway et al. 2016](#)). The seven bright neutron-star dippers compared by [Díaz Trigo et al. \(2006\)](#) span orbital periods from 0.8 h in 4U 1916–053 to 21 h in X 1624–490. The longer-period systems generally have larger binary separations and accretion discs. The sample also includes X 1254–690, MXB 1659–298, EXO 0748–676, 4U 1746–371, and 4U 1323–62 ([Díaz Trigo et al. 2006](#)). Figure 1.4 shows a 0.6–10 keV light curve and hardness ratio of 4U 1916–053 observed with the European Photon Imaging Camera pn detector (EPIC-pn) on board XMM-

Newton (Boirin et al. 2004). A hardness ratio is the count rate in a higher-energy band divided by that in a lower-energy band. It rises as the count rate falls in Fig. 1.4, showing that the spectrum becomes harder during a dip.

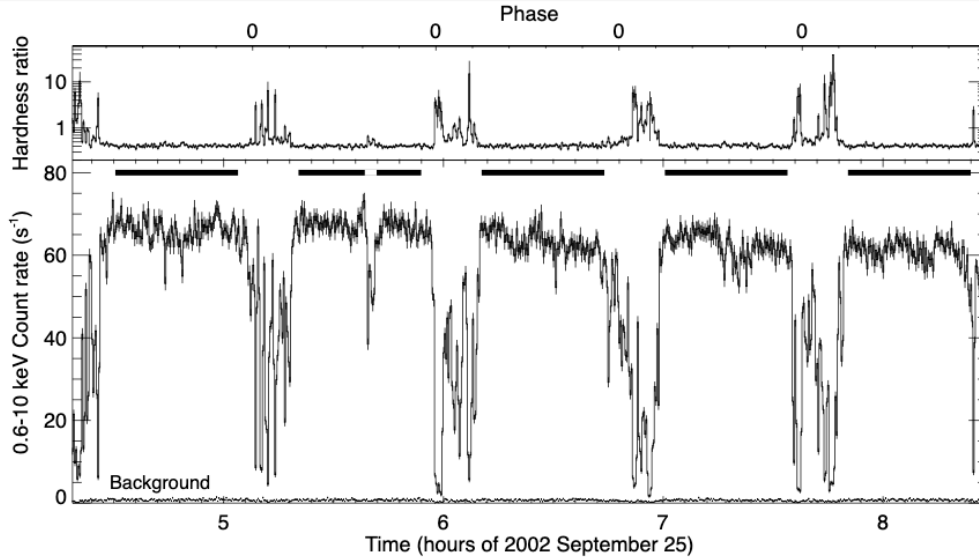


FIG. 1.4— XMM-*Newton*/EPIC-pn 0.6–10 keV light curve (lower panel) and hardness ratio (upper panel) of 4U 1916–053. Dips appear as count-rate reductions accompanied by a higher hardness ratio. Orbital phase follows the ephemeris of Chou et al. (2001). Adapted from Boirin et al. (2004).

In a simple neutral-absorption approximation, the observed and intrinsic spectra are related by $F_{\text{obs}}(E) = F_{\text{int}}(E) \exp[-\tau(E)]$. Here, E is the photon energy, $F_{\text{obs}}(E)$ and $F_{\text{int}}(E)$ are the observed and intrinsic spectral fluxes, and $\tau(E)$ is the optical depth. The optical depth is $\tau(E) = N_{\text{H}}\sigma(E)$, where N_{H} is the equivalent hydrogen column density and $\sigma(E)$ is the photoelectric absorption cross section per hydrogen atom. Because $\sigma(E)$ decreases with energy, low-energy X-rays are absorbed more strongly. This explains the increase in hardness ratio during dips.

Dips vary from one orbit to the next. Their depth, duration, and substructure can change, and the phase of deepest absorption can shift. This behaviour requires an absorber that changes with time and is not uniform around the disc. Frank et al. (1987) linked these changes to the impact of the accretion stream on the outer disc. The stream follows an approximately ballistic trajectory from L_1 (Lubow & Shu 1975). At the outer disc rim, the impact can form a vertically extended, azimuthally asymmetric bulge (Frank et al. 1987). Figures 1.2 and 1.5 show this geometry. Frank et al. also proposed that an ionisation or thermal instability could produce cool, dense clouds within hotter gas. This two-phase structure could account for multiple dip timescales and changes in hardness ratio (Frank et al. 1987; Krolik et al. 1981).

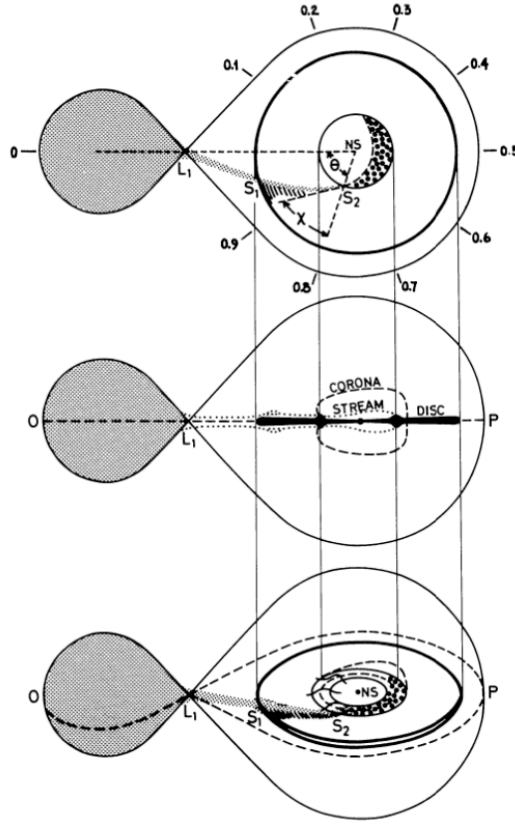


FIG. 1.5— Stream-disc geometry proposed for a high-inclination LMXB. The panels show the donor, accretion stream, disc, and corona in plan and side views. Adapted from [Frank et al. \(1987\)](#).

When the outer-disc bulge crosses the line of sight, it absorbs more low-energy X-rays. The count rate falls and the hardness ratio increases ([White & Swank 1982](#); [Frank et al. 1987](#)). Changes in the azimuthal extent of the absorber can shift the dip phase and alter its shape from one orbit to the next. More recent work models the dipping gas as a disc-associated photoionised absorber. High-resolution spectra constrain its ionisation and line-of-sight velocity ([Trueba et al. 2020](#); [Sharma 2025](#)). Time- and flux-resolved XMM-Newton spectra of 4U 1624–490, for example, find a multiphase, clumpy bulge associated with a highly ionised disc atmosphere ([Caruso et al. 2025](#)).

Figure 1.6 compares six neutron-star dippers observed with XMM-Newton. Some sources show narrow, deep dips, while others show broad or highly fragmented dipping activity ([Díaz Trigo et al. 2006](#)). Inclination determines how close the line of sight passes to the disc plane, but dipping also requires gas that extends far enough above the disc to intercept that line of sight. The vertical extent can reflect the pressure-supported scale height of the outer disc atmosphere and can be increased where the accretion stream disturbs the disc rim ([Jimenez-Garate et al. 2002](#); [Armitage & Livio 1998](#)). Density inhomogeneities in this gas can then produce differences in dip depth and substructure. Time-resolved spectroscopy also shows that the absorber changes during dips and can

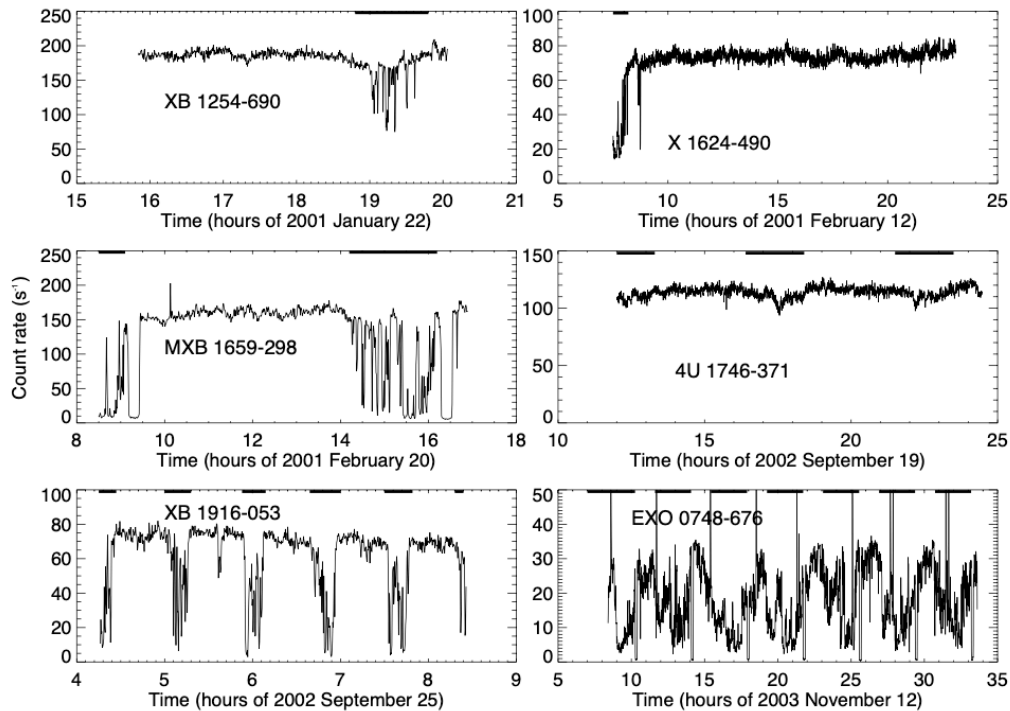


FIG. 1.6— EPIC-pn 0.6–10 keV light curves of six dipping NS LMXBs, illustrating their range of dip depths, durations, and substructure. Thick bars mark the intervals used for dip spectra. Adapted from [Díaz Trigo et al. \(2006\)](#).

become less ionised ([Díaz Trigo et al. 2006](#)).

1.2.3 Disc atmospheres and winds

Here, a disc *atmosphere* means ionised gas above the disc that is bound or quasi-static. Illumination by the central X-ray source heats the disc surface and can inflate the vertical scale height, producing a stratified structure in temperature, density, and ionisation state ([Jimenez-Garate et al. 2002](#); [Neilsen & Degenaar 2023](#)). In such models, multiple ionic species can coexist at a given radius because the atmosphere is vertically layered, and the transition to a hotter, more tenuous coronal region occurs at higher scale heights ([Jimenez-Garate et al. 2002](#)).

Observationally, disc atmospheres can imprint narrow absorption lines (and sometimes recombination emission), often without a strong systematic velocity shift. Their detectability depends strongly on geometry: at high inclination the line of sight is more likely to pass through equatorial disc-associated plasma, while lower-inclination views may miss it ([Díaz Trigo & Boirin 2015](#); [Ponti et al. 2015](#); [Neilsen & Degenaar 2023](#)). An atmosphere is distinguished from a disc wind by whether the gas has a measurable outflow velocity.

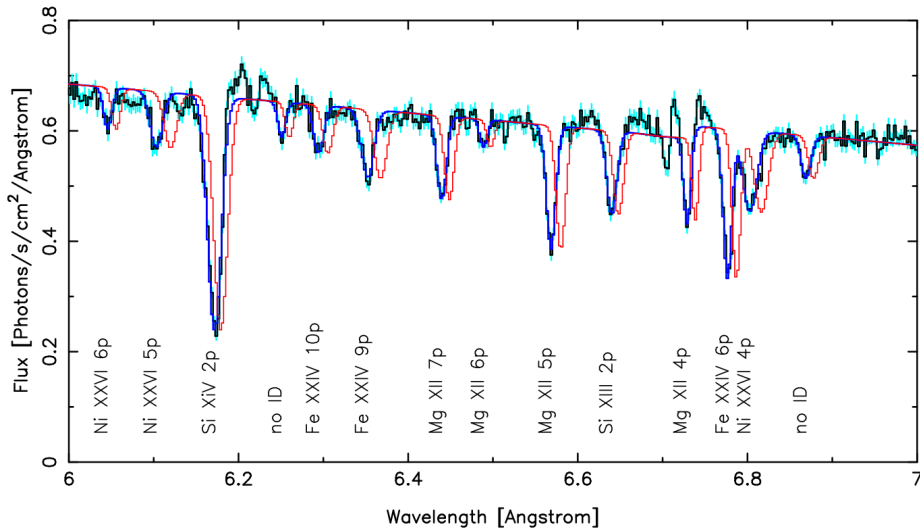


FIG. 1.7— *Chandra*/HETGS spectrum of the black-hole LMXB GRO J1655–40. The labelled ionic transitions form a dense absorption-line spectrum from highly ionised gas. The blue curve is the model at the measured blueshift, while the red curve places the same transitions at their laboratory wavelengths. Adapted from [Miller et al. \(2006\)](#).

Disc winds are outflowing components identified through a systematic blueshift of their line centroids. Typical velocities in LMXBs range from a few hundred to a few thousand km s^{-1} , although faster outflows have been reported ([Neilsen & Degenaar 2023](#); [Muñoz-Darias et al. 2026](#)). GRO J1655–40 provides a clear example: its many narrow lines (Fig. 1.7) constrain both the plasma conditions and the outflow kinematics ([Miller et al. 2006](#)).

Neutron-star LMXBs can also launch winds. For instance, repeated *Chandra* observations of GX 13+1 found blueshifted absorption from a persistent, multicomponent disc wind ([Allen et al. 2018](#)). A later *Chandra*/NICER/VLA campaign detected the wind throughout the source’s Z track while a variable radio jet was also present ([Rogantini et al. 2025](#)). XRISM/Resolve then revealed a slow, stratified, Compton-thick wind, and *Chandra* observations in the preceding days traced the increase in its column density ([XRISM Collaboration 2025](#); [Neilsen et al. 2026](#)).

XRISM has also measured winds in other neutron-star systems. Resolve found an outflow velocity of about $200\text{--}320 \text{ km s}^{-1}$ in 4U 1624–49, resolved the vertical velocity structure of the wind in Her X-1, and characterised outflowing photoionised plasma in Cir X-1 ([Díaz Trigo et al. 2026](#); [Kosec et al. 2025](#); [Tsujiimoto et al. 2025](#)).

The number of detections depends on how strictly a wind is defined. A recent compilation lists eight black-hole or black-hole-candidate systems with confident high-ionisation wind detections and 13 neutron-star systems with confident detections of photoionised plasma. Six of the neutron-star systems show significant outflow velocities; the other seven are

consistent with disc atmospheres (Muñoz-Darias et al. 2026). A uniform archival survey detected Fe K wind lines in five of 42 black-hole LMXB candidates, all of which are dippers (Parra et al. 2024). The six neutron-star systems with measured wind velocities span orbital periods from 8.35 h to 24.1 d.¹ The high-inclination dipper 4U 1624–49 is one of these six systems and provides a useful reference point at 20.89 h.

Several physical mechanisms can contribute to launching disc winds.

X-ray irradiation can Compton-heat the disc surface, causing the gas to expand. At sufficiently large radii, thermal motion can overcome the gravitational binding and the gas can escape; at smaller radii, it remains a bound atmosphere (Begelman et al. 1983; Woods et al. 1996; Done et al. 2018). The characteristic scale is the Compton radius,

$$R_{\text{IC}} \simeq \frac{GM\mu m_{\text{p}}}{kT_{\text{IC}}},$$

where the thermal energy of gas at the Compton temperature T_{IC} is comparable to its gravitational binding energy per particle. Thermal winds are generally expected when the disc extends beyond about $0.1\text{--}0.2R_{\text{IC}}$ and the irradiating luminosity is high enough (Begelman et al. 1983; Woods et al. 1996; Done et al. 2018).

Radiation pressure from electron scattering reduces the effective gravity near the Eddington limit. It can strengthen a thermal wind and allow launching from smaller radii, whereas line driving is usually weak once the gas is highly ionised. Magnetic pressure gradients or magnetocentrifugal acceleration can instead lift gas from radii where thermal driving is ineffective. Magnetic driving is therefore considered for dense or fast flows, although present spectra do not always distinguish it uniquely from thermal–radiative driving, and the mechanisms can operate together (Neilsen & Degenaar 2023; Higginbottom et al. 2020). For example, thermal–radiative driving has been modelled for the wind in GX 13+1 (Tomaru et al. 2020).

The Compton temperature is set by the energy distribution of the irradiating continuum. A stronger hard X-ray contribution raises T_{IC} and reduces R_{IC} . The same continuum controls the balance of heating and cooling, the thermal stability of the gas, and its ionic populations (Done et al. 2018; Higginbottom et al. 2017). Changes in absorption lines can therefore reflect a change in the visibility or ionisation of the gas; they do not necessarily show that the wind has disappeared or that its mass-loss rate has changed.

¹The six systems are AX J1745.6–2901 (8.35 h), 4U 1624–49 (20.89 h), Her X-1 (40.8 h), 2S 0921–63 (216.2 h), Cir X-1 (398.4 h), and GX 13+1 (577.6 h) (Muñoz-Darias et al. 2026).

1.3 4U 1916–053

4U 1916–053 lies at the short-period end of the dipper population. Its compact orbit makes it a useful comparison with longer-period dippers, which generally have larger accretion discs. This is also relevant to the conditions under which neutron-star LMXBs can launch winds. [Fijma et al. \(2023\)](#) detected transient UV P-Cygni profiles in HST/COS spectra of the 111-min neutron-star LMXB UW CrB, indicating outflowing gas in a short-period system. With its 50-min orbit, 4U 1916–053 tests whether similar outflow signatures occur in an even more compact dipper.

1.3.1 System properties

4U 1916–053, also known as XB 1916–053, is a persistent neutron-star low-mass X-ray binary viewed at high inclination. It is an ultracompact system and the shortest-period known LMXB dipper. The deep absorption dips recur at $P_{\text{orb}} \simeq 3000.6\text{--}3000.7$ s (~ 50 min), and the source also produces Type-I X-ray bursts ([White & Swank 1982](#); [Homer et al. 2001](#)). The absence of full eclipses constrains the inclination to be high but not edge-on; the *EXOSAT* analysis of [Smale et al. \(1988\)](#) obtained $i < 79^\circ$ for the assumed binary geometry.

Long-baseline dip timing reveals evolution beyond a strictly constant ephemeris. The X-ray and optical periods differ by approximately one per cent, and the long-term timing solution gives a positive orbital-period derivative of order $\dot{P} \sim 1.5 \times 10^{-11} \text{ s s}^{-1}$ ([Chou et al. 2001](#); [Homer et al. 2001](#); [Iaria et al. 2015](#)).

The 50-min orbit and the He/N-dominated optical spectrum both support a He-rich donor ([Nelemans et al. 2006](#)). Source-specific studies based on photospheric radius-expansion bursts give distance estimates near 9 kpc. [Gambino et al. \(2019\)](#) quote 9.3 ± 1.4 kpc from the analysis of Yoshida (1993), while [Trueba et al. \(2020\)](#) adopt 9.0 ± 1.3 kpc from [Galloway et al. \(2008\)](#).² This thesis therefore adopts 9.1 kpc as the fiducial H-poor distance for the burst analysis.

1.3.2 Long-standing variability of the dips

The dip morphology of 4U 1916–053 has never been strictly repeatable. The three *EXOSAT* observations analysed by [Smale et al. \(1988\)](#) provide an especially direct demonstration. The 1983 observation showed narrow dips occupying less than ten per cent of the orbital cycle. In 1985 May, two broad dips, each lasting up to approximately 800 s, occurred within each 50-min cycle, leaving almost no interval free of dipping. In 1985 October, the first three hours showed no pronounced dips before narrow primary and

²The Multi-INstrument Burst ARchive (MINBAR) catalogue lists 7.6 ± 0.4 kpc under its H-poor calibration ([Galloway et al. 2020](#)). The quoted 0.4-kpc uncertainty is the error reported for this catalogue value. It does not include the full systematic spread among distance estimates based on radius-expansion bursts.

secondary features appeared. The obscuration pattern changed within only a few binary cycles. Irregular dip morphology is therefore not unique to the observations analysed here.

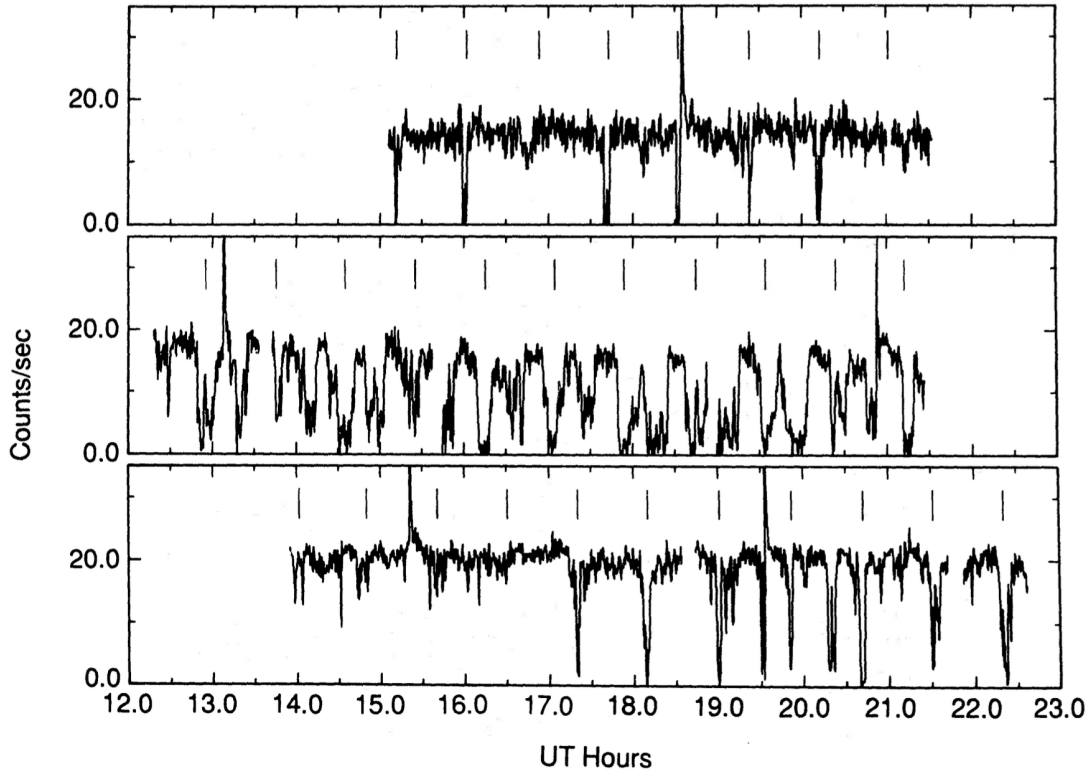


FIG. 1.8— *EXOSAT*/ME 1.5–10 keV light curves of 4U 1916–053 from 1983 September (top), 1985 May (middle), and 1985 October (bottom), binned at 30 s. The panels show narrow, broad or nearly continuous, and initially absent then recurrent dips, respectively. Bursts are truncated and data gaps are caused by detector-array swaps. Adapted from [Smale et al. \(1988, Fig. 1\)](#).

Longer observations reveal additional modulation without establishing one unique clock. RXTE timing linked changes in dip shape and phase to a 3.9087 ± 0.0008 -d beat period ([Chou et al. 2001](#)), whereas [Homer et al. \(2001\)](#) obtained a 4.74 ± 0.05 -d quasi-period that still required phase adjustments between cycles. [Hu et al. \(2008\)](#) measured a 4.87-d modulation of the dip width. These results all establish variability on a roughly 4–5-d time-scale, but they should not be treated as a single exact ephemeris. Possible links to disc precession are discussed in Section 1.4.1.

1.3.3 Previous high-resolution spectroscopy

The long-known dips in 4U 1916–053 motivated observations with modern X-ray spectrometers to identify the absorbing ions and compare the spectra in and out of dips. [Boirin et al. \(2004\)](#) separated persistent and dipping intervals in a 2002 *XMM-Newton* observation. They detected Fe XXV and Fe XXVI absorption in the persistent spectrum.

During dips, the Fe XXVI line weakened below detection and an absorption edge shifted to lower energy, consistent with a decrease in ionisation.

After excluding bursts and dips, [Iaria et al. \(2006\)](#) analysed the first-order *Chandra*/HETGS spectrum of the persistent emission. They detected Ne X, Mg XII, Si XIV, and S XVI, and confirmed the iron lines. Under their interpretation of the line widths as bulk or turbulent motion, they inferred a photoionised absorber at about 4×10^{10} cm, near the outer disc edge.

Using three *Chandra*/HETGS spectra, [Trueba et al. \(2020\)](#) compared absorption in persistent and dipping intervals. The main highly ionised component in the persistent spectra was redshifted by approximately 220–290 km s^{−1}, whereas the dip lines were consistent with no shift. Two spectra also showed evidence for a more strongly redshifted component.

These studies established the presence of ionised plasma associated with the disc, but their Fe K measurements lacked the microcalorimeter resolution now provided by *Resolve*. The coordinated XMM-*Newton* and XRISM observations combine soft-band RGS spectroscopy with high-resolution *Resolve* measurements of the Fe K lines. Together, they constrain multiple ionisation phases and test whether the measured line-centroid shifts are consistent with a bound atmosphere or show evidence for an outflow.

1.4 Time variability

X-ray binaries vary on time-scales from milliseconds to months and longer. Millisecond-to-second changes are generally associated with the inner accretion flow and, in neutron-star systems, the boundary layer ([Belloni 2010](#); [van der Klis 2004](#)). In high-inclination systems, variability on orbital time-scales of tens of minutes to days can also arise when vertically extended outer-disc gas passes through the line of sight and produces absorption dips ([Bahramian & Degenaar 2023](#); [Díaz Trigo & Boirin 2015](#)). Changes in outer-disc orientation or structure can produce still longer modulation. Type I X-ray bursts add seconds-to-minutes episodes of intense irradiation in neutron-star binaries ([Lewin et al. 1993](#); [Galloway & Keek 2017](#)). Dips and bursts provide complementary probes in this thesis: dips trace changes in line-of-sight absorption, while bursts test whether a brief irradiation event perturbs the surrounding accretion flow.

1.4.1 Superorbital modulation and precession

Superorbital variability consists of (quasi-)periodic modulation on time-scales longer than the orbital period, usually days to months. [Kotze & Charles \(2012\)](#) examined 25 X-ray binaries for which superorbital periods had been reported. The 16 LMXBs in their sample have reported periods spanning approximately 2.6–250 d. Some signals are stable, whereas others drift or appear intermittently. The period and stability depend on the proposed mechanism and on the binary and disc properties. Tidal forces can cause a

tilted disc to precess, with the rate depending partly on the mass ratio and disc size. Irradiation can also warp the disc, depending on the irradiation strength and the viscous response of the disc (Charles et al. 2008; Wijers & Pringle 1999; Ogilvie & Dubus 2001). Superorbital modulation can appear in the X-ray flux, spectral hardness, or absorption. Disc tilt, warping, or precession can change the column density along the line of sight, while changes in mass transfer can alter the intrinsic emission or disc structure. In a high-inclination system, even a small change in disc orientation can alter how much material crosses the line of sight and hence modify the dips.

Two geometrically distinct forms of disc precession may be relevant. During nodal precession, a tilted disc precesses around the binary orbital axis. This rotates the line of nodes, where the disc intersects the binary orbital plane, and changes the viewing angle to the disc. During apsidal precession, the major axis of an eccentric disc rotates within the orbital plane. This changes the orientation of non-axisymmetric disc structure relative to the observer (Retter et al. 2002). Figure 1.9 illustrates these geometries.

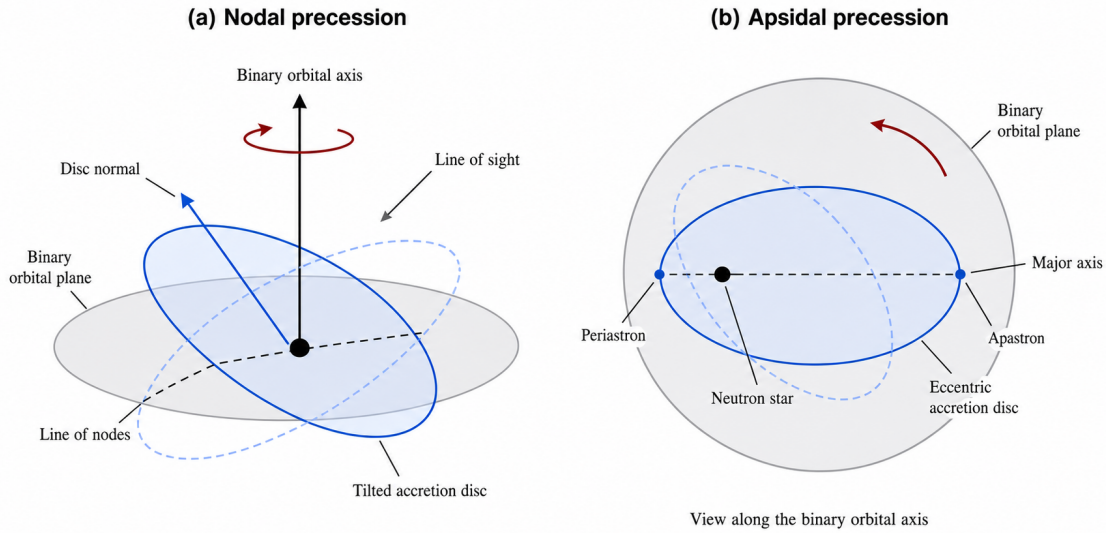


FIG. 1.9— Schematic comparison of nodal and apsidal disc precession. Grey denotes the fixed binary orbital plane, dark blue the accretion disc, and the light-blue dashed outline a later orientation of the disc. Left: nodal precession changes the orientation of a tilted disc and rotates its line of nodes relative to the observer. Right: apsidal precession rotates the major axis of an eccentric disc within the binary orbital plane. The diagram is not to scale. The schematic was first drawn by the author and then polished using ChatGPT Image 2.0.

In compact binaries, disc precession is often inferred from superhumps, photometric modulations with periods slightly offset from the orbital period. Positive superhumps have periods slightly longer than the orbital period and are commonly associated with prograde apsidal precession. Negative superhumps have slightly shorter periods and are commonly associated with retrograde nodal precession (Retter et al. 2002; Hu et al. 2008).

4U 1916–053 shows long-term periodicities in its dipping behaviour. [Hu et al. \(2008\)](#) found a coherent variation in dip width consistent with nodal precession. Figure 1.10 shows the dip dispersion folded on the superorbital period.

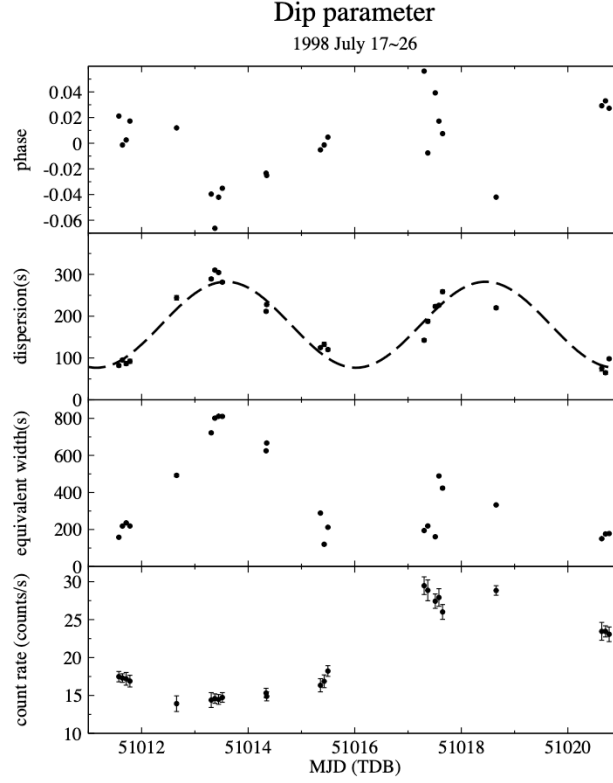


FIG. 1.10— Dip-dispersion modulation in 4U 1916–053 folded on the superorbital period. Figure from [Hu et al. \(2008\)](#).

A disc is warped when its tilt changes with radius. In luminous X-ray binaries, irradiation can exert a reaction torque that drives or maintains such a warp, which may then precess ([Pringle 1996](#); [Wijers & Pringle 1999](#); [Ogilvie & Dubus 2001](#)). Other mechanisms can also produce superorbital variability. Changes in the donor’s mass-transfer rate or in the inward transport of matter can produce long-term modulation ([Charles et al. 2008](#); [Kotze & Charles 2012](#)), and a hierarchical third body has been proposed for 4U 1820–30 ([Chou & Grindlay 2001](#)).

In a high-inclination neutron-star LMXB, measurements over a superorbital cycle can test whether absorber column density, ionisation, and centroid shift vary with phase. A repeatable phase dependence would support modulation tied to the viewing orientation, whereas irregular changes could instead reflect variations in the gas or intrinsic continuum. Such a comparison can help distinguish line-of-sight effects from intrinsic or stochastic variability.

1.4.2 Type-I bursts and burst oscillations

Disc precession changes the line of sight over days. Type-I X-ray bursts instead expose the accretion flow to a brief increase in irradiation. They provide a complementary short-timescale test of how the accretion flow responds. Type-I X-ray bursts are thermonuclear flashes occurring on the surface of accreting neutron stars. These events are triggered by the unstable ignition of accreted H/He at the base of the fuel layer (Lewin et al. 1993; Galloway & Keek 2017). The burning front typically engulfs the stellar surface within $\lesssim 10$ s (Galloway et al. 2020), producing light curves with a rapid rise (~ 1 – 10 s) and a multi-second to multi-minute decay. Figure 1.11 shows this characteristic shape over a range of burst durations.

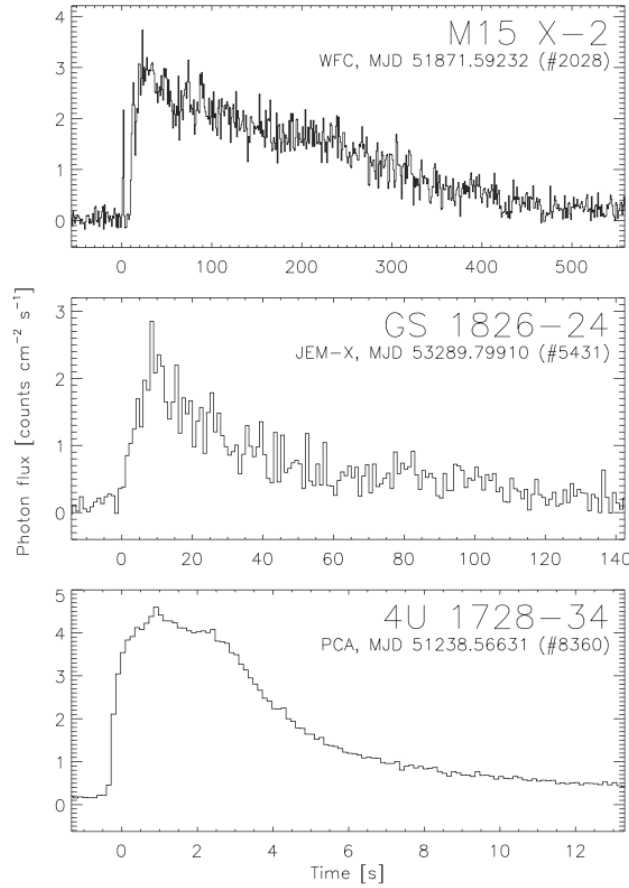


FIG. 1.11— Examples of Type-I X-ray burst light curves, showing rapid rises followed by slower decays over a range of durations. From top to bottom, the bursts are from M15 X-2 observed with BeppoSAX/WFC, GS 1826-24 with INTEGRAL/JEM-X, and 4U 1728-34 with RXTE/PCA. Figure from Galloway et al. (2020).

Time-resolved burst spectra are generally well modelled by a blackbody that undergoes rapid heating followed by cooling. Peak colour temperatures typically reach $kT_{\text{bb}} \sim 1$ – 3 keV (e.g. Lewin et al. 1993; Galloway & Keek 2017). During the peak, the luminosity

can approach the Eddington limit,

$$L_{\text{Edd}} = \frac{4\pi GMc}{\kappa}, \quad (1.1)$$

which is approximately $3 \times 10^{38} \text{ erg s}^{-1}$ for a $1.4 M_{\odot}$ neutron star (Galloway et al. 2020). At this limit, radiation pressure can lift the photosphere and produce photospheric radius expansion (PRE). Comparing an assumed, composition-dependent Eddington luminosity with the measured bolometric Eddington flux then gives a distance estimate, $D = (L_{\text{Edd}}/4\pi F_{\text{Edd}})^{1/2}$ (Lewin et al. 1993; Galloway & Keek 2017).

The intense burst radiation can also perturb the surrounding accretion flow. In the variable persistent-emission method, the normalisation of the pre-burst persistent spectrum is allowed to change during the burst (Wörpel et al. 2013). The observed spectrum is written as

$$F(E, t) = f_a(t) F_{\text{pers}}(E) + F_{\text{bb}}(E, t), \quad (1.2)$$

where $F_{\text{pers}}(E)$ is the fixed-shape persistent spectrum, $F_{\text{bb}}(E, t)$ is the burst blackbody component, and $f_a(t)$ is a scaling factor for the persistent emission. Values of $f_a > 1$ can reflect burst-induced changes such as reprocessing, Comptonisation, or a transient change in the accretion flow (see Degenaar et al. 2018, for a review).

The time-resolved spectra also test for PRE. A PRE detection would indicate an Eddington-limited phase and show whether the measured f_a response occurred during Eddington-level irradiation.

Burst oscillations. A subset of Type-I bursts exhibit nearly coherent brightness modulations at several hundred Hz. These signals trace rotating surface-brightness asymmetries and may arise from a spreading hotspot or a global surface mode. In sources with independently known spin frequencies, the dominant burst-oscillation frequency is usually at or close to the spin frequency, although offsets and frequency drifts of a few hertz can occur (Watts 2018; Bilous & Watts 2019). Recent overviews include Watts (2018) and Di Salvo et al. (2024), while Bilous & Watts (2019) present a uniform RXTE search. A burst dynamical power spectrum is formed by calculating Fourier power spectra in a sequence of overlapping short time windows. It shows how the power changes with time and frequency, allowing transient oscillations and frequency drifts to be identified (Fig. 1.12).

For 4U 1916–053, Galloway et al. (2001) detected an oscillation in one of ten bursts observed with RXTE. The signal was strongest at 269.3 Hz approximately 1 s after burst onset and drifted to about 272 Hz over the following 4 s; no comparable signal was found in the other nine bursts. A repeat detection would test whether this signal recurs. The burst-rise morphology and any amplitude or frequency evolution could constrain ignition latitude and flame spreading, which may affect disc irradiation (Spitkovsky et al. 2002; Cooper & Narayan 2007; Maurer & Watts 2008). This is the search motivation, not a result of this thesis.

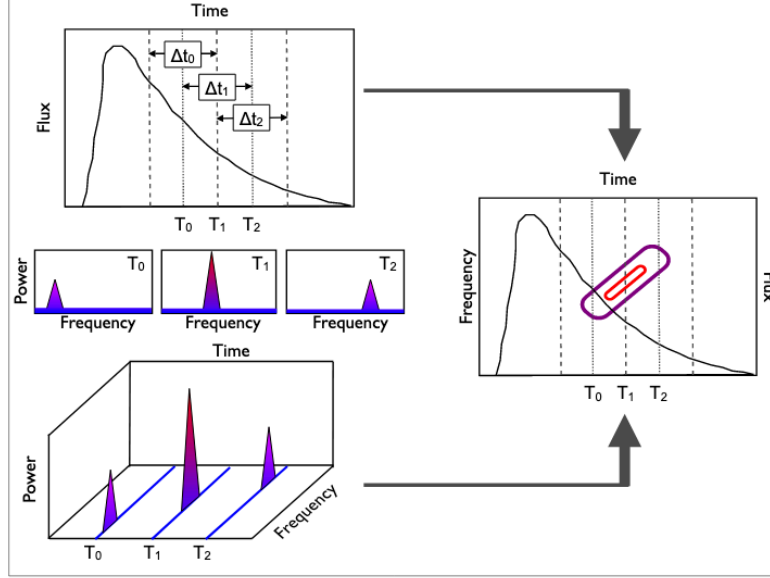


FIG. 1.12— Construction of a burst dynamical power spectrum from overlapping time windows. Adapted from [Watts \(2012\)](#).

1.5 High-resolution spectroscopy

Both intrinsic continuum changes and energy-dependent absorption can alter the broad-band spectral shape of a dipping LMXB. At CCD resolution, a broad-band spectrum may not distinguish between these effects. Resolved absorption lines and edges provide separate constraints on the absorber. Line strengths and ionic ratios constrain column density and ionisation, while line centroids constrain line-of-sight motion. These measurements can help distinguish a bound disc atmosphere from an outflow.

This thesis uses high-resolution spectroscopy to study the ionisation and kinematics of disc-associated plasma in 4U 1916–053. *XRISM/Resolve* provides high-resolution, non-dispersive spectra of highly ionised Fe K absorption. Its closed gate valve strongly attenuates the response below about 2 keV, and its effective area is lower than that of EPIC-pn. *XMM-Newton* therefore provides complementary information: RGS resolves soft-band features from lower- and intermediate-ionisation gas, while EPIC-pn supplies the broad-band continuum and the count-rate information used to select the time intervals. Together, these instruments constrain the absorber’s ionisation, column density, and line-centroid shifts over complementary energy bands.

1.5.1 *XMM-Newton*: RGS and EPIC

XMM-Newton has three co-aligned grazing-incidence telescopes with high throughput. Its calibrated energy range is approximately 0.1–10 keV. Two telescopes feed the

Reflection Grating Spectrometers (RGS), while all three illuminate the European Photon Imaging Camera (EPIC) focal planes (Jansen et al. 2001; den Herder et al. 2001) (Fig. 1.13).

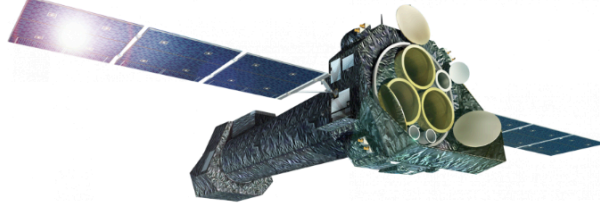


FIG. 1.13— Artist's impression of the XMM-Newton satellite. Credit: ESA (via ESA Science & Technology) (ESA Science & Technology 2016).

RGS delivers dispersive spectroscopy in the soft X-ray band ($\sim 0.35\text{--}2.5$ keV; $\sim 5\text{--}35$ Å). The use of absorption lines to constrain disc-associated plasma was introduced in Section 1.2.1. In the RGS band, features from abundant elements provide diagnostics of ionisation and column density (den Herder et al. 2001; Díaz Trigo & Boirin 2015). Its typical first-order resolution of $\Delta\lambda \sim 0.06$ Å (FWHM) separates narrow absorption features from broad continuum curvature.

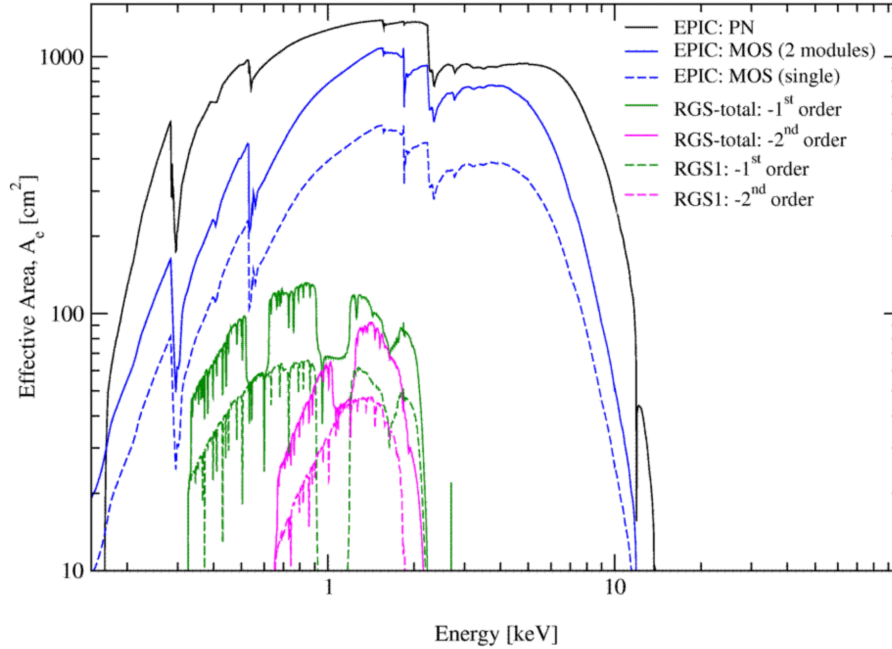


FIG. 1.14— On-axis net effective area of the XMM-Newton telescopes folded through EPIC and RGS responses. Adapted from the XMM-Newton Users Handbook (ESA XMM-Newton Science Operations Centre 2025a).

EPIC complements RGS with broad-band CCD spectroscopy and timing through two

MOS cameras and one pn camera (Turner et al. 2001; Strüder et al. 2001). Its high throughput supports time-resolved spectroscopy and the selection of intervals in variable sources. Figure 1.14 shows the on-axis net effective area of EPIC and RGS.

Figure 1.17 compares the pre-launch line sensitivity of RGS and *Resolve*.

1.5.2 XRISM/*Resolve*

The X-ray Imaging and Spectroscopy Mission (XRISM) was designed for high-resolution imaging and spectroscopy across the 0.3–12 keV band. The *Resolve* gate valve has remained closed, shifting its nominal bandpass to approximately 1.7–12 keV and reducing the effective area at soft energies (XRISM SOC-PVO 2022). This thesis uses *Resolve* over 2–10 keV (XRISM Science Data Center 2026a). Figure 1.15 shows the payload configuration: two co-aligned telescopes feed the *Resolve* microcalorimeter spectrometer and the *Xtend* soft X-ray imager. *Xtend* provides simultaneous wide-field imaging spectroscopy with a field of view of $\sim 38' \times 38'$ over ~ 0.4 –13 keV, complementing *Resolve* by characterising the surrounding field and local background while *Resolve* targets high-resolution line diagnostics.

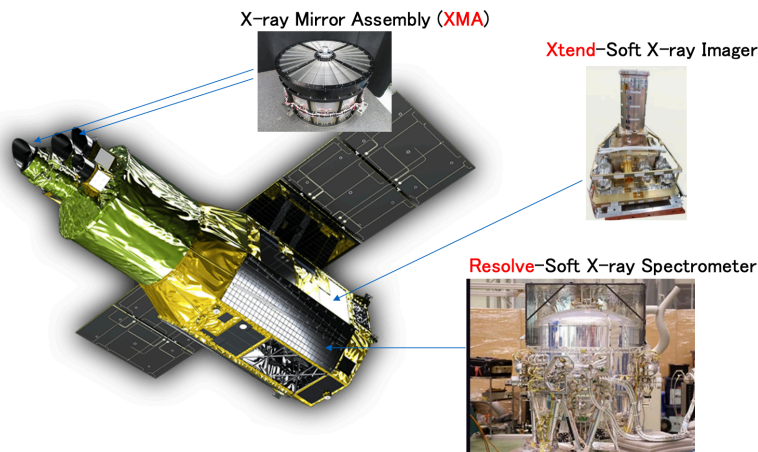


FIG. 1.15— Schematic view of the XRISM satellite and payload (*Resolve* and *Xtend*). Adapted from the XRISM Quick Reference (v2.1), Fig. 2 (XRISM SOC-PVO 2022).

Resolve measures photon energies thermally with a microcalorimeter array. It is non-dispersive and maintains approximately constant energy resolution across the bandpass, which aids centroid measurements in crowded spectral regions. Dedicated calibration hardware, including modulated X-ray sources, tracks the in-flight energy scale for velocity measurements (XRISM SOC-PVO 2022). Each absorbed photon produces a temperature rise that is converted into a pulse height (Fig. 1.16) (JAXA XRISM Project 2025).

Resolve achieves high spectral resolution over a $\sim 3' \times 3'$ field of view using a 6×6 pixel array. The mission requirement of $\Delta E \leq 7$ eV (FWHM) at 6 keV allows precise

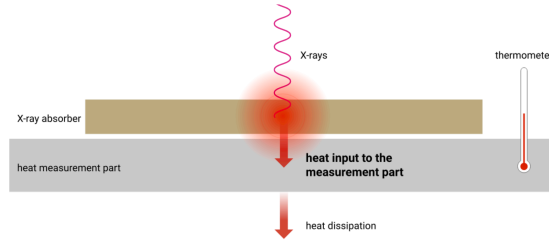


FIG. 1.16— Microcalorimeter measurement concept: photon absorption produces a measurable temperature rise. Adapted from the XRISM technology overview (JAXA) ([JAXA XRISM Project 2025](#)).

measurements of Doppler shifts and intrinsic line broadening, particularly for Fe XXV and Fe XXVI in the Fe K band.

1.5.3 NuSTAR

NuSTAR uses two co-aligned focusing telescopes, FPMA and FPMB, to measure the hard X-ray spectrum over 3–79 keV ([Harrison et al. 2013](#)). In this thesis, its high-energy coverage constrains continuum curvature during construction of the broad-band continuum model. It does not provide the high-resolution line constraints used in the persistent and dip absorber fits.

1.6 Research Goals

Time-, phase-, and morphology-resolved studies treat dipping intervals as recurrent sight lines through disc-associated plasma. They examine the structure of the stream-impact region, its relation to a persistent disc atmosphere, and the kinematics of the absorbing gas ([Boirin et al. 2004](#); [Díaz Trigo et al. 2006](#); [Xiang et al. 2009](#); [Psaradaki et al. 2018](#); [Trueba et al. 2020](#); [Rojas-Martínez 2024](#); [Caruso et al. 2025](#)).

The central question is:

How do the ionisation structure, column density, and line-of-sight kinematics of the disc-associated plasma differ between persistent and dip intervals as the observed dip morphology evolves, and do these measurements show evidence for a disc wind in 4U 1916–053 or remain consistent with a bound disc atmosphere?

The EPIC-pn light curve defines three time-selected morphology intervals, with persistent and dip selections in each. We compare the corresponding RGS and *Resolve* spectra with the same continuum and three-zone absorber model to measure changes in column density, ionisation, and line-centroid shifts. RGS constrains the soft-band absorption, *Resolve* measures the highly ionised Fe K lines, and EPIC-pn defines the timing selections

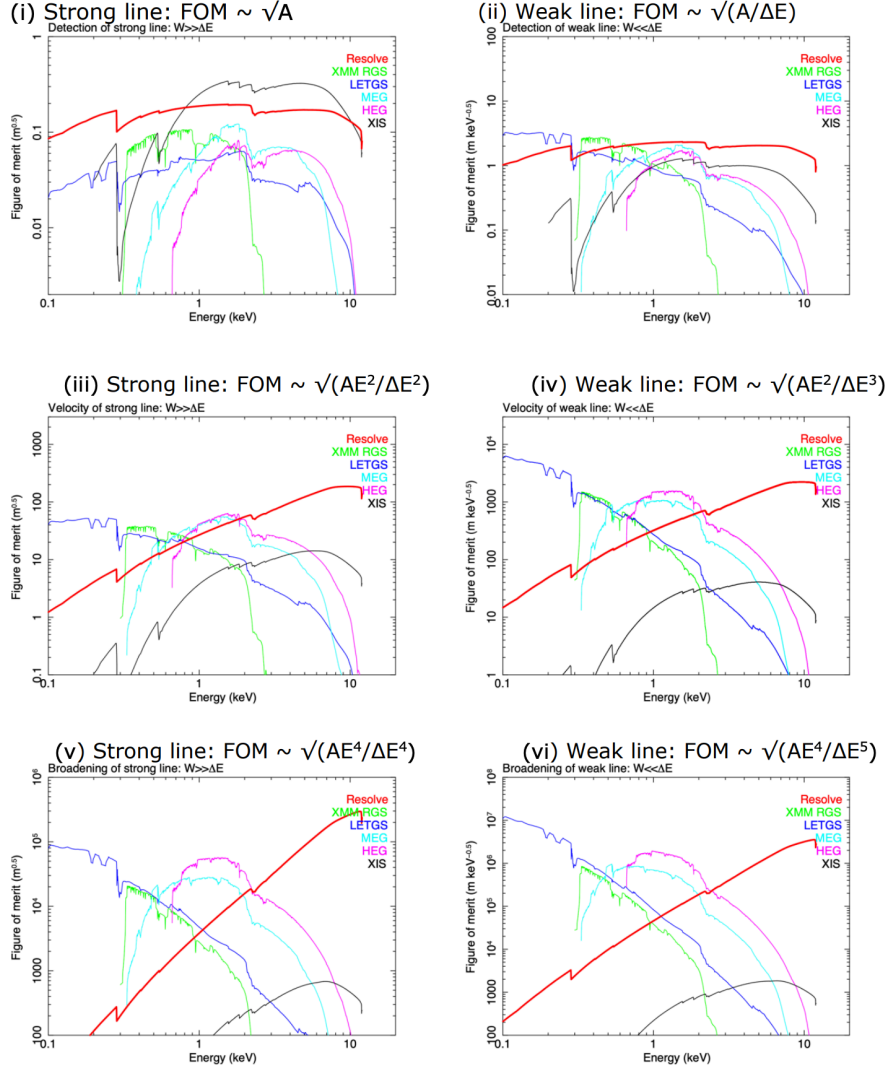


FIG. 1.17— Pre-launch figures of merit for spectral-line measurements. From top to bottom, the rows compare line detection, centroid (velocity) measurement, and line-broadening measurement. The left and right columns show strong and weak lines, defined by whether the equivalent width is larger or smaller than the instrumental energy resolution. Larger values indicate greater sensitivity. The curves compare nominal *Resolve* performance with XMM-Newton/RGS, *Chandra* gratings, and Suzaku/XIS. The *Resolve* curves assume the planned 0.3–12 keV response and do not include the attenuation caused by the closed gate valve. Figure from the XRISM Quick Reference (v2.1), Fig. 17 (XRISM SOC-PVO 2022).

and broad-band continuum. NuSTAR constrains the high-energy continuum curvature but does not enter the persistent and dip absorber fits.

The EPIC-pn observation also contains three Type-I bursts, two of which occur near boundaries between the morphology intervals. We analyse them as a secondary test of whether a brief period of intense irradiation changes the persistent emission or contributes to the subsequent dip-morphology evolution. The burst-oscillation search tests whether the reported 269–272 Hz signal recurs; see Section 1.4.2 for the scientific motivation.

Chapter 2

Observations and Data Reduction

2.1 Observations

The coordinated data set contains observations of 4U 1916–053 with *XMM-Newton*, XRISM, and NuSTAR. *XMM-Newton*/EPIC-pn supplies the high-throughput broad-band timing and spectral data, *XMM-Newton*/RGS the soft X-ray grating spectra, and XRISM/*Resolve* the high-resolution, non-dispersive hard-band spectra. The NuSTAR observation extends the continuum coverage to hard X-rays and constrains the curvature of the Comptonised component.

The observations are summarised in Table 2.1. The durations listed there are nominal observation durations where available. Cleaned exposures depend on the final good-time intervals (GTIs), detector selections, and time selections, and are therefore quoted in the reduction sections where they are relevant.

TABLE 2.1— Observation log for the data sets used in this thesis. Durations are the elapsed observation spans listed in the mission archives; screened or dead-time-corrected exposures are quoted in the text where relevant.

Mission	Instrument	ObsID	Start (2024)	End (2024)	Duration
<i>XMM-Newton</i>	EPIC-pn/ RGS	0934990201	Apr 16 04:03:41	Apr 17 09:17:01	105.2 ks
XRISM	<i>Resolve</i>	300039010	Apr 15 10:32:04	Apr 18 06:26:04	244.4 ks
NuSTAR	FPMA/ FPMB	30902017002	Apr 16 08:56:07	Apr 16 22:01:12	47.1 ks

EPIC-pn defines the morphology sectors and persistent/dip selections and provides the broad-band spectra used to construct the continuum. The simultaneous RGS and *Resolve* spectra then provide the soft- and hard-band line diagnostics, respectively.

2.1.1 XMM-Newton

XMM-Newton observed 4U 1916–053 under ObsID 0934990201 from 2024 April 16 to 2024 April 17 (Jansen et al. 2001). The EPIC-pn camera was operated in Timing/FastTiming mode, which is well suited to this source because it preserves high time resolution while collapsing one imaging coordinate (Strüder et al. 2001). The resulting light curve separates persistent emission, dips, and Type-I X-ray bursts. The RGS instruments observed simultaneously, providing grating spectra in the soft X-ray band (den Herder et al. 2001). The reduction follows the standard SAS workflow for EPIC-pn Timing-mode and RGS analysis, with source-specific choices described below.

2.1.2 XRISM

XRISM observed 4U 1916–053 under ObsID 300039010 from 2024 April 15 to April 18 (Tashiro 2022; XRISM Science Team 2020). *Resolve* was operated in PX_NORMAL mode, with an archive exposure of 113.9 ks. Its non-dispersive microcalorimeter response above 2 keV makes *Resolve* the primary data set for the Fe-K analysis, without relying on

grating dispersion for the highly ionised iron features. The *Resolve* reduction follows the XRISM ABC Guide and Quick-Start Guide, with additional screening applied to match the science goal and the behaviour of the observation ([XRISM Guest Observing Facility and Science Data Center 2024](#); [XRISM Science Data Center 2026b](#)).

2.1.3 NuSTAR

NuSTAR observed 4U 1916–053 during the coordinated observing window (ObsID 30902017002), with FPMA/FPMB archive exposures of 23.2/23.0 ks. Its hard-band spectrum was used to constrain the broad-band continuum. The observation covers only part of the XMM-Newton window and does not provide the high-resolution line diagnostics available from RGS and *Resolve*, so it was not included in the six persistent and dip absorber fits.

2.2 XMM-Newton Data Reduction

The XMM-Newton data were reduced with the Science Analysis Software (SAS v22.1.0), using a Current Calibration File (CCF) index generated on 2025 September 28 and following the ABC Guide where appropriate ([NASA/GSFC XMM-Newton Guest Observer Facility 2025](#)). The reduction produced exposure-corrected EPIC-pn light curves, sector-matched EPIC-pn and RGS spectra for the persistent and dip selections, and a response for each extraction. The command-level history is kept in the reduction notes; this chapter records the choices that determine the scientific products. A compact command record and the exact product-level calibration provenance are given in [Appendix A](#).

2.2.1 EPIC-pn Timing-Mode Products

The EPIC-pn event list was filtered with `evselect`; the same task was used to extract the light curves and spectra. Single and double events were retained with `PATTERN<=4`. Standard event-quality filtering used `#XMMEA_EP` for light curves and `FLAG==0` for the spectral products. The spectra were extracted over 0.3–10 keV, whereas the source light curves used to define the time selections covered 0.3–12 keV. A separate 10–12 keV off-source light curve was used to screen for high background, as described below.

The source and off-source diagnostic regions were defined in RAWX coordinates. The source region was RAWX 28–48, centred on the one-dimensional count distribution of the Timing-mode data. RAWX 5–8 was retained as a restricted off-source region for background checks, while RAWX 5–10 was used for the high-energy flare diagnostic because it provides more counts at large separation from the source-dominated columns. Neither off-source region was subtracted from the science products.

Soft-proton flaring was assessed from a high-energy background light curve extracted

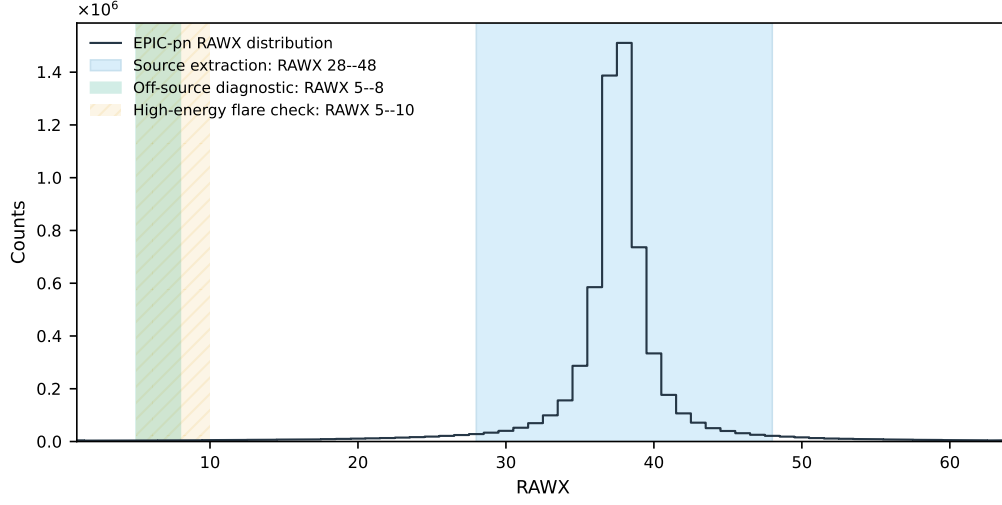


FIG. 2.1— EPIC-pn Timing-mode RAWX distribution. The source region is RAWX 28–48; RAWX 5–8 and RAWX 5–10 are off-source diagnostic regions.

away from the source. The diagnostic light curve used single-pixel events (`PATTERN==0`) in the 10–12 keV band from RAWX 5–10. I tested rate-based and sigma-clipping GTI definitions during the reduction, but some background-region choices showed evidence for source or burst leakage and could remove genuine burst time. For this reason, `tabgtigen` was used to create a GTI that removes only the obvious terminal high-background interval. Events with `TIME <= 8.29727e8` were retained, where `TIME` is the XMM-Newton Mission Elapsed Time (MET), measured in seconds from 1998 January 1 00:00:00 TT. This leaves a screened EPIC-pn exposure of approximately 96.1 ks, compared with the 105.2 ks nominal observation duration.

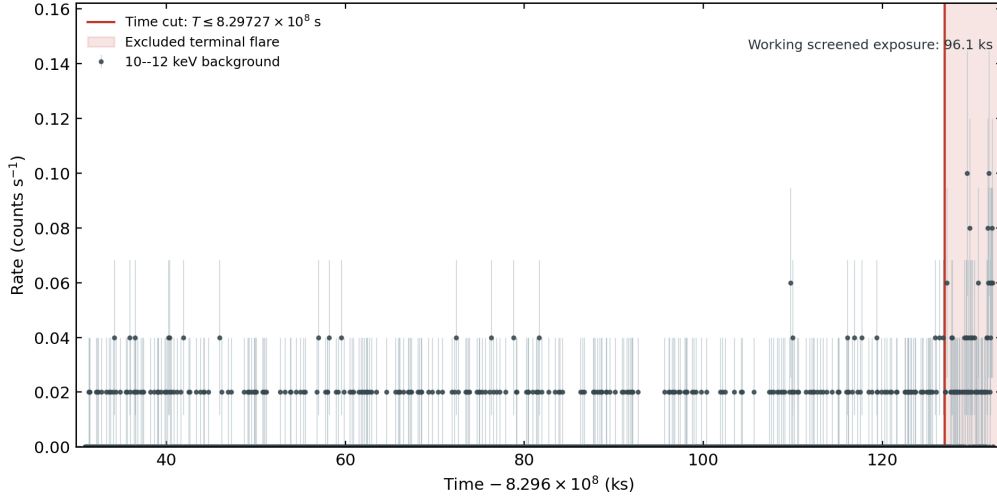


FIG. 2.2— High-energy EPIC-pn background light curve plotted against XMM-Newton Mission Elapsed Time (MET). The shaded terminal flare is excluded by retaining events with $T \leq 8.29727 \times 10^8$ s.

Broad-band source light curves were corrected with `epiclccorr` for dead time, GTI and exposure losses, and instrumental effects such as vignetting, bad pixels, and PSF losses. No background light curve was supplied, so these products were not background-subtracted. The same extraction and correction procedure was used for persistent, dip, and burst selections. The main interval-selection light curve was binned at 20 s. This binning is short compared with the $\simeq 50$ min dipping cycle of 4U 1916–053, but still gives sufficient counts to identify persistent emission, dips, and burst intervals robustly.

EPIC-pn spectra were extracted from the same time-filtered event list for the full observation and for each persistent or dip GTI. Off-source products were kept for quality control but were not associated with the source spectra during fitting. Spectral areas were computed with `backscale`; redistribution matrices and ancillary response files were generated with `rmfgen` and `arfgen`. Pile-up was tested with `epatplot`. Where the pattern distribution suggested that pile-up could affect the spectrum, a second extraction excluding the central RAWX columns was inspected as a consistency check. This check is not used as a separate result unless it changes the adopted spectral products.

2.2.2 RGS Products

The RGS1 and RGS2 data were processed with the standard `rgsproc` workflow. Before extracting the persistent and dip spectra, the cross-dispersion–wavelength images were inspected with the source and off-source extraction boundaries overlaid, as recommended in the XMM-*Newton* ABC Guide ([NASA/GSFC XMM-Newton Guest Observer Facility 2025](#)). This check verifies that the source trace is enclosed by the adopted source aperture and that the diagnostic off-source strips do not intersect the bright trace. The two instruments were inspected separately because their CCD gaps occur at different wavelengths.

Corrected RGS light curves were produced with `rgslccorr`, using the first and second spectral orders where useful for timing inspection. They provide a soft-band check on the EPIC-pn selections, but do not define independent selection boundaries. First-order spectra were extracted with `rgsspectrum` for RGS1 and RGS2, and the corresponding response matrices were generated with `rgsrmfgen`. For the persistent and dip products, the event lists were filtered with the pn-defined GTIs before extraction.

The RGS1 and RGS2 spectra were combined with `rgscombine` only for visual checks. The quantitative fits retain them as separate, simultaneous SPEX regions, preserving the response of each instrument and allowing candidate features to be checked in both spectrometers.

2.3 XRISM *Resolve* Data Reduction

The XRISM/*Resolve* data were reduced with HEASoft 6.34, XRISM software release 21Aug2024 V001, and the CALDB set recorded in the cleaned event product

(Appendix A). The resulting products are cleaned *Resolve* event files, 2–10 keV light curves for checking the persistent/dip selection, and spectra whose responses use the same time and detector selections (XRISM Guest Observing Facility and Science Data Center 2024).

2.3.1 Event Processing and Screening

The *Resolve* data were first reprocessed with **xpipeline** through stages 1–3. The cleaned event products were then screened further before extracting science products.

Additional filters were applied following the *Resolve* analysis guidance and the inspection of this observation. Pixel-pixel coincident events were screened using pulse-shape and status selections, and anomalous Ls events were removed by retaining events with $ITYPE < 4$. A barycentric correction was applied with **barycen** using the spacecraft orbit file and the sky position of 4U 1916–053. The *Resolve* timing products therefore use barycentric times.

The cleaned event file was then filtered in XSELECT. High particle-background periods were excluded with $CORTIME > 8$, and off-pointing intervals were removed with $ANG_DIST < 0.1$. Pixel 27 was excluded because it showed anomalous behaviour in the inspected products; pixel 12 is the out-of-field calibration pixel and is absent from the cleaned celestial-event data. The exact pixel expression is recorded in Appendix A. The resulting event file was used to create 2–10 keV light curves and persistent and dip spectra.

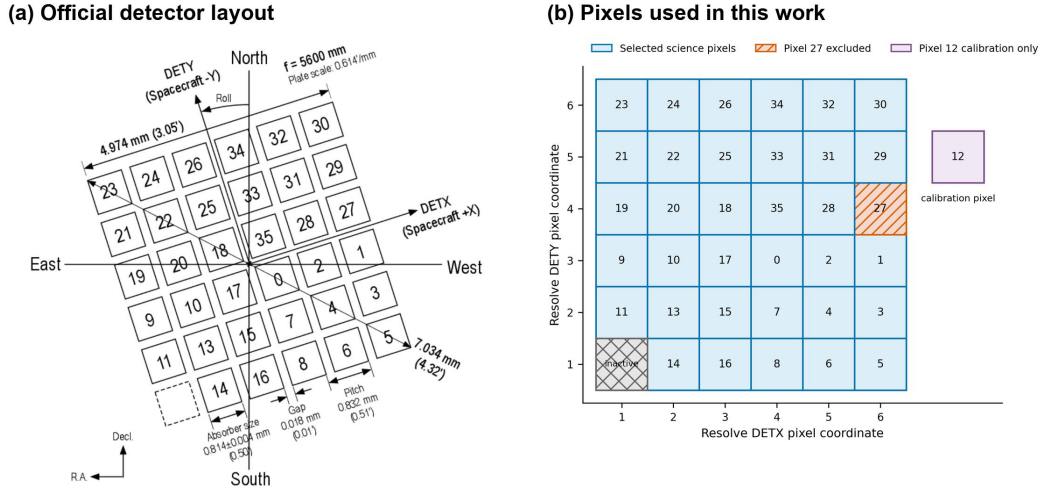


FIG. 2.3— *Resolve* detector layout and pixel selection. Panel (a) shows the official focal-plane layout (XRISM Science Data Center 2026a); panel (b) marks the 34 selected science pixels. Pixel 27 is excluded for irregular gain behaviour and pixel 12 is the out-of-field calibration pixel.

2.3.2 Spectra, Responses, and Timing Products

Resolve 2–10 keV light curves were extracted from the screened event file with XSELECT using 5 s bins. They were used only to verify how the pn-defined intervals appear in *Resolve*; they did not redefine the pn selections.

Spectra were extracted from the cleaned and time-filtered *Resolve* event files, with the high-resolution grade selection used in the current analysis. Separate spectra were produced for persistent and dip intervals. These selections were checked against the EPIC-pn timing structure.

For each *Resolve* spectrum, the response was generated using the same event and detector selections. Redistribution matrix files were produced with `rslmkrmf`. Exposure maps were computed with `xaexpmap`, and ancillary response files were generated with `xaarfgen` for a point source at the position of 4U 1916–053. The ARF generation used the detector-region file that excludes pixel 27. A separate response is required because the GTIs and detector selection determine the effective exposure and response of each spectrum.

2.4 Background Treatment

Background products were retained for quality control but were not subtracted from the light curves or spectra used in the analysis. Over the instrument-specific energy ranges retained for fitting, 4U 1916–053 is strongly source dominated. In the EPIC-pn broad-band check, the off-source rate was approximately 350 times lower than the source rate, while the Timing-mode off-source columns could still contain a small contribution from the source wings. Subtracting this weak and potentially source-contaminated signal would add statistical noise and could remove genuine source photons. No background was therefore subtracted from the persistent, dip, or burst products from any instrument. Off-source products were used only to identify detector-background problems, such as the terminal EPIC-pn flare, and to confirm that the fitted bands remained source dominated.

2.5 Time and Count-Rate Selection

The high-throughput pn light curve defines the persistent (non-dip) intervals, dip intervals, and sector boundaries that are then transferred to RGS and *Resolve*. These empirical selections do not assume an absorber geometry. Type-I bursts are removed from the persistent and dip selections and processed separately.

2.5.1 EPIC-pn Reference Light Curve

The persistent, dip, and burst intervals were defined first from the background-unsubtracted EPIC-pn 0.3–12 keV light curve, because pn provides the highest-

throughput timing view of the coordinated observation. Persistent emission, dips, and burst intervals were identified from the 20 s light curve shown in Fig. 2.4.

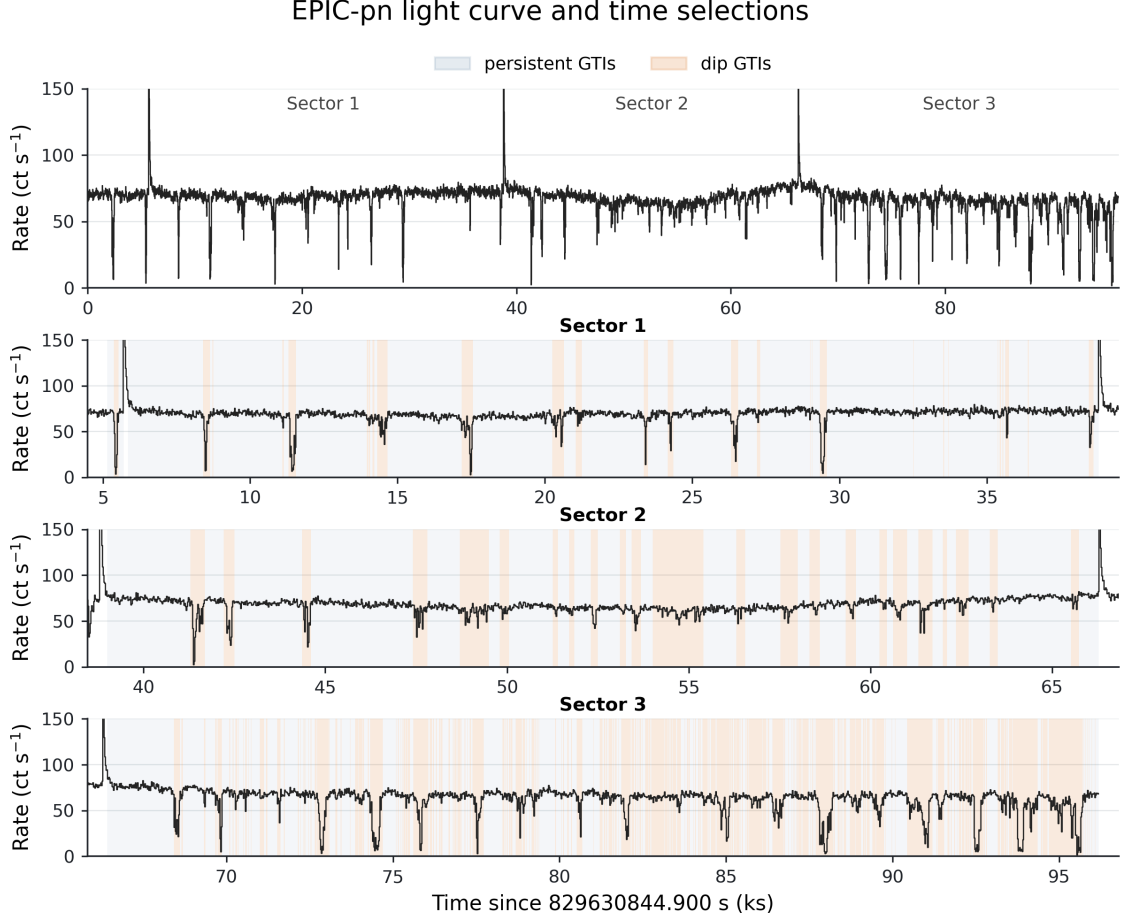


FIG. 2.4— Background-unsubtracted EPIC-pn 0.3–12 keV light curve binned at 20 s. The top panel shows the full screened observation and the lower panels the three sectors; blue and orange shading marks persistent and dip GTIs. Burst peaks extend above the displayed 150 counts s^{-1} limit.

The three sectors are time-selected morphology intervals. They were chosen because the dip morphology changes with time: sector 1 contains relatively isolated dips, sector 2 shows broader and more structured attenuation, and sector 3 is dominated by frequent deep dips. Keeping these intervals separate prevents spectra with visibly different dip behaviour from being averaged together. Table 2.2 summarises the sector boundaries and selection logic. Exact GTI filenames, interval counts, and summed durations are reported once in Table A.1.

TABLE 2.2— EPIC-pn sector definitions and persistent/dip selection criteria. Times are mission elapsed times in seconds.

Sector	Time span	Selection basis
1	829636000–829669646	Manual windows; dips are relatively isolated.
2	829669850–829697131	Manual windows; dips are broader and structured.
3	829697275–829727035	Rate threshold; persistent >65 counts s^{-1} and dip <65 counts s^{-1} .

2.5.2 Transfer to RGS and *Resolve*

The pn-defined persistent and dip intervals were transferred to the RGS and *Resolve* data¹ and intersected with the valid GTIs of each high-resolution instrument. The persistent/dip label is therefore inherited from pn, whereas RGS and *Resolve* contribute photons only during their own valid observing times; neither instrument is reclassified using an independent count-rate threshold. The resulting alignment is shown at observation-wide and sector-resolved scales in Figs. 2.5 and 2.6, respectively.

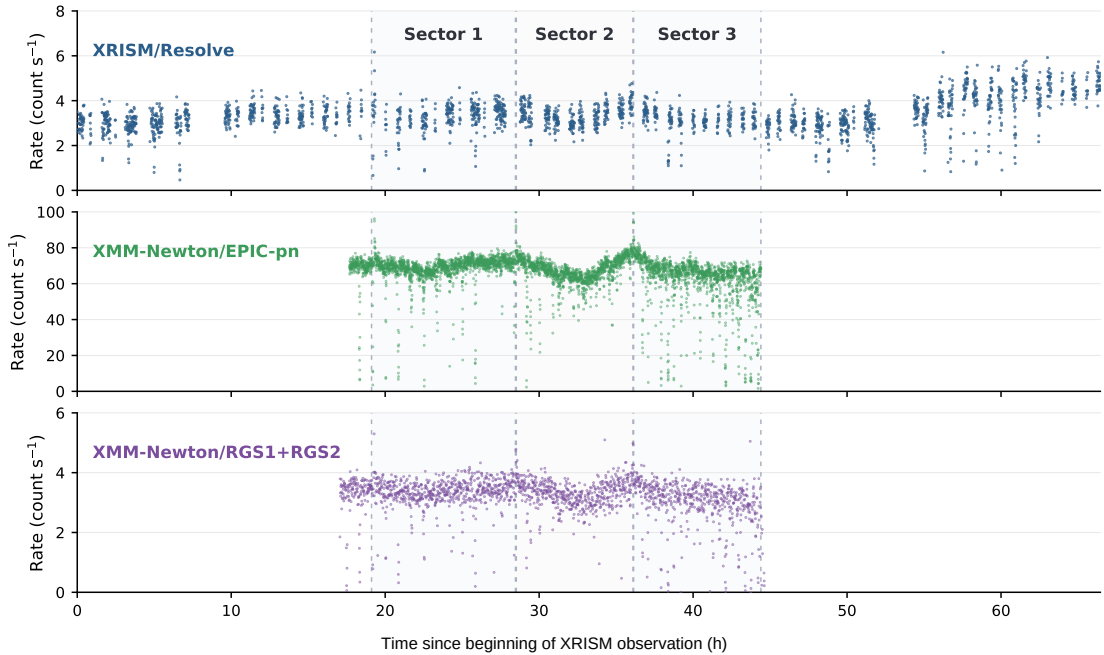


FIG. 2.5— Simultaneous XRISM/*Resolve*, XMM-Newton/EPIC-pn, and combined RGS1+RGS2 light curves shown on a common elapsed-time axis. Shading and dashed lines mark the pn-defined sectors; gaps in *Resolve* coverage arise mainly from the low-Earth orbit and screening.

¹During thesis revision, it was found that the XMM-Newton times had not been barycentrically corrected, unlike the XRISM times, giving a formal relative offset of approximately 1 min. A check with a 1-min relative shift left the sector boundaries unchanged and changed the mean *Resolve* count rates of the accumulated persistent and dip selections only slightly. Since the spectra integrate over much longer intervals and bursts were excluded, this offset is unlikely to affect the spectral results.

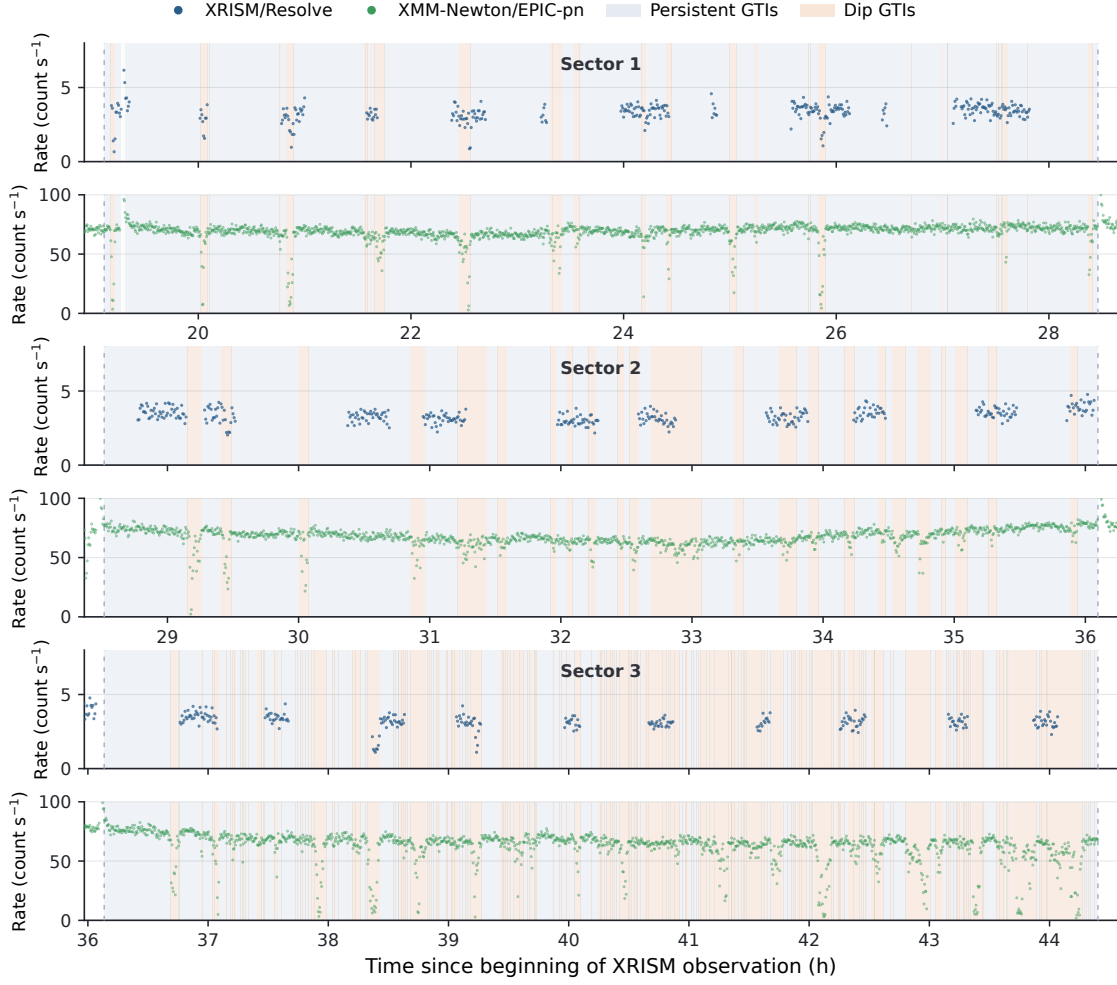


FIG. 2.6— Sector-by-sector XRISM/*Resolve* and XMM-Newton/EPIC-pn light curves. Pale blue and orange bands mark the pn-defined persistent and dip GTIs, respectively, on the displayed elapsed-time axis.

The final sector-matched data set contains the pn timing and continuum reference together with sector-matched persistent and dip spectra from RGS and *Resolve*. Burst intervals are excluded before the spectra are accumulated, and NuSTAR is used only during continuum construction. Chapters 3 and 4 analyse the absorber with the RGS and *Resolve* spectra.

2.5.3 Burst Detection, Duration, and Spectral Products

Three type-I X-ray bursts are identified in a 1-s, 0.3–10 keV EPIC-pn light curve. The timing analysis is used mainly to define reproducible windows: the burst intervals removed from the persistent and dip spectra, the peak, tail, and time-resolved products used for the burst spectral tests, and the event lists used for the burst-oscillation search. The three burst light curves and the fitted timing points are shown in Fig. 4.9. For

each burst, the pre-burst level is measured over -50 to -15 s relative to the first 1-s onset bin. The net peak rate is defined as the observed peak rate minus this pre-burst level, and the onset is the centre of the first 1-s bin above 5 per cent of the net peak. A linearly interpolated within-bin crossing is retained only as a diagnostic and is not used to anchor the spectral bins or oscillation windows. The decay is fitted over 300 s after the peak to constrain the late tail, while the shorter D_5 interval defined below gives the adopted burst duration. Four decay functions are compared for two purposes: the single exponential, power law, and power law plus one-sided Gaussian follow the MINBAR duration comparison, while the additive two-exponential model is an additional empirical model for the measured EPIC-pn decays. The fits are run with custom Python scripts using NumPy, Astropy, and `scipy.optimize.least_squares`; for each successful model, $\text{BIC} = \chi^2 + k \ln n$, where k is the number of free parameters and n is the number of fitted points.

The adopted burst duration, D_5 , is the time from the 5-per-cent onset to the downward crossing at which the fitted net burst rate reaches 5 per cent of the net peak. The MINBAR-style measurement is retained so that the burst durations can be compared with catalogue values and historical measurements (Galloway et al. 2020). For the products used in the spectral analysis, the additive two-exponential model is used only as an empirical description of the measured EPIC-pn decay. It has the lowest BIC for all three bursts and defines the adopted extraction windows. Its fitted form is

$$R_{\text{net}}(t) = C + A_1 \exp(-t/\tau_1) + A_2 \exp(-t/\tau_2), \quad (2.1)$$

where the two terms provide an empirical description of the faster and slower parts of the measured decay, without a separate physical interpretation. The resulting duration measurements and decay-model comparisons are presented in Section 4.6.1.

For the time-averaged spectral comparison, the peak interval contains the two brightest adjacent 1-s bins. The tail begins at the end of that 2-s interval and ends at the adopted two-exponential D_5 . The exact extraction windows are listed in Table 2.3. Each pre-burst candidate window is intersected with the corresponding persistent GTI and explicitly excludes the dip GTIs. For bursts 1 and 2 the outer window is therefore longer than 300 s, because two retained persistent pieces are needed to accumulate 300 s of non-dip exposure; burst 3 uses one continuous 300-s interval. The resulting pre-burst spectrum is used only to constrain the persistent source model.

Figure 2.7 locates the three bursts in the observation and shows the time intervals used for the burst products.

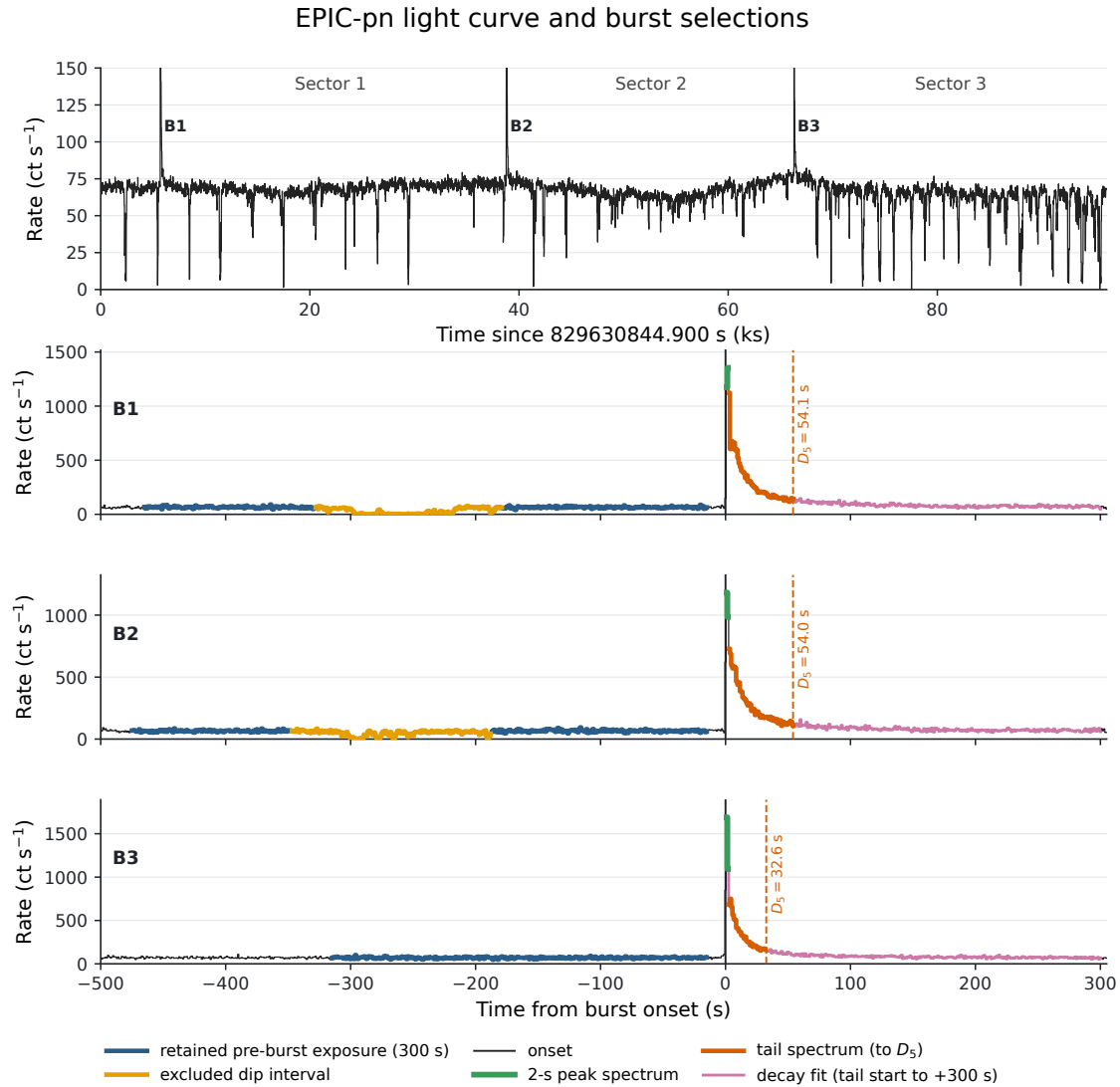


FIG. 2.7— EPIC-pn 0.3–10 keV light curve and adopted burst selections. The top panel locates bursts B1–B3 within the three morphology sectors. The lower panels show the 1-s light curves relative to each burst onset. Blue marks the retained 300-s persistent pre-burst exposure; yellow marks excluded dip intervals; green marks the 2-s peak spectrum; orange marks the tail spectrum from the end of the peak interval to D_5 ; and magenta marks the decay-fitting interval from the start of the tail to 300 s after onset. Solid black lines mark the burst onsets and dashed orange lines mark D_5 .

The 2-s EPIC-pn peak spectra use RAWX 28–48 with the central RAWX = 38 column removed. The Burst 1 source centroid lies near RAWX = 37.8, so RAWX = 38 is the central detector column. It is removed for the peak products because the full-strip Burst 1 peak showed a marginal hard-band double-event excess, whereas the one-column excision retained 1834 of 2527 all-pattern events and brought the hard-band pattern ratios into agreement with the expected values. The time-averaged pre-burst and tail spectra use the full RAWX 28–48 strip. All science spectra retain FLAG==0, PATTERN<=4, and PI=300:10000. For `epatplot`, the diagnostic event lists are generated from all

TABLE 2.3— EPIC-pn time windows used for the time-averaged burst spectral products.

Burst	Pre-burst candidate window	Retained selection	Peak interval (2 s)	Tail interval
B1	829636083.400– 829636533.400	Sector 1 persistent; dips excluded	829636548.900– 829636550.900	829636550.900–829636602.469
B2	829669172.400– 829669632.400	Sector 1 persistent; dips excluded	829669647.900– 829669649.900	829669649.900–829669701.421
B3	829696820.400– 829697120.400	Sector 2 persistent; dips excluded	829697135.900– 829697137.900	829697137.900–829697168.008

Notes. All boundaries are spacecraft **TIME** values in seconds. The pre-burst column gives the outer candidate window; events are additionally restricted to the listed persistent sector GTI and explicitly exclude dip GTIs, yielding 300 s of retained pre-burst exposure.

event patterns because the task compares the observed and expected single- and double-event pattern fractions. Pile-up would appear as a single-event deficit together with a double-event excess. With the central column removed, the peak `epatplot` single-event fractions range from 0.998 to 1.002 in 0.5–2 keV and from 0.986 to 1.002 in 2–10 keV; the corresponding double-event fractions range from 1.065 to 1.100 and from 1.056 to 1.130. There is no significant combination of a single-event deficit and double-event excess, so the peak analysis uses this core-excised region.

For the time-resolved PRE test, the same core-excised region is used in every bin. The first 10 s from onset are divided into 1-s intervals, the next 10 s into 2-s intervals, and the remaining decay to D_5 into 4-s intervals, with the last bin truncated at D_5 . This gives 24, 24, and 19 spectra for bursts 1–3. One response pair is generated per burst and shared by its time bins because the detector mode, event selection, energy range, and RAWX region are unchanged.

2.5.4 Burst-Oscillation Search

A separate high-time-resolution search uses one full- D_5 event list per burst with the same core-excised EPIC-pn selection as the time-resolved PRE products. The search is run with a custom Python script using Astropy FITS I/O and NumPy FFT routines. Times are measured from the onset-bin centre defined in Section 2.5.3. The headers retain **TIMESYS=TT** and **TIMEREF=LOCAL**. No barycentric correction was applied, so the quoted frequencies are Fourier-bin centres calculated from local spacecraft times.

The events are binned at 1 ms. For each valid window, the NumPy FFT power is converted to the Leahy normalisation, $P = 2|F|^2/N_{\text{ph}}$, where F is the Fourier amplitude and N_{ph} is the number of events in the window. We use 0.5-s windows stepped by 0.25 s and 2-s windows stepped by 0.5 s, targeting short-lived or evolving structure and weaker coherent signals, respectively. Both configurations cover $100 \leq \nu < 500$ Hz, with Fourier spacings of 2 and 0.5 Hz. A pre-defined 260–280 Hz band is the primary test because the previous *RXTE* signal drifted from 269.3 to about 272 Hz (Galloway et al. 2001); the

full band is retained as a secondary blind search. Windows with less than 95 per cent GTI exposure are excluded. This removes three windows in each configuration around the 0.179-s gap near the peak of burst 3.

For Poisson noise, the single-trial tail probability of a Leahy power P is $\exp(-P/2)$. A family-wise false-alarm probability is the probability of at least one noise peak anywhere in the searched windows and Fourier bins. We set $\alpha = 10^{-3}$ and convert it to

$$P_{\text{thr}} = -2 \ln \left(\frac{\alpha}{N_{\text{tr}}} \right), \quad (2.2)$$

This is a Bonferroni trial correction: N_{tr} counts all tested Fourier bins across valid windows, configurations, and bursts. Separate thresholds are used for the pre-defined and blind searches. The independent-trial count is conservative because neighbouring windows overlap.

To quantify the sensitivity of the non-detections, we calculate analytic 95 per cent upper limits on the fractional-rms amplitude of a coherent sinusoid. For each burst and window length, the largest 260–280 Hz Leahy power in the highest-count valid window is converted to a signal-power upper limit using the non-central χ^2 distribution. The fractional rms is then $r_{\text{rms}} = \sqrt{P_{\text{sig},95}/N_{\text{ph}}}$ (Bilous & Watts 2019). The reported limits include corrections for the average FFT-bin response and the 1-ms time binning; background dilution is not included.

Chapter 3

Spectral Analysis

This chapter describes the SPEX setup, continuum, absorber model, and fitting sequence. Chapter 4 reports the fitted spectra and parameters, and Chapter 5 discusses their physical interpretation.

3.1 Spectral Fitting Framework

3.1.1 SPEX Data Preparation and Fit Setup

The source spectra are modelled with SPEX v3.08.01, using the official SPEX tools and model definitions as the reference for component names and parameter conventions (SPEX Development Team 2026a,b). The OGIP-style spectra and response files are converted to SPEX format with `trafo`. No background spectra are attached because, as described in Chapter 2, the source count rate is much higher than the background over the fitted energy ranges. For each morphology sector, the persistent and dip count-rate selections are fitted as separate data sets. Within each data set, instrument regions can be fitted simultaneously while retaining their own responses and energy ranges. In the final absorber fits, these regions are *Resolve*, RGS1, and RGS2; pn is not included. To avoid confusion with internal SPEX bookkeeping, “sector 1–3” throughout this thesis refers only to the temporal divisions defined from the pn light curve.

The broad-band continuum check uses EPIC-pn over 0.6–10 keV, RGS over 7–35 Å with energies below 0.49 keV ignored, and NuSTAR over 3–50 keV. The final fits use *Resolve* over 2–10 keV and the same RGS range. NuSTAR therefore tests the high-energy curvature during continuum construction but is not included in the final absorber fits.

3.1.2 Fit Statistic

The fits use the Cash statistic, which is appropriate for Poisson-distributed spectral data and avoids the need to impose Gaussian assumptions on low-count bins (Cash 1979). Optimal or fixed rebinning is applied where stated, but the statistical comparison uses the Poisson likelihood for the retained spectral bins.

3.2 Continuum Modelling

The intrinsic continuum model was selected through trial fits. Initial two-component models, `bb+pow` and `dbb+pow`, both gave unsatisfactory fits. In the `dbb+pow` fit, the power law dominated while the disc-blackbody contribution was very weak. The adopted model therefore retains both thermal components and uses Comptonisation for the hard continuum:

$$C(E) = \text{bb} + \text{dbb} + \text{comt}. \quad (3.1)$$

The blackbody component represents thermal emission associated with the neutron star, the disc-blackbody component describes the multi-temperature accretion-disc emission, and `comt` accounts for the Comptonised continuum. Earlier studies of 4U 1916–053 used similar thermal components. The 0.6–10 keV XMM-*Newton* spectrum was described with a disc blackbody plus a power law by Boirin et al. (2004), while the 1.9–9 keV Chandra/HETG spectra were fitted with `bb+dbb` by Trueba et al. (2020). The `bb+dbb` model used by Trueba et al. (2020) motivated the inclusion of both thermal components in the present model, although their 1.9–9 keV HETG energy range did not constrain the broad-band hard continuum. The broad-band BeppoSAX spectrum was instead well described by a blackbody plus a cut-off power law (Church et al. 1998). This cut-off indicates that the hard continuum turns over at high energies instead of continuing as a power law. Such curvature is consistent with Comptonisation by electrons at a finite temperature.

`comt` is therefore used instead of a simple `pow` component. It describes the Comptonisation of seed photons in a hot plasma through the seed-photon temperature, electron temperature, and optical depth (SPEX Development Team 2026a). Unlike a power law, `comt` includes the seed-photon turnover and the high-energy curvature associated with a finite electron temperature. The soft-X-ray data constrain the low-energy continuum, while the auxiliary NuSTAR coverage constrains the hard-band curvature.

A low-temperature `hot` component is placed in front of the full model to describe neutral foreground absorption in the Galaxy. Although `hot` is a collisional-ionisation-equilibrium transmission model, fixing its temperature at 10^{-6} keV makes the gas effectively neutral, allowing it to reproduce cold absorption (SPEX Development Team 2026a). This foreground component is distinct from the additional low-temperature `hot` component introduced only in the dip fits. The latter represents cold or weakly ionised material local to the binary and modifies only the direct continuum during dipping. This separation follows previous analyses in which changes in local neutral absorption accompanied the stronger changes in the ionised absorber (Díaz Trigo et al. 2006). For example, independent dust-scattering measurements of 4U 1624–490 showed that a substantial part of the total absorbing column can be local to the binary rather than Galactic (Xiang et al. 2007).

The continuum is determined separately for each of the three sectors. Within a given sector, the persistent and dip spectra share the same intrinsic continuum parameters, and the *Resolve* and RGS regions are coupled to the same continuum solution. Continuum parameters are not coupled between different sectors, allowing for the observed long-timescale spectral evolution. The Galactic `hot` column density is tied across all morphology sectors and the persistent and dip selections because the foreground column cannot vary during the observation. In contrast, the column densities of the `pion` zones and the dip-local `hot` component are allowed to vary between the persistent and dip selections.

TABLE 3.1— SPEX model components used in the fitting framework. The table summarises the model roles; numerical parameter values are reported in the Results chapter.

Component	Role in the model	Main parameters followed in this thesis
hot (Galactic)	Neutral Galactic foreground absorption	Column density N_{H} ; temperature fixed at the low-temperature neutral limit
hot (dip-local)	Additional cold absorption local to the binary	Column density N_{H} for the dip selections; applied only to the directly viewed continuum
bb	Blackbody continuum	Persistent thermal component; a separate bb with free temperature and normalisation describes each burst
dbb	Disc-blackbody continuum	Inner-disc temperature proxy and normalisation; not treated as a unique radius without further assumptions
comt	Comptonised continuum	Seed temperature, electron temperature/optical-depth parameters, and normalisation where allowed by the data
pion	Photoionised absorber zone	N_{H} , $\log \xi$, velocity broadening, velocity shift, and covering assumptions
Scattered continuum	A fraction of the intrinsic continuum that bypasses the local absorbers	Same spectral shape as the direct continuum; normalisation fixed to $f_{\text{sc}} = 0.09$ relative to the direct continuum
$f_a C_{\text{pers}}$	Variable persistent emission during bursts	Pre-burst continuum shape fixed; one fitted factor scales the bb , dbb , and comt normalisations together

3.3 Photoionised Absorber Modelling

3.3.1 pion Models

The absorber is modelled with the SPEX **pion** component. Earlier XMM-*Newton* analyses of dipping LMXBs, including 4U 1916–053, used **xabs** to link the ionic columns through a pre-calculated photoionisation balance (Díaz Trigo et al. 2006). The **pion** component instead recalculates the ionisation and thermal balance as the ionising continuum changes and as the radiation is processed by successive absorber zones. This treatment accommodates the different sector continua and the use of several absorbing layers. Trueba et al. (2020) likewise used **pion** for the *Chandra*/HETG spectra of 4U 1916–053, noting its self-consistent response to the incident continuum. In H-like ions, the $\text{Ly}\alpha$ feature comprises the closely spaced $\text{Ly}\alpha_1$ and $\text{Ly}\alpha_2$ transitions, which **pion** models separately. This treatment is relevant here because the Fe XXVI doublet ratio and line-centroid shifts are used in the absorber analysis. The final absorber fits therefore use **pion**.

Each **pion** zone is described primarily by a hydrogen-equivalent column density N_{H} , an ionisation parameter $\log \xi$, a velocity broadening parameter, and a velocity shift. These parameters are model-dependent and are therefore interpreted only within the adopted continuum and absorber geometry. The additional low-temperature dip-local **hot** component is included in the dip model to account for local cold absorption that is not represented by the ionised zones.

3.3.2 Saturated Lines and Scattered Continuum

The *Resolve* spectra indicate that the Fe XXVI $\text{Ly}\alpha_1$ and $\text{Ly}\alpha_2$ line ratio is inconsistent with the optically thin ratio. For optically thin absorption, the $\text{Ly}\alpha_1/\text{Ly}\alpha_2$ equivalent-width ratio is expected to be approximately 2:1 because the oscillator strength of $\text{Ly}\alpha_1$ is about twice that of $\text{Ly}\alpha_2$. Once $\text{Ly}\alpha_1$ saturates, its equivalent width grows more slowly with column density and the ratio decreases towards unity. The absorption troughs also remain above zero flux. To account for the non-zero residual flux in the saturated line cores, we include a second copy of the intrinsic continuum with the same spectral shape as $C(E)$ but a normalisation fixed to $f_{\text{sc}} = 0.09$ of the direct continuum. This component represents photons that reach the observer without passing through the local photoionised and dip absorbers, for example after scattering or reprocessing elsewhere in the system.

The fixed value of 0.09 comes from an analysis of the full XRISM spectrum by Peluso et al. (in preparation). With this value, the continuum-plus-absorber fit reproduces the observed Fe XXVI doublet ratio while retaining non-zero residual flux in the saturated line cores. The fraction is kept fixed to limit degeneracy between f_{sc} and N_{H} . It should therefore be regarded as a modelling assumption supported by the line profiles, not as an independent measurement of the scattering efficiency or geometry. Scattering or obscuration has also been considered in previous broad-band dip spectra of 4U 1916–053 (Church et al. 1998).

Thus, both continuum paths are attenuated by the Galactic foreground, whereas only the direct path passes through the local ionised absorber and, during dips, the additional cold absorber. The complete model expressions for the persistent and dip spectra and their parameter-coupling strategy are given in Section 3.4.

3.3.3 Multi-Zone Structure

The adopted absorber model contains three `pion` components. This multi-zone structure allows gas with different ionisation states and columns to contribute to the same spectrum. The zones were introduced sequentially: first the continuum and cold absorption were fitted, then one photoionised zone was added, and subsequent zones were retained when they reduced related absorption residuals and gave stable parameter estimates. This procedure avoids assigning a separate plasma component to every isolated residual. Because a complete sequence of formal one-, two-, and three-zone model-comparison statistics was not retained, the procedure defines the adopted model rather than proving that three zones are a unique decomposition.

The static `xabs` model was retained for exploratory checks, but its ionisation balance is read from a pre-calculated grid generated for an assumed spectral energy distribution. It is therefore not used for the final parameters, for which the sector-dependent ionising continua and the transmission through successive zones are handled self-consistently with `pion` (SPEX Development Team 2026a).

3.3.4 Velocity Shifts and Line Broadening

Velocity information is treated cautiously. In SPEX, the absorber redshift or velocity-shift parameter tests whether the lines are displaced relative to the systemic frame, while the broadening parameter captures unresolved turbulent, thermal, or instrumental contributions not otherwise separated by the model. A measured shift is not by itself identified as an outflow or inflow.

The available spectra did not independently constrain the centroid shift or velocity width of the lower-ionisation `pion 3` component. These parameters were therefore fixed at $v_{\text{shift}} = 0$ and $\sigma_v = 100 \text{ km s}^{-1}$ in all morphology sectors and the persistent and dip selections. The fixed zero defines the adopted fit rather than a measurement that this component is stationary; the present data cannot distinguish rest from a small velocity shift.

3.4 Analysis Strategy

The model-component names and roles used throughout this fitting sequence are defined in Table 3.1. The models below are written in compact SPEX syntax, with multiplication indicating the order in which the continuum is modified by the transmission components.

3.4.1 Persistent Spectra

The persistent spectra define the reference continuum and absorber solution for each sector. The sector-matched *Resolve*, RGS1, and RGS2 spectra are fitted simultaneously, with the continuum parameters coupled between instruments. The continuum is determined independently for the three sectors, whereas the Galactic foreground column is common to all morphology sectors and the persistent and dip selections. The model for the persistent spectra is summarised as

$$\text{hot}_{\text{Gal}} \times [(\text{pion}_1 \times \text{pion}_2 \times \text{pion}_3) + f_{\text{sc}}] \times (\text{bb} + \text{dbb} + \text{comt}).$$

Here $f_{\text{sc}} = 0.09$. The `pion` term represents the direct continuum transmitted through the three photoionised zones, while the f_{sc} term is the fixed scattered continuum that bypasses those local absorbers. Both paths are attenuated by the Galactic `hot` component.

The continuum and foreground absorption are constrained first. The `pion` column densities, ionisation parameters, velocity broadenings, and velocity shifts are allowed to vary where they are constrained by the data; parameters that cannot be constrained independently are fixed or coupled consistently across the simultaneously fitted regions. The fitted persistent solution is the sector-specific baseline for the corresponding dip spectrum.

Fit quality is assessed using the ratio of the measured C-statistic to the value expected by SPEX, $C_{\text{obs}}/C_{\text{exp}}$. A ratio close to unity indicates that the measured statistic is close

to the expectation for the fitted model. Unless stated otherwise, parameter uncertainties are quoted as one-parameter 68 per cent ($\Delta C = 1$) profile intervals calculated after the final continuum and absorber structure has been selected.

3.4.2 Dip Spectra

The dip spectra are fitted with the same sector-specific intrinsic continuum as the persistent spectra. Thus, within a given sector, the temperatures, shape parameters, and normalisations of `bb`, `dbb`, and `comt` are fixed to the corresponding persistent solution. This isolates changes in the line-of-sight material from changes in the underlying emission. The model for the dip spectra is summarised as

$$\text{hot}_{\text{Gal}} \times [(\text{pion}_1 \times \text{pion}_2 \times \text{pion}_3 \times \text{hot}_{\text{dip}}) + f_{\text{sc}}] \times (\text{bb} + \text{dbb} + \text{comt}).$$

Here `hotdip` is an additional low-temperature component representing cold material local to the binary. The three `pion` zones used for the dip spectra retain the same ordering and component definitions as in the persistent model, but their N_{H} , $\log \xi$, velocity broadening, and velocity shift are allowed to differ from the persistent values where supported by the data.

The scattered fraction remains fixed at 9 per cent. As in the persistent model, this continuum copy is attenuated by the Galactic foreground but bypasses the local `pion` zones; in the dip model it also bypasses the additional cold absorber. The persistent–dip comparison tests whether increased cold absorption and changes in the ionised line-of-sight gas can explain the dip spectra without a change in the intrinsic continuum. Numerical differences between the persistent and dip absorber parameters are reported in Chapter 4. The dip fits use the same fit-quality and uncertainty conventions as the persistent fits.

3.4.3 Burst Spectra

The burst spectra are fitted with the variable persistent-emission, or f_a , method introduced by [Wörpel et al. \(2013\)](#) and tested on a larger burst sample by [Wörpel et al. \(2015\)](#) (see Section 1.4.2). Rather than subtracting the pre-burst spectrum as a fixed background, this method retains the persistent emission in the source model and allows its normalisation to change during the burst. The model used here is

$$M_{\text{burst}}(E, t) = \text{hot}_{\text{Gal}}(E) [B_{\text{burst}}(E, t) + f_a(t)C_{\text{pers}}(E)], \quad (3.2)$$

where B_{burst} is a separate SPEX `bb` component and $C_{\text{pers}} = \text{bb} + \text{dbb} + \text{comt}$ is the independently fitted local pre-burst continuum. For bursts 1 and 2 the pre-burst fits are initialised from the sector-1 persistent solution, and burst 3 is initialised from the sector-2 solution. In these baseline fits, Galactic N_{H} is fixed, the `comt` electron temperature, optical depth, and normalisation are fixed at 16 keV, 5.56, and 0.1, respectively, and

its seed temperature is coupled to the persistent `bb` temperature. The temperatures and normalisations of the persistent `bb` and `dbb` components are free. These `comt` values come from the broad-band continuum fit including NuSTAR and remain fixed in all three pre-burst fits because `pn` alone does not constrain them independently.

In the SPEX implementation, the persistent `dbb` normalisation is used as the scaling anchor. The `bb` and `comt` normalisations are coupled to it at their pre-burst ratios, so that all three persistent components change by the same factor while their relative contributions remain fixed. The fitted scale is therefore calculated as

$$f_a = \frac{N_{\text{dbb}}(t)}{N_{\text{dbb},0}}, \quad (3.3)$$

where $N_{\text{dbb},0}$ is the pre-burst normalisation. The `comt` seed temperature remains coupled to the persistent `bb` temperature, rather than to the burst blackbody. The Galactic `hot` component is fixed to the corresponding pre-burst absorption solution.

All fits use C-statistics over 0.6–10 keV. Fits including 0.3–0.59 keV were consistently poor, so this interval was excluded throughout. The 2-s peak spectra use the SPEX optimal-binning command, whereas the lower-rate time-averaged tails and all time-resolved spectra use a fixed binning factor of eight over the fitting band. For comparisons with bolometric burst measurements, only the fitted burst blackbody is integrated over 0.01–100 keV; the quoted burst flux excludes the scaled persistent contribution. A free power law was added to each time-averaged tail to test for additional non-blackbody emission. It reduced the BIC for bursts 1 and 2 but drove f_a to approximately 0.15 or zero, while its normalisation converged to zero for burst 3. The power law is not adopted because it is degenerate with the scaled persistent continuum and is not consistently constrained across the three bursts. The reported peak, tail, and time-resolved results use one burst blackbody plus the standard $f_a C_{\text{pers}}$ term.

The time-resolved sequence leaves three quantities free in each interval: the burst-blackbody temperature, its normalisation, and the common persistent scaling f_a . The intrinsic burst-blackbody flux is evaluated over 0.01–100 keV. The apparent blackbody radius is calculated as

$$R_{\text{bb}} = D \left(\frac{F_{\text{bol}}}{\sigma_{\text{SB}} T_{\text{bb}}^4} \right)^{1/2}, \quad (3.4)$$

using the adopted distance of 9.1 kpc, based on the estimates discussed in Section 1.3.1 (Gambino et al. 2019; Trueba et al. 2020; Galloway et al. 2008). This is an apparent colour radius under the adopted model; no colour-correction, gravitational-redshift, or anisotropy correction is applied.

The same fit-quality and 68 per cent profile-error conventions are used here. Profile errors are calculated for all free parameters in the time-averaged pre-burst, peak, and tail fits, but only for the two time-resolved bins that define the peak evolution of each burst. The remaining time-resolved bins and the 0.01–100 keV model flux have no separately profiled uncertainties.

The parameter f_a is a phenomenological measure of changes in the persistent spectral contribution. Although $f_a > 1$ can be consistent with enhanced accretion, reprocessing, or a coronal response during the burst, it is not by itself a direct measurement of the instantaneous accretion rate ([Speicher et al. 2024](#)).

Chapter 4

Results

The results are presented in observation-first order. The light curve defines the morphology sectors and persistent/dip selections, followed by the continuum, the persistent and dip spectra, and their cross-sector comparison. The final section reports the burst timing, spectroscopy, and oscillation search. Chapter 5 considers the physical interpretation.

4.1 Sector-dependent Dip Morphology

The 20-s EPIC-pn light curve defines the time selections (Fig. 2.4); the 100-s representation in Fig. 4.1 is used only to expose the longer-timescale morphology. Sector 1 shows isolated minima close to successive relative phase-zero markers. Sector 2 contains broader attenuation that is less clearly separated from the surrounding variability, whereas sector 3 shows repeated, structured low-rate excursions, often more than once within an orbital cycle. The adopted dip GTIs account for 10.9, 28.9, and 49.7 per cent of the classified EPIC-pn exposure in sectors 1–3, respectively. These fractions describe the selections, not a model-independent dip duty cycle: sectors 1 and 2 were selected manually, whereas sector 3 uses a count-rate threshold. The sectors are therefore empirical morphology intervals. They provide the basis for testing whether the absorber properties follow the timing evolution.

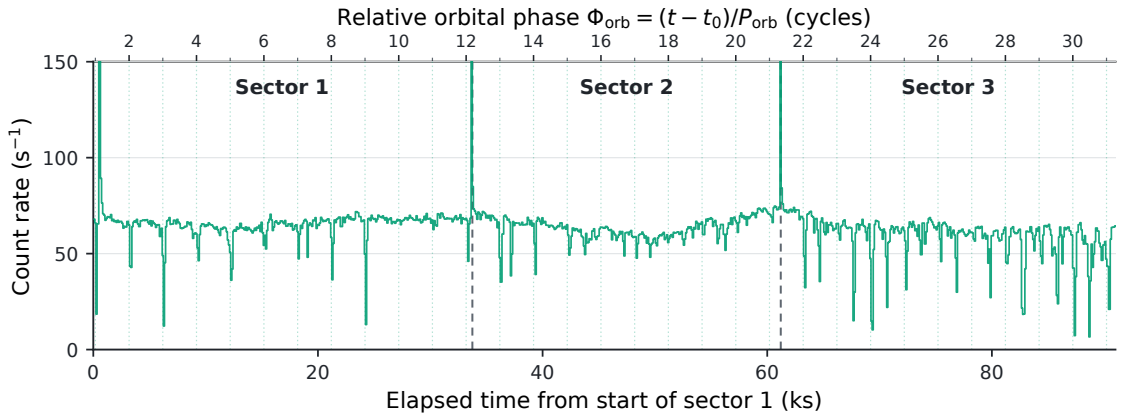


FIG. 4.1— Background-unsubtracted EPIC-pn 0.3–12 keV light curve corrected with `epiclccorr` and binned at 100 s. The upper axis gives relative orbital phase for $P_{\text{orb}} = 3000.65126$ s (Iaria et al. 2015), anchored to the first dip minimum. Green dotted lines mark integer cycles and grey dashed lines delimit the three morphology sectors. The ordinate is limited to 150 counts s^{-1} .

4.2 Broad-Band Continuum Baseline

4.2.1 Overall fit and component fluxes

The broad-band persistent continuum was first fitted without the local photoionised absorbers to determine its baseline curvature. The model is the Galactic foreground

absorption multiplied by the three-component continuum defined in Section 3.2, $\text{hot} \times (\text{bb} + \text{dbb} + \text{comt})$. Figure 4.2 shows this fit jointly in the EPIC-pn, RGS1, RGS2, and NuSTAR detector count spaces. The different vertical normalisations of the spectra primarily reflect their responses, effective areas, and fitted cross-normalisations; they should not be read as direct instrument-to-instrument flux differences.

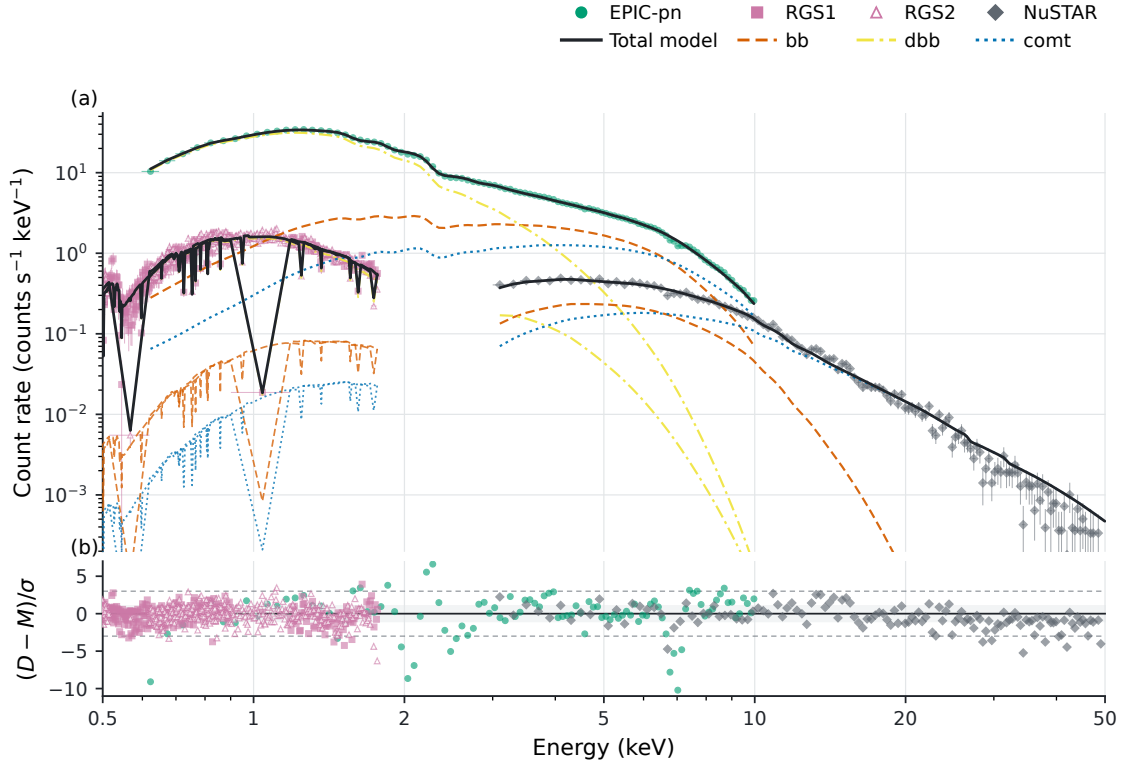


FIG. 4.2— Broad-band persistent-emission continuum baseline. Panel (a) shows EPIC-pn (green circles), RGS1 and RGS2 (filled and open magenta symbols), and NuSTAR (grey diamonds) with the response-folded $\text{hot} \times (\text{bb} + \text{dbb} + \text{comt})$ model. Black is the total model; dashed vermilion, dash-dotted yellow, and dotted blue show bb , dbb , and comt . Panel (b) shows $(D - M)/\sigma$. No local pion absorber is included. The sharp downward excursions in the RGS curves result from reductions in effective area at inoperative CCDs, chip boundaries, and bad detector columns or pixels; they are not intrinsic continuum absorption features (ESA XMM-Newton Science Operations Centre 2025b).

The response-folded components dominate different energy ranges. dbb is largest in the soft band below about 3 keV, bb supplies most of the intermediate-energy continuum at roughly 3–7 keV, and comt becomes dominant above approximately 7 keV. The NuSTAR spectrum constrains the hard curvature, so the Comptonised component is not determined only by the upper end of the EPIC-pn band. The continuum model reproduces the broad spectral shape but leaves structured residuals, particularly across the RGS band. We model these features with local photoionised absorption in Section 4.3, rather than with additional continuum curvature.

4.2.2 Cross-instrument consistency

The EPIC-pn normalisation is fixed to unity in the fit shown in Figure 4.2. Relative to this reference, the best-fitting constants are 0.926 for RGS1, 0.944 for RGS2, and 1.161 for NuSTAR. The two RGS values agree to within approximately 2 per cent, while the NuSTAR data were restricted to the sector-1 time range, but their usable GTIs began about 2.7 h after the start of the XMM-*Newton* sector-1 interval. The spectra are therefore contemporaneous but do not have identical time coverage. The larger NuSTAR constant may reflect both XMM-*Newton*–NuSTAR cross-calibration offsets (Madsen et al. 2017) and differences in this coverage, and is not interpreted on its own as evidence for source variability.

4.3 Persistent and Dip Spectroscopy Across Sectors

4.3.1 Persistent spectra across all sectors

Table 4.1 lists the continuum and absorber parameters, while Fig. 4.3 shows the spectra, total models, and residuals on matched axes.

Figure 4.3 places the three persistent spectra on the same energy ranges and axes. The RGS spectra provide the soft-band coverage below 2 keV, whereas *Resolve* supplies the 2–10 keV spectrum and resolves the Fe-K absorption structure in panels (b), (d), and (f). The response-folded total models describe the broad count-space spectra, whereas the contribution of each absorber zone is separated in Figs. 4.5 and 4.6.

In the persistent fits, the highest-ionisation zone, **pion 1**, also had the largest fitted column. Its N_{H} values were 26^{+14}_{-12} , 23^{+5}_{-4} , and $60.5^{+35}_{-23} \times 10^{22} \text{ cm}^{-2}$ in sectors 1–3, with $\log \xi = 3.827^{+0.046}_{-0.054}$, $3.645^{+0.060}_{-0.057}$, and $4.15^{+0.17}_{-0.16}$. The intermediate-ionisation **pion 2** columns were much smaller, $0.83\text{--}1.30 \times 10^{22} \text{ cm}^{-2}$, but this zone had a positive centroid shift in every sector: 1656^{+110}_{-73} , 1579^{+110}_{-95} , and $1734^{+110}_{-130} \text{ km s}^{-1}$. The lower-ionisation **pion 3** component had $N_{\text{H}} = 0.25\text{--}1.52 \times 10^{22} \text{ cm}^{-2}$ and $\log \xi = 2.73\text{--}3.038$. Because the available statistics did not constrain its centroid or width, these parameters were fixed at $v_{\text{shift}} = 0$ and $\sigma_v = 100 \text{ km s}^{-1}$ in all three sectors. The zero shift is therefore not a measurement of stationary gas, and this zone provides no independent kinematic constraint. The sector-3 persistent absorber uncertainties are retained in the comparison, but are interpreted cautiously because the corresponding error scans did not all share a verified common minimum.

4.3.2 Dip spectra across all sectors

Each dip spectrum was fitted with its sector-matched persistent continuum held fixed. The same three **pion** zones were retained, together with an additional low-temperature **hot** component local to the dip model. Table 4.2 reports the resulting parameters and

Persistent and Dip Spectroscopy Across Sectors

TABLE 4.1— Best-fitting model parameters for the persistent spectra in sectors 1–3.

Component and parameter	Unit	Sector 1	Sector 2	Sector 3
Foreground absorption				
hot 1				
N_{H}	10^{22} cm^{-2}		$0.321^{+0.006}_{-0.006}$	
Continuum				
bb				
norm	10^{16} m^2	$(3.00^{+0.1}_{-0.03}) \times 10^{-9}$	$(2.57^{+0.18}_{-0.18}) \times 10^{-9}$	$(3.34^{+0.85}_{-0.48}) \times 10^{-9}$
kT	keV	$1.603^{+0.011}_{-0.021}$	$1.626^{+0.023}_{-0.026}$	$1.640^{+0.120}_{-0.099}$
dbb				
norm	10^{16} m^2	$(3.48^{+0.30}_{-0.29}) \times 10^{-9}$	$(2.77^{+0.41}_{-0.37}) \times 10^{-9}$	$(4.11^{+1.1}_{-0.66}) \times 10^{-9}$
kT_{in}	keV	$1.255^{+0.002}_{-0.034}$	$1.299^{+0.028}_{-0.002}$	$1.260^{+0.044}_{-0.044}$
comt				
norm	$10^{44} \text{ ph s}^{-1} \text{ keV}^{-1}$	0.100 ^f	0.100 ^f	0.100 ^f
kT_0	keV	1.603 ^t	1.626 ^t	1.640 ^t
kT_1	keV	16.0 ^f	16.0 ^f	16.0 ^f
τ	–	5.56 ^f	5.56 ^f	5.56 ^f
Photoionised absorber				
pion 1				
N_{H}	10^{22} cm^{-2}	26.1^{+14}_{-12}	$22.8^{+4.9}_{-3.9}$	60.5^{+35}_{-23}
$\log \xi$	10^{-9} W m	$3.827^{+0.046}_{-0.054}$	$3.645^{+0.060}_{-0.057}$	$4.15^{+0.17}_{-0.16}$
σ_v	km s^{-1}	200^{+0}_{-29} b	165^{+13}_{-12}	$176^{+24}_{-25} \text{ b}$
v_{shift}	km s^{-1}	128^{+60}_{-21}	134^{+18}_{-18}	55^{+38}_{-38}
pion 2				
N_{H}	10^{22} cm^{-2}	$1.30^{+0.94}_{-0.38} \text{ a}$	$0.83^{+0.36}_{-0.19}$	$1.30^{+0.98}_{-0.64}$
$\log \xi$	10^{-9} W m	$3.42^{+0.22}_{-0.17}$	$3.35^{+0.11}_{-0.13}$	$3.52^{+0.25}_{-0.28}$
σ_v	km s^{-1}	$100^{+100}_{-0} \text{ b}$	200^{+0}_{-97} b	$200^{+0}_{-100} \text{ b}$
v_{shift}	km s^{-1}	1656^{+110}_{-73}	1579^{+110}_{-95}	1734^{+110}_{-130}
pion 3				
N_{H}	10^{22} cm^{-2}	$0.39^{+0.93}_{-0.22}$	$0.25^{+0.22}_{-0.14}$	$1.52^{+0.95}_{-0.65}$
$\log \xi$	10^{-9} W m	$2.84^{+0.23}_{-0.22}$	$2.73^{+0.12}_{-0.13}$	$3.038^{+0.095}_{-0.098}$
σ_v	km s^{-1}	100 ^f	100 ^f	100 ^f
v_{shift}	km s^{-1}	0 ^f	0 ^f	0 ^f
Fit statistic	C/dof	2864.98/2624	3005.18/2924	1797.50/1642

Notes. Column densities were converted from the native SPEX unit 10^{28} m^{-2} to 10^{22} cm^{-2} . Positive v_{shift} denotes a redshift in the adopted SPEX convention. All quoted uncertainties are 68 per cent ($\Delta C = 1$) profile intervals expressed in the same unit as the corresponding best-fitting value and rounded to approximately two significant figures. Superscripts indicate parameters that were fixed (f), tied to another parameter (t), or reached an imposed fitting boundary (b). The **pion 3** centroid and width were fixed because the available spectra did not constrain them; $v_{\text{shift}} = 0$ is therefore not a measurement of stationary gas. The **a** entry uses the adopted sector-1 **pion 2** column-density centre with the final freeze-other-zone error widths. The **comt** seed temperature was tied to the **bb** temperature. The foreground **hot 1** column is common to all three sectors; its uncertainty is adopted from the sector-2 error search. The sector-3 absorber error scans did not all share a verified common minimum; those uncertainties should therefore be treated with caution.

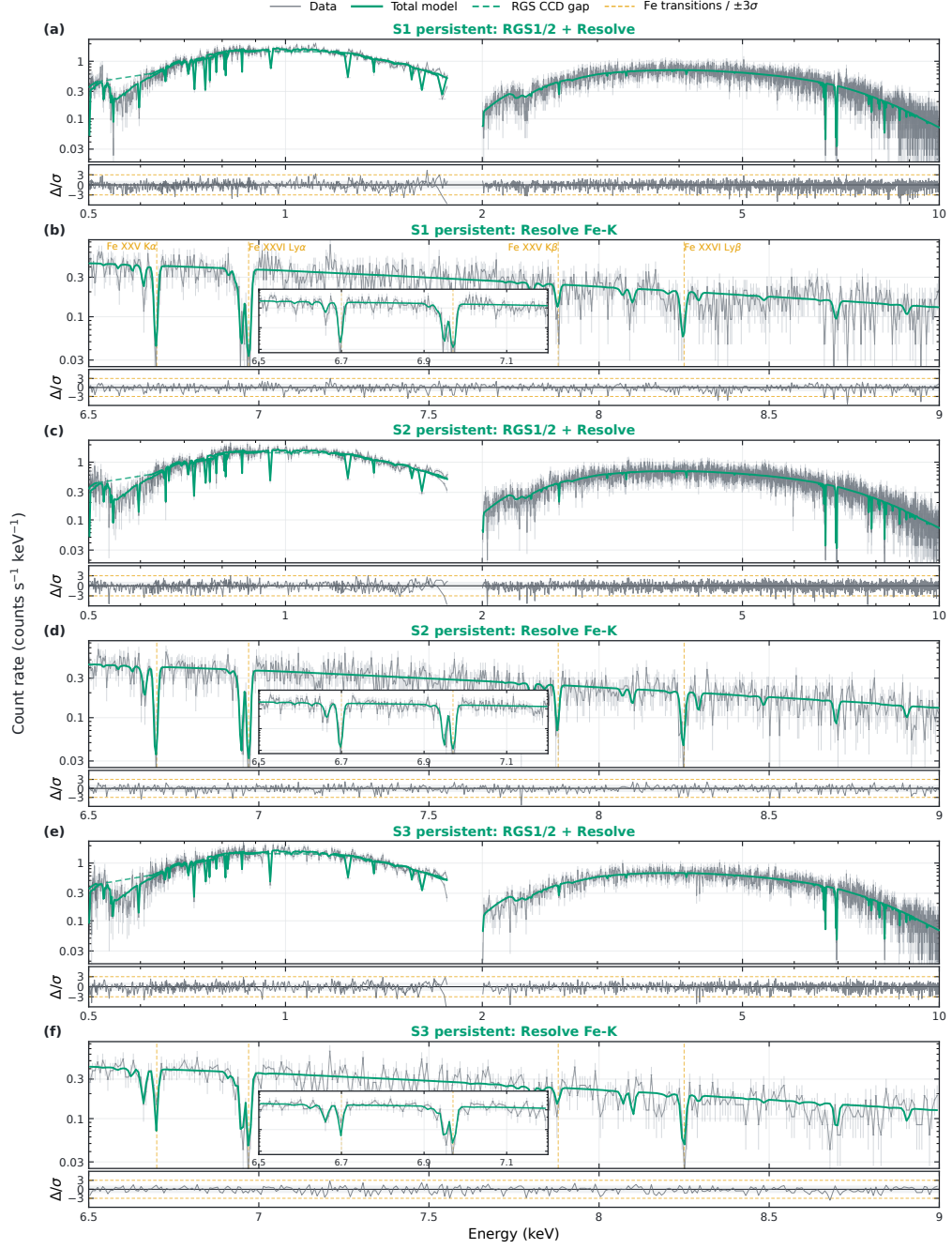


FIG. 4.3— Persistent spectra and total models for sectors 1–3. For each sector, the left panel shows the 0.5–10 keV RGS1, RGS2, and *Resolve* spectra; the right panel shows the 6.5–9.0 keV *Resolve* Fe-K band with a 6.5–7.2 keV inset. Grey denotes data, green the model, and the lower panels show $(D - M)/\sigma$. Orange lines mark Fe XXV/XXVI rest energies; green dashed segments bridge excluded RGS CCD gaps.

identifies fixed, boundary-limited, and one-sided constraints.

Figure 4.4 uses the same arrangement and energy ranges as the persistent comparison. The sector-specific persistent continua are held fixed as described in Section 3.4, so the figure displays the dip spectra under the matched continuum and absorber framework.

The dip **pion 1** column densities were 42.0 ± 2.0 , $21.8^{+1.4}_{-1.3}$, and $72.5^{+1.7}_{-1.8} \times 10^{22} \text{ cm}^{-2}$ in sectors 1–3. Their ionisation parameters ranged from $\log \xi = 3.495$ to 4.170 and did not follow a single monotonic sector trend. The **pion 1** centroid shift was $414^{+37}_{-34} \text{ km s}^{-1}$ in sector 1, $157^{+68}_{-79} \text{ km s}^{-1}$ in sector 2, and reached the imposed zero-velocity boundary in sector 3. By contrast, **pion 2** remained positively shifted in all three dip spectra, with values of 1736 ± 140 , 1734^{+94}_{-100} , and $1507^{+57}_{-61} \text{ km s}^{-1}$.

The dip **pion 3** columns were $0.81^{+0.43}_{-0.38}$, $0.78^{+0.34}_{-0.27}$, and $3.13^{+0.82}_{-0.72} \times 10^{22} \text{ cm}^{-2}$, while their ionisation parameters were $2.52^{+0.11}_{-0.16}$, $2.665^{+0.089}_{-0.086}$, and $2.792^{+0.036}_{-0.037}$. As in the persistent fits, the available statistics did not constrain the **pion 3** centroid or width, which were fixed at $v_{\text{shift}} = 0$ and $\sigma_v = 100 \text{ km s}^{-1}$. These fixed values do not distinguish stationary material from a weakly shifted component. The additional dip-local **hot** component was not detected: its sector columns are reported as one-sided upper limits of 3.2×10^{-3} , 1.4×10^{-2} , and $4.9 \times 10^{-4} \times 10^{22} \text{ cm}^{-2}$.

4.3.3 *Resolve* and RGS model-to-continuum signatures

The complementary high-resolution bandpasses separate the spectral signatures of the three fitted zones. Figures 4.5 and 4.6 show the individual and combined absorber models divided by the corresponding continuum model for all three persistent sector fits. The RGS comparison uses diagnostic line windows, while the *Resolve* comparison retains two continuous Fe-K intervals. The layout shows each zone and its sector-to-sector changes without compressing the Fe-K structure into one broad panel. This “transmission” view is a model decomposition, not the instrumental throughput nor a data-to-model residual. A ratio of unity marks an unchanged continuum, whereas downward departures identify where the adopted absorber model removes flux.

TABLE 4.2— Best-fitting dip-spectrum parameters for sectors 1–3.

Component and parameter	Unit	Sector 1	Sector 2	Sector 3
Foreground absorption				
hot 1				
N_{H}	10^{22} cm^{-2}		$0.321^{+0.006}_{-0.006}$	
Continuum fixed to the corresponding persistent spectrum				
bb (fixed)				
norm	10^{16} m^2	$(3.00^{+0.1}_{-0.03}) \times 10^{-9}$	$(2.57^{+0.18}_{-0.18}) \times 10^{-9}$	$(3.34^{+0.85}_{-0.48}) \times 10^{-9}$
kT	keV	$1.603^{+0.011}_{-0.021}$	$1.626^{+0.023}_{-0.026}$	$1.640^{+0.120}_{-0.099}$
dbb (fixed)				
norm	10^{16} m^2	$(3.48^{+0.30}_{-0.29}) \times 10^{-9}$	$(2.77^{+0.41}_{-0.37}) \times 10^{-9}$	$(4.11^{+1.1}_{-0.66}) \times 10^{-9}$
kT_{in}	keV	$1.255^{+0.002}_{-0.034}$	$1.299^{+0.028}_{-0.002}$	$1.260^{+0.044}_{-0.044}$
comt (fixed)				
norm	$10^{44} \text{ ph s}^{-1} \text{ keV}^{-1}$	0.100	0.100	0.100
kT_0	keV	1.603	1.626	1.640
kT_1	keV	16.0	16.0	16.0
τ	—	5.56	5.56	5.56
Photoionised absorber				
pion 1				
N_{H}	10^{22} cm^{-2}	$42.0^{+2.0}_{-2.0}$	$21.8^{+1.4}_{-1.3}$	$72.5^{+1.7}_{-1.8}$
$\log \xi$	10^{-9} W m	$3.495^{+0.082}_{-0.072}$	$3.55^{+0.12}_{-0.095}$	$4.170^{+0.068}_{-0.067}$
σ_v	km s^{-1}	$100^{+12\text{b}}_{-0}$	134^{+34}_{-34}	167^{+23}_{-20}
v_{shift}	km s^{-1}	414^{+37}_{-34}	$157^{+68\text{c}}_{-79}$	$0^{+32\text{b}}_{-0}$
pion 2				
N_{H}	10^{22} cm^{-2}	$1.83^{+1.0}_{-0.79}$	$1.61^{+0.51}_{-0.44}$	$1.12^{+0.68}_{-0.49}$
$\log \xi$	10^{-9} W m	$3.34^{+0.23}_{-0.22}$	$3.11^{+0.12}_{-0.098}$	$3.49^{+0.18}_{-0.16}$
σ_v	km s^{-1}	$200^{+0}_{-100} \text{ b}$	200^{+0}_{-36} b	$100^{+50\text{b}}_{-0}$
v_{shift}	km s^{-1}	1736^{+140}_{-140}	1734^{+94}_{-100}	1507^{+57}_{-61}
pion 3				
N_{H}	10^{22} cm^{-2}	$0.81^{+0.43}_{-0.38}$	$0.78^{+0.34}_{-0.27}$	$3.13^{+0.82}_{-0.72}$
$\log \xi$	10^{-9} W m	$2.52^{+0.11}_{-0.16}$	$2.665^{+0.089}_{-0.086}$	$2.792^{+0.036}_{-0.037}$
σ_v	km s^{-1}	100^{f}	100^{f}	100^{f}
v_{shift}	km s^{-1}	0^{f}	0^{f}	0^{f}
Dip-local cold absorption				
hot 2				
N_{H}	10^{22} cm^{-2}	$< 3.2 \times 10^{-3}$	$< 1.4 \times 10^{-2}$	$< 4.9 \times 10^{-4}$
Fit statistic	C/dof	1866.11/1560	2850.88/2586	2836.90/2700

Notes. Column densities are in 10^{22} cm^{-2} ; positive v_{shift} denotes redshift. The quoted uncertainties are 68 per cent ($\Delta C = 1$) profile intervals expressed in the listed units and rounded to about two significant figures. The continuum components were held fixed at the corresponding persistent solutions. Their persistent-fit uncertainties are listed for reference and were not re-profiled in the dip fits. Symbols f and b mark fixed and boundary-limited values. The common **hot 1** column and its uncertainty (from sector 2 persistent) are shown once. Marker c identifies the adopted earlier complete sector-2 **pion 1** velocity scan. The **pion 3** centroid and width were fixed because the available spectra did not constrain them; $v_{\text{shift}} = 0$ is not a measured stationary velocity. The dip-local **hot 2** entries are one-sided upper limits.

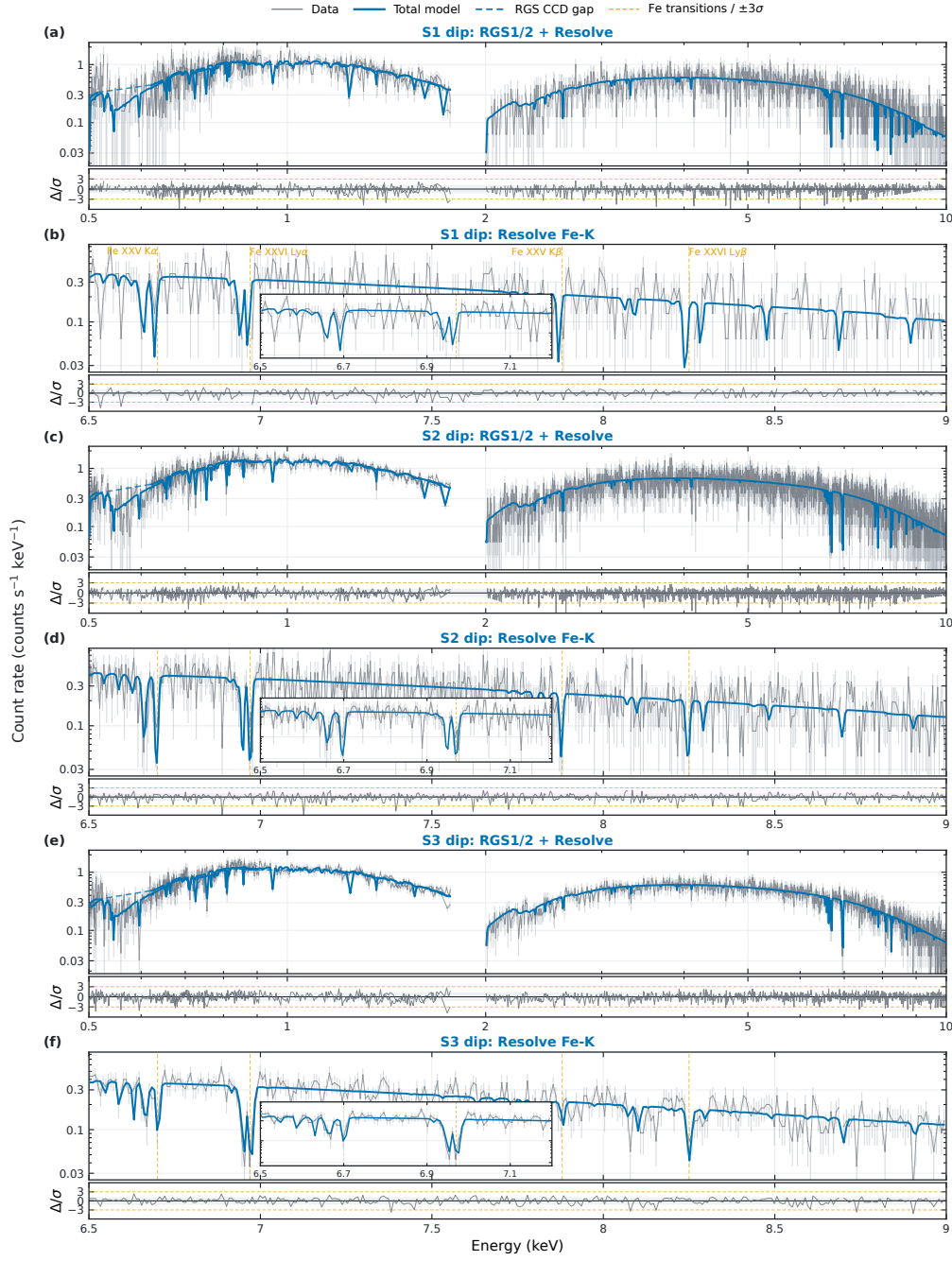


FIG. 4.4— Dip spectra and total models for sectors 1–3, arranged as in Fig. 4.3. Grey denotes data, blue the model, and the lower panels show $(D - M)/\sigma$. Orange lines mark Fe XXV/XXVI rest energies; blue dashed segments bridge excluded RGS CCD gaps.

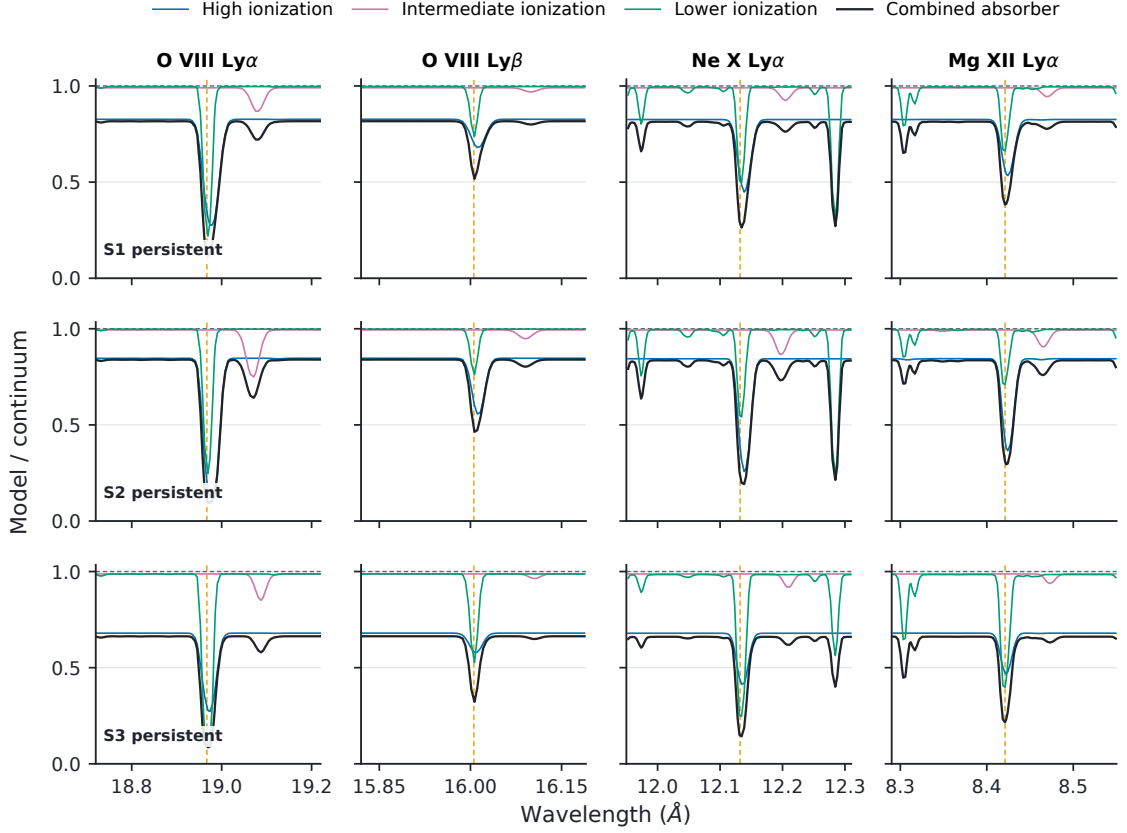


FIG. 4.5— RGS absorber model-to-continuum ratios for persistent sectors 1–3 (rows) around O VIII, Ne X, and Mg XII lines (columns). Blue, magenta, and green denote π ion 1–3; black is the combined absorber. Dashed lines mark unity and laboratory wavelengths.

The same principal line families were present in the adopted model for all three persistent sectors, while their fitted depths and centroid positions changed with sector. The π ion 2 troughs were displaced towards longer wavelength in the RGS windows and lower energy in the *Resolve* intervals, consistent with the positive fitted shifts reported above. The higher-order O and Fe transitions provide additional model diagnostics beyond the strongest Ly α and He α complexes. Their depths depend jointly on column density and ionisation, however, and these component profiles are not independent measurements of individual line strengths or absorber geometry.

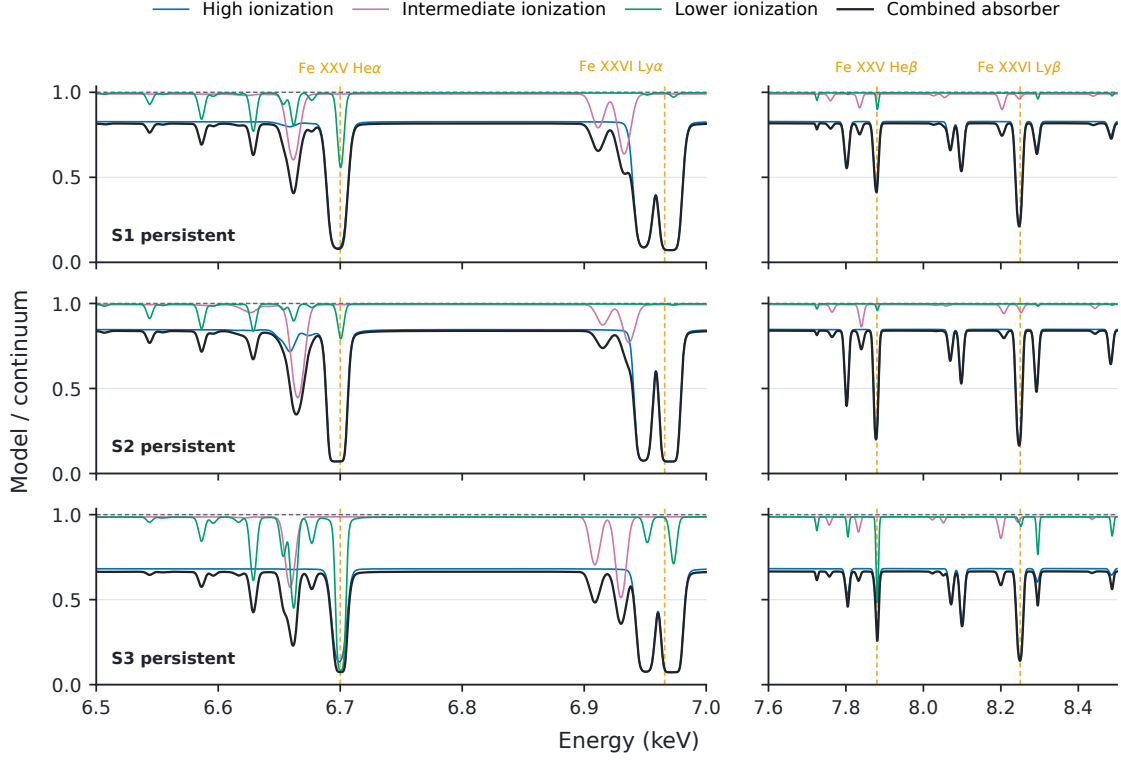


FIG. 4.6— *Resolve* absorber model-to-continuum ratios for persistent sectors 1–3 (rows) over 6.5–7.0 and 7.6–8.5 keV (columns). Colours follow Fig. 4.5; black is the combined absorber, and dashed lines mark unity and Fe XXV/XXVI laboratory energies.

4.4 Persistent-to-Dip and Cross-Sector Comparison

4.4.1 Column density and ionisation

Figure 4.7 compares the persistent and dip measurements with their asymmetric uncertainties. The three zones did not respond to the dip selections in the same way.

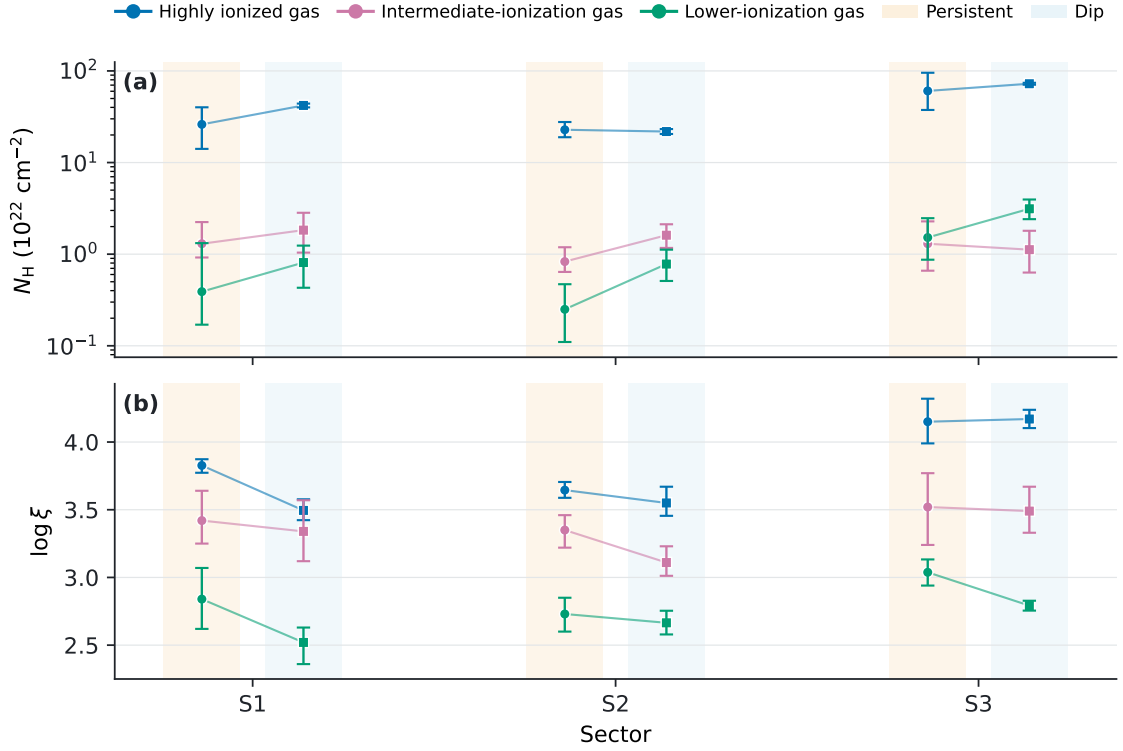


FIG. 4.7— Absorber column density (a) and ionisation parameter (b) across the three sectors. Colours denote **pion** 1–3. Pale yellow and pale blue bands mark the persistent and dip selections, respectively; circles and squares provide the same persistent/dip distinction. Error bars are 68 per cent ($\Delta C = 1$) profile intervals. The sector-3 persistent caveat is given in Table 4.1.

For the lower-ionisation **pion** 3 zone, the dip best-fitting column was higher than the persistent value by factors of 2.1, 3.1, and 2.1 in sectors 1–3. The 68 per cent ($\Delta C = 1$) profile intervals nevertheless overlapped in sector 1 and marginally in sector 3; only the sector-2 intervals were separated. The corresponding $\log \xi$ values decreased by 0.32, 0.07, and 0.246. The sector-3 intervals were clearly separated, the sector-1 intervals nearly touched, and the sector-2 intervals overlapped. Thus the adopted model showed a recurrent best-fit tendency towards more, less-ionised **pion** 3 material in the dip selections, but the strength of that tendency was not uniform across all three sectors.

Neither **pion** 1 nor **pion** 2 showed the same persistent-to-dip change in every sector. The **pion** 1 column increased in sectors 1 and 3 but was nearly unchanged in sector 2, while its ionisation decreased in sector 1 and remained consistent within the quoted intervals in sectors 2 and 3. **pion** 2 had overlapping column-density intervals in all three sectors; its largest best-fit ionisation decrease occurred in sector 2.

4.4.2 Line-of-sight velocity shifts

Figure 4.8 separates the persistent and dip measurements of the fitted centroid shift. Positive v_{shift} denotes a redshift in the adopted SPEX convention. Only **pion 1** and **pion 2** are shown because the available statistics did not constrain the **pion 3** velocity. It was fixed at zero and therefore provides no independent kinematic measurement.

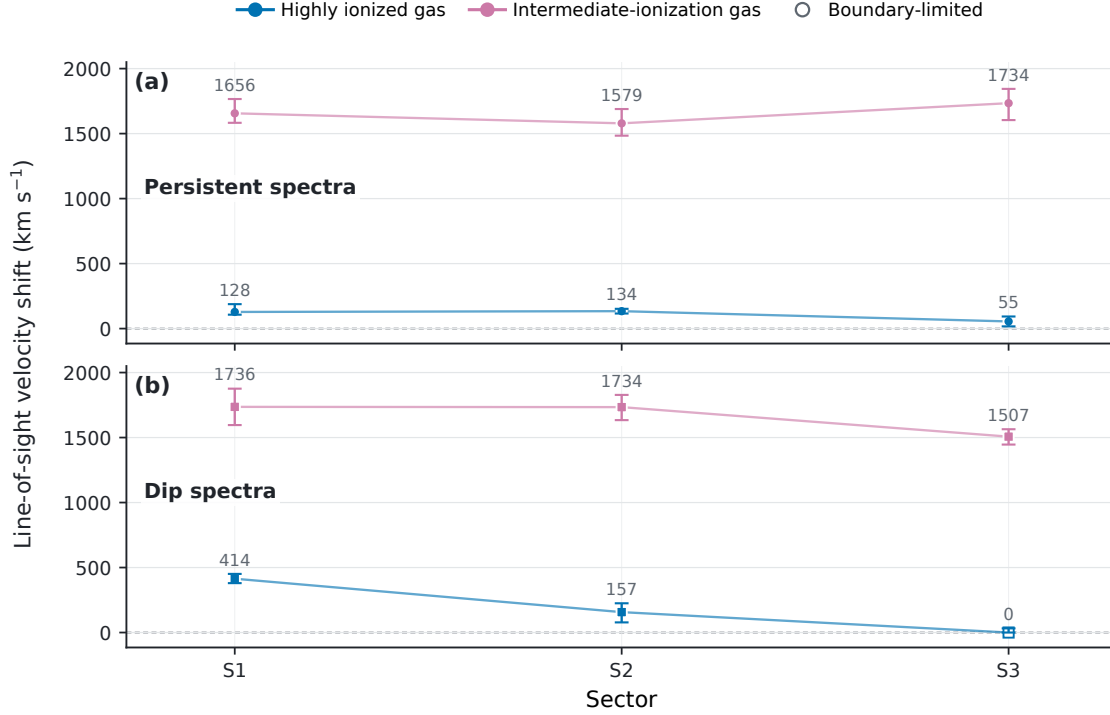


FIG. 4.8— Line-of-sight velocity shifts for persistent (a) and dip (b) spectra. Blue and magenta denote **pion 1** and **2**; labels and error bars give the best fits and 68 per cent ($\Delta C = 1$) intervals. The open sector-3 square is boundary limited; dashed lines mark zero velocity.

The intermediate-ionisation **pion 2** component was positively shifted in all six spectra. Its persistent values spanned 1579–1734 km s^{-1} and its dip values spanned 1507–1736 km s^{-1} ; every quoted 68 per cent ($\Delta C = 1$) profile interval remained above zero. Persistent and dip intervals overlapped in sectors 1 and 2. In sector 3 the dip interval lay below the quoted persistent interval, although that comparison inherits the sector-3 persistent error-scan caveat.

The **pion 1** shifts were smaller and showed different persistent-to-dip behaviour across the three sectors. The persistent values were 55–134 km s^{-1} . In the dip spectra, sector 1 increased to $414^{+37}_{-34} \text{ km s}^{-1}$, sector 2 remained at $157^{+68}_{-79} \text{ km s}^{-1}$, and sector 3 reached the zero-velocity boundary. No kinematic trend was assigned to **pion 3** because its centroid and width were fixed. The velocity widths are retained in Tables 4.1 and 4.2, but are not plotted because most **pion 2** widths and several **pion 1** widths reached imposed boundaries. The width measurements show no robust persistent-to-dip trend.

4.4.3 Fit robustness, degeneracies, and non-detections

The comparisons above are conditional on the adopted three-zone model, the sector-matched fixed dip continua, and the fixed scattered fraction $f_{sc} = 0.09$. The **pion** 3 velocities and widths were fixed, several other widths were boundary limited, and the dip-local **hot** columns were upper limits rather than detections. These quantities were therefore excluded from physical trend claims. Some initial sector 2 dip error searches were started from a solution approximately $\Delta C = 14$ above a subsequently identified lower minimum and consequently returned spuriously large intervals. Repeating the affected searches around the lower- C solution, with neighbouring absorber columns fixed where required, yielded finite profile intervals for the reported parameters. Boundary-limited, fixed, and upper-limit cases remain explicitly identified in Table 4.2. The sector 3 persistent absorber intervals remain less secure because their error scans did not all share a verified common minimum. Finally, because an explicit model-comparison statistic for removing each **pion** zone is not tabulated here, the figures characterise the parameter behaviour within the adopted architecture rather than an independent detection significance for every zone in every spectrum.

4.5 Summary of the Persistent and Dip Results

The 100-s EPIC-pn light curve evolves from relatively isolated dips in sector 1 to broader attenuation in sector 2 and repeated, structured dips in sector 3. Within the adopted three-zone absorber model, the lower-ionisation **pion** 3 component had a higher best-fitting column and lower best-fitting ionisation in every dip spectrum, although the persistent and dip 68 per cent ($\Delta C = 1$) profile intervals were not separated in every sector. The intermediate-ionisation **pion** 2 component retained a positive 1507–1736 km s^{−1} centroid shift in all persistent and dip fits, whereas the **pion** 1 shifts were smaller and showed no uniform persistent-to-dip trend. The physical interpretations of these measured model trends are considered in Chapter 5.

4.6 Burst Timing and Spectroscopy

The EPIC-pn observation contains three Type-I bursts. This section reports their timing and durations, the search for the previously reported ~ 270 -Hz signal, and their time-averaged and time-resolved spectra. The spectral fits test whether burst irradiation produces a short-lived change in the persistent continuum; because bursts 2 and 3 lie near morphology-interval boundaries, their timing is also compared with the later dip evolution.

4.6.1 Burst timing and usable exposure

All three events occurred on 2024 April 16. Their measured onsets, peak rates, and durations are summarised in Table 4.3. Bursts 1 and 2 rise to their 1-s count-rate maxima in 2 and 1 s, respectively; burst 3 also rises within 1 s. The peak 1-s bin of burst 3 contains a 0.179-s GTI gap and therefore has 0.821 s of exposure. For the timing fit, the measured rate in this bin is divided by 0.821. This assumes that the mean rate during the gap was the same as during the observed part of the bin. Table 4.3 reports the uncorrected rate of 1387 counts s⁻¹; the count rate during the gap is unknown.

TABLE 4.3— Timing measurements for the three EPIC-pn bursts.

Burst	Onset (UTC)	Rise (s)	Peak rate	MINBAR-style D_5 (s)	Adopted D_5 (s)
B1	06:34:39.216	2	1354	$47.1^{+0.8}_{-1.5}$	$54.1^{+1.5}_{-2.6}$
B2	15:46:18.216	1	1182	$47.5^{+0.8}_{-1.6}$	$54.0^{+1.4}_{-2.7}$
B3	23:24:26.216	1	1387	$32.7^{+0.4}_{-1.0}$	$32.6^{+0.8}_{-1.5}$

Notes. Onset times are UTC on 2024 April 16. The MINBAR-style column is the D_5 obtained from the lowest-BIC model among the single-exponential, power-law, and power-law-plus-Gaussian families. The adopted column uses the additive two-exponential fit; uncertainties are the 16th–84th percentiles of 300 bootstrap realisations. The burst-3 peak rate is the unscaled value from its incompletely exposed 1-s bin.

The additive two-exponential BIC values are 374.0, 344.2, and 377.8 for bursts 1–3. They are lower than the next-best tested models by approximately 178, 144, and 53, respectively. The corresponding D_5 values of 54.1, 54.0, and 32.6 s define the burst spectral products. For comparison, the observed cumulative-count T_{90} values, integrated until the fitted decay returns to within 2σ of the pre-burst level, are 116.4, 111.0, and 76.1 s. These longer T_{90} values describe the weak late tail and are not used as the peak/tail extraction boundary.

All three burst intervals are excluded before the persistent and dip sector spectra are formed. Table 2.2 reports the remaining pn GTI durations for those persistent and dip selections.

4.6.2 Burst-oscillation search

The two-window search contains 326,400 blind frequency–time trials and 17,145 trials in the pre-defined 260–280 Hz band. Equation 2.2 therefore gives three-burst family-wise thresholds of $P_{\text{thr}} = 39.207$ and 33.314, respectively. No power in either search family exceeds its threshold. The largest blind-search power is 29.437 in burst 2, at 226.5 Hz and 9.0 s after onset in a 2-s window. The largest power in the pre-defined band is also in burst 2: $P = 18.610$ at 262.5 Hz and 11.0 s in a 2-s window. Table 4.4 lists the maximum from each burst.

Figure 4.10 shows the 0.5-s dynamical spectra, with the previously reported 269.3–272 Hz interval marked for reference. No feature in this band exceeds the corresponding threshold. The vertical blank strip near the burst 3 peak is produced by rejecting windows

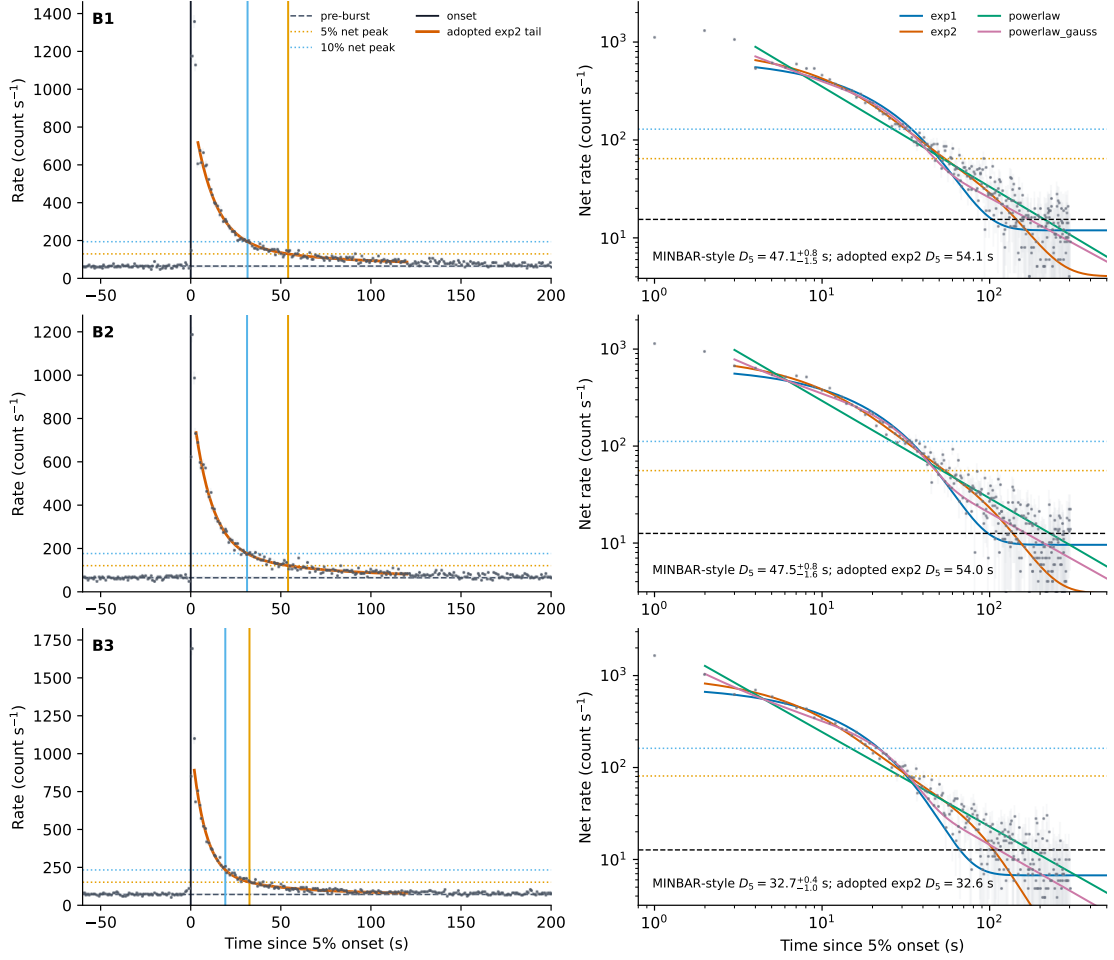


FIG. 4.9— Burst-duration measurements from the 1-s EPIC-pn light curves. Left panels show the pre-burst level, 10- and 5-per-cent thresholds, adopted two-exponential decay, and D_{10}/D_5 crossings. Right panels compare the fitted tail models in logarithmic coordinates.

that intersect the GTI gap; the same exclusion is included in the trial count. The companion 2-s search yields the maxima in Table 4.4, but also remains below threshold.

TABLE 4.4— Maximum Leahy powers in the burst-oscillation search.

Burst	Blind search, $100 \leq \nu < 500$ Hz	Pre-defined search, 260–280 Hz
B1	27.121 (205.0 Hz, 21.00 s, 2 s)	15.876 (272.0 Hz, 44.25 s, 0.5 s)
B2	29.437 (226.5 Hz, 9.00 s, 2 s)	18.610 (262.5 Hz, 11.00 s, 2 s)
B3	26.922 (185.5 Hz, 10.00 s, 2 s)	17.735 (269.5 Hz, 12.50 s, 2 s)

Notes. Each entry gives $P_{\max}$ followed in parentheses by frequency, window-centre time from onset, and window duration. The family-wise thresholds across all three bursts and both window configurations are 39.207 for the blind search and 33.314 for the pre-defined 260–280 Hz search.

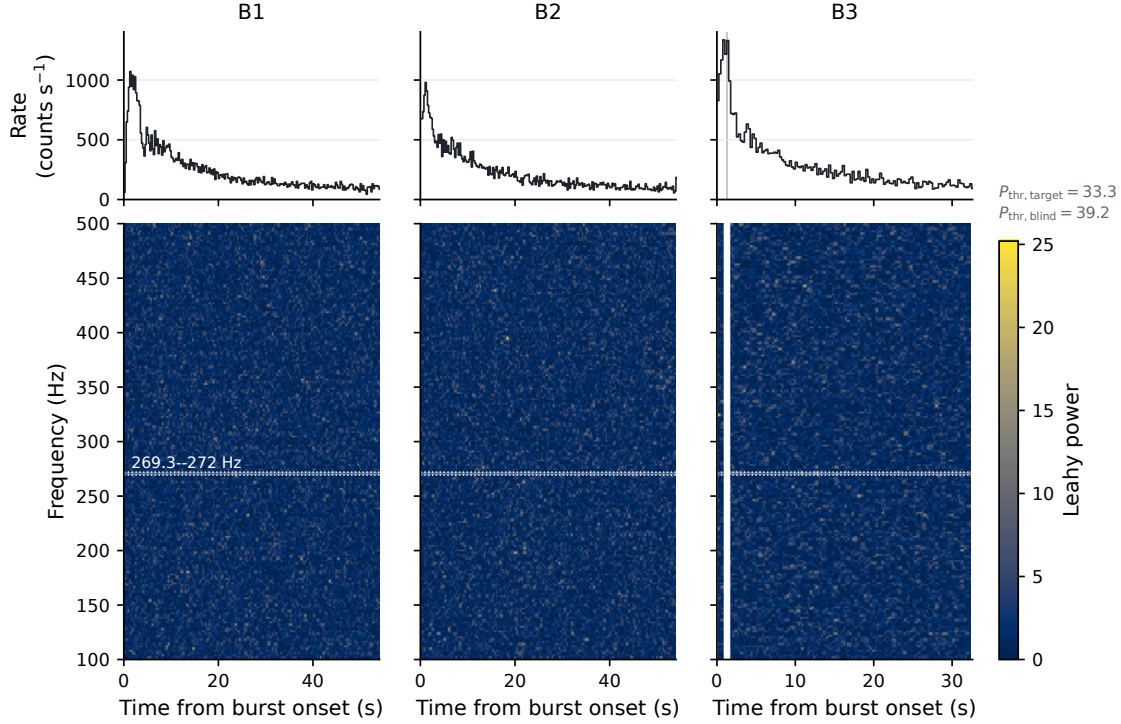


FIG. 4.10— Trial-corrected burst-oscillation search in three core-excised 0.3–10 keV EPIC-pn event lists from ObsID 0934990201. The upper panels show 0.25-s, GTI-exposure-corrected count rates over the adopted D_5 intervals. The lower panels show Leahy-normalised dynamical power spectra from 0.5-s windows stepped by 0.25 s over $100 \leq \nu < 500$ Hz. The lightly shaded band marks the 269.3–272 Hz interval of the previous *RXTE* signal (Galloway et al. 2001). The grey interval in burst 3 marks the 0.179-s GTI gap; windows with less than 95 per cent exposure are excluded. The global thresholds, including both window configurations and all three bursts, are $P_{\text{thr,target}} = 33.314$ and $P_{\text{thr,blind}} = 39.207$. No feature passes either threshold.

No statistically significant burst oscillation is detected in this three-burst sample. In the highest-count valid 0.5-s windows, the analytic 95 per cent fractional-rms upper limits are 20.7, 20.8, and 19.1 per cent for bursts 1–3, respectively. These windows provide the closest comparison with the 0.5-s windows used for the historical *RXTE* detection. The limits are above the reported 6–12 per cent rms amplitude, so these data were not sensitive enough to test whether a signal of that strength recurred. The corresponding 2-s limits are 10.7, 16.4, and 19.1 per cent for a signal coherent throughout the full 2-s window.

4.6.3 Pre-burst continua and time-averaged burst spectra

The independent pre-burst fits are listed in Table 4.5. Burst 1 has the largest fit-statistic offset: $C = 101.46$ for 75 degrees of freedom, compared with the SPEX expectation of approximately 79.1 ± 12.6 , or 1.8 standard deviations. The three solutions are retained as phenomenological persistent baselines for the corresponding burst fits.

TABLE 4.5— Independent pre-burst persistent-continuum fits.

Component and parameter	Unit	Burst 1	Burst 2	Burst 3
Persistent continuum				
bb				
norm	10^{16} m^2	$(2.25^{+0.44}_{-0.39}) \times 10^{-9}$	$(1.49^{+0.31}_{-0.27}) \times 10^{-9}$	$(2.77^{+0.45}_{-0.40}) \times 10^{-9}$
kT	keV	$1.517^{+0.067}_{-0.059}$	$1.697^{+0.086}_{-0.077}$	$1.508^{+0.056}_{-0.051}$
dbb				
norm	10^{16} m^2	$(1.45^{+0.19}_{-0.16}) \times 10^{-9}$	$(1.28^{+0.15}_{-0.14}) \times 10^{-9}$	$(1.69^{+0.20}_{-0.18}) \times 10^{-9}$
kT_{in}	keV	$1.382^{+0.049}_{-0.047}$	$1.442^{+0.048}_{-0.046}$	$1.343^{+0.045}_{-0.043}$
Fit statistic	C/dof	101.46/75	60.57/75	53.15/75

Notes. Normalisations are SPEX model normalisations in 10^{16} m^2 . Quoted uncertainties are 68 per cent ($\Delta C = 1$) profile intervals.

Table 4.6 gives the adopted f_a fits to the 2-s peak and full D_5 tail spectra. At the 2-s peaks, the burst-blackbody temperatures are $3.08^{+0.26}_{-0.21}$, $3.17^{+0.29}_{-0.24}$, and $2.88^{+0.23}_{-0.19}$ keV for bursts 1–3. The corresponding intrinsic 0.01–100 keV burst-blackbody fluxes are 2.450, 2.228, and $2.330 \times 10^{-8} \text{ erg cm}^{-2} \text{ s}^{-1}$, and the fitted persistent scales are $f_a = 6.28$, 5.47, and 6.53. These fluxes are 2-s averages and do not represent the shorter instantaneous maxima.

 TABLE 4.6— Time-averaged burst fits with the adopted $\text{bb}_{\text{burst}} + f_a C_{\text{pers}}$ model.

Component and parameter	Unit	Burst 1	Burst 2	Burst 3
Peak interval (2 s)				
bb_{burst}				
norm	10^{16} m^2	$(2.61^{+0.50}_{-0.44}) \times 10^{-8}$	$(2.13^{+0.43}_{-0.38}) \times 10^{-8}$	$(3.27^{+0.63}_{-0.55}) \times 10^{-8}$
kT	keV	$3.08^{+0.26}_{-0.21}$	$3.17^{+0.29}_{-0.24}$	$2.88^{+0.23}_{-0.19}$
F_{bol}	$10^{-8} \text{ erg cm}^{-2} \text{ s}^{-1}$	2.4499	2.2281	2.3295
Persistent-continuum scaling				
f_a	–	$6.28^{+0.50}_{-0.49}$	$5.47^{+0.45}_{-0.44}$	$6.53^{+0.54}_{-0.53}$
Fit statistic	C/dof	61.73/64	82.42/63	85.28/64
Full-tail interval				
bb_{burst}				
norm	10^{16} m^2	$(2.63^{+0.14}_{-0.14}) \times 10^{-8}$	$(2.85^{+0.17}_{-0.16}) \times 10^{-8}$	$(3.92^{+0.24}_{-0.23}) \times 10^{-8}$
kT	keV	$1.624^{+0.024}_{-0.023}$	$1.500^{+0.023}_{-0.022}$	$1.505^{+0.024}_{-0.023}$
F_{bol}	$10^{-8} \text{ erg cm}^{-2} \text{ s}^{-1}$	0.1900	0.1494	0.2088
Persistent-continuum scaling				
f_a	–	$1.566^{+0.054}_{-0.054}$	$1.490^{+0.053}_{-0.053}$	$1.336^{+0.071}_{-0.070}$
Fit statistic	C/dof	301.76/232	300.97/232	299.39/232

Notes. Normalisations are SPEX model normalisations in 10^{16} m^2 . F_{bol} is the intrinsic burst-blackbody flux integrated over 0.01–100 keV. Parameter uncertainties are 68 per cent ($\Delta C = 1$) profile intervals; no separately profiled flux uncertainties were calculated.

The full-tail spectra have lower temperatures and persistent scaling factors than the peak spectra. The full-tail spectra average over the changing temperature during cooling. The C-statistics of the three adopted tail fits are approximately 2.7–3.0 standard deviations above their SPEX expected values. Adding a power law lowers the BIC for bursts 1 and

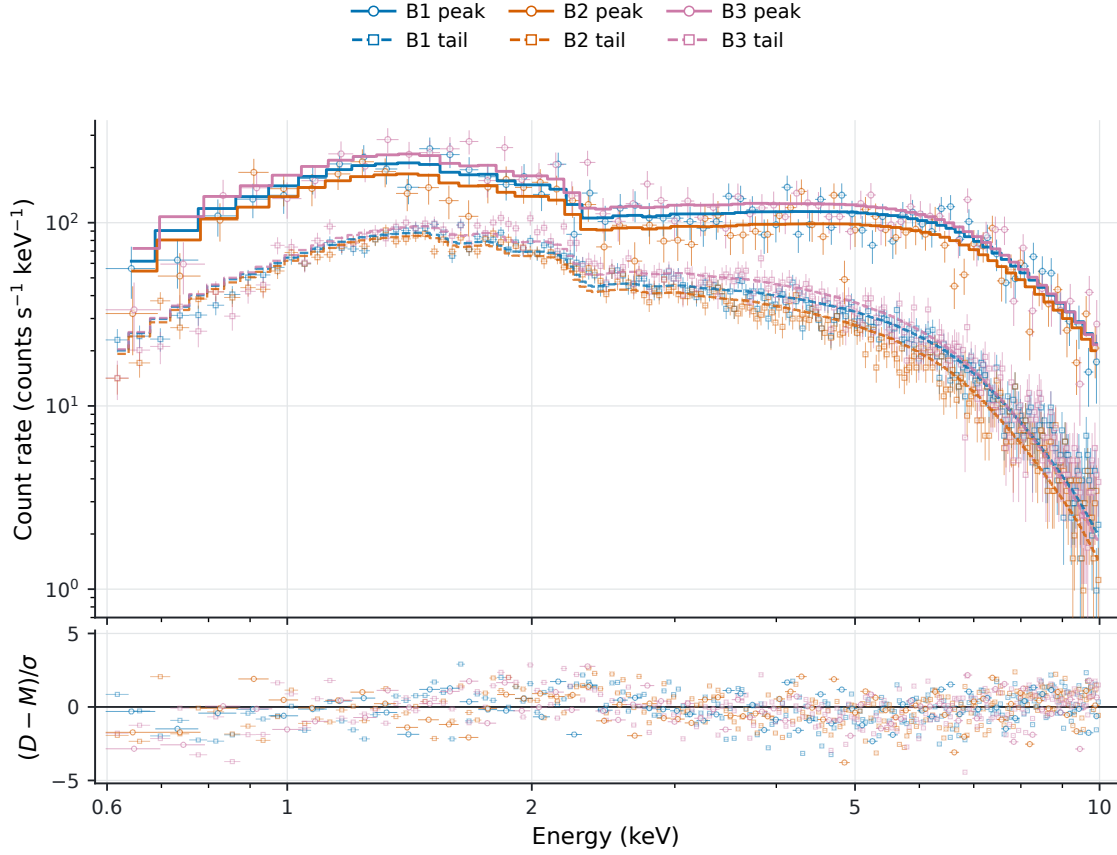


FIG. 4.11— Time-averaged 0.6–10 keV burst spectra and response-folded models. Blue, orange, and pink denote bursts 1–3. Circles and solid curves show the 2-s peaks; squares and dashed curves show the full D_5 tails. The lower panel shows $(D - M)/\sigma$.

2 but makes it strongly degenerate with the scaled persistent continuum, driving f_a to approximately 0.15 or zero; for burst 3 its normalisation converges to zero. Because of this degeneracy, Table 4.6 reports the fits without the additional power law.

4.6.4 Time-resolved evolution and PRE test

The 67 time-resolved spectra show the rapid rise and subsequent cooling more directly. The one-second burst-blackbody maxima are reported in Table 4.7. Relative to the 2-s averages, the fitted one-second flux maxima are larger by factors of 1.71, 1.15, and 1.18 for bursts 1–3. For bursts 1 and 2, however, the model burst-blackbody maximum occurs one bin after the raw count-rate maximum because the fit redistributes flux between the burst blackbody and the f_a -scaled persistent component. The instantaneous component peak therefore depends on this spectral decomposition.

Each burst has only one time bin within 10 per cent of its maximum fitted burst-blackbody flux; none shows a resolved near-constant peak-flux plateau. For burst 1, flux and temperature reach their maxima in the same bin. In the preceding bin,

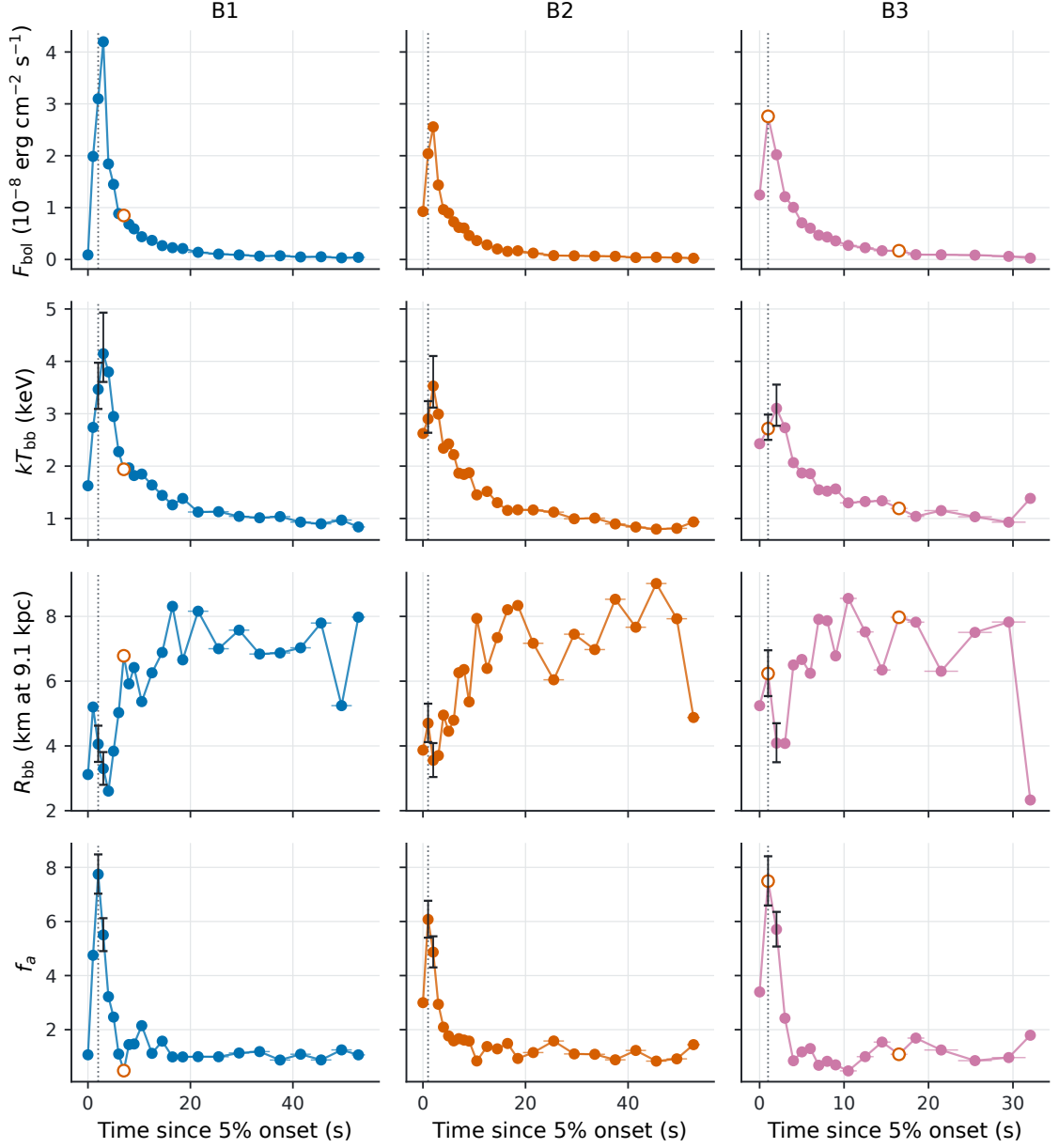
Time-resolved burst blackbody and f_a evolution


FIG. 4.12— Time-resolved EPIC-pn burst evolution. Columns show bursts 1–3; rows show burst-blackbody flux, temperature, apparent radius at 9.1 kpc, and f_a . Dotted lines mark count-rate maxima, black error bars are 68 per cent intervals for peak-defining bins, and open orange-edged symbols mark fractional exposure below 0.97.

TABLE 4.7— One-second burst-blackbody maxima from the time-resolved f_a fits.

Burst	Time from onset (s)	F_{bol}	kT_{bb} (keV)	R_{bb} (km)	f_a
B1	3.0	4.197	$4.147^{+0.783}_{-0.540}$	$3.30^{+0.51}_{-0.50}$	$5.50^{+0.62}_{-0.60}$
B2	2.0	2.557	$3.529^{+0.573}_{-0.411}$	$3.56^{+0.53}_{-0.52}$	$4.87^{+0.58}_{-0.56}$
B3	1.0	2.759	$2.716^{+0.267}_{-0.215}$	$6.24^{+0.72}_{-0.70}$	$7.49^{+0.92}_{-0.90}$

Notes. F_{bol} is in units of 10^{-8} erg cm $^{-2}$ s $^{-1}$ and has no separately profiled uncertainty. Apparent radii were calculated at the adopted distance of 9.1 kpc.

$kT_{\text{bb}} = 3.466^{+0.507}_{-0.375}$ keV and $R_{\text{bb}} = 4.06^{+0.57}_{-0.55}$ km, whose intervals overlap those of the maximum bin. Burst 2 behaves similarly: between the two peak-defining bins, the radius decreases from $4.70^{+0.60}_{-0.58}$ to $3.56^{+0.53}_{-0.52}$ km while the fitted flux is still rising from 2.040 to 2.557×10^{-8} erg cm $^{-2}$ s $^{-1}$. Neither burst shows the expansion and contraction at approximately constant flux expected for PRE.

From 1 to 2 s after the onset of burst 3, the best-fitting radius changes from $6.24^{+0.72}_{-0.70}$ to $4.09^{+0.61}_{-0.59}$ km, while the temperature changes from $2.716^{+0.267}_{-0.215}$ to $3.103^{+0.453}_{-0.333}$ keV. The radius intervals do not overlap, but the temperature intervals do, and the fitted burst-blackbody flux decreases by approximately 27 per cent, from 2.759 to 2.019×10^{-8} erg cm $^{-2}$ s $^{-1}$, rather than remaining constant. In addition, the first of these intervals has only 0.802 s of exposure because of a GTI gap. Burst 3 also lacks the near-constant flux required for PRE. None of the three bursts is therefore classified as PRE.

Expected C-statistics are available for 66 of the 67 time-resolved fits, and $C_{\text{obs}}/C_{\text{exp}}$ remains close to unity overall. The first burst-1 onset bin was run as an initial manual test and has no saved expected statistic. The burst-3 peak lies towards the upper end of the observed fit-statistic ratios and also carries the GTI-gap caveat.

The 1–2 s rises and $D_5 = 32.6\text{--}54.1$ s durations identify all three events as ordinary short type-I bursts rather than intermediate-duration bursts. MINBAR gives a historical mean PRE-burst flux of $(3.06 \pm 0.35) \times 10^{-8}$ erg cm $^{-2}$ s $^{-1}$ for 4U 1916–053 (Galloway et al. 2020). The 2-s peak averages in Table 4.6 correspond to 0.80, 0.73, and 0.76 of this reference for bursts 1–3. The one-second fitted blackbody maxima correspond to 1.37, 0.84, and 0.90 times the same value. The one-second fitted blackbody flux of burst 1 exceeds the historical mean PRE flux, but its radius and temperature evolution does not support a PRE classification. The fitted value also depends on the division between the burst blackbody and the f_a -scaled persistent continuum. We therefore use the historical PRE flux as an empirical irradiation reference for the measured f_a response.

Chapter 5

Discussion

The discussion begins with the changing dip morphology and the absorber geometry that might account for both the timing and spectral results. The velocity shifts are then tested against a pure gravitational-redshift interpretation. The final sections discuss the bursts and compare 4U 1916–053 with other compact dipping systems.

5.1 Synthesis of the Main Findings

The coordinated XMM-*Newton* and XRISM data set was divided into three time-selected morphology sectors using the EPIC-pn light curve. Sector 1 contains relatively isolated dips, sector 2 shows broader attenuation, and sector 3 contains frequent, structured dips, often more than once per orbital cycle (Fig. 4.1). The selected dip intervals account for 10.9, 28.9, and 49.7 per cent of the classified EPIC-pn exposure in sectors 1–3. Each sector contributes one persistent and one dip selection, giving six spectral data sets in total.

Within the adopted model, the absorber is represented by three photoionised components. The highly ionised **pion 1** component has the largest column density, the intermediate-ionisation **pion 2** component has a recurrent positive velocity shift, and the lower-ionisation **pion 3** component shows a recurrent best-fitting increase in column density and decrease in ionisation during dips, although the persistent and dip profile intervals do not separate in every sector.

The most significant spectroscopic result is the recurrent positive velocity shift of the intermediate-ionisation absorber. Within the adopted three-zone model, **pion 2** is redshifted by 1507–1736 km s^{−1} in all three sectors and in both the persistent and dip selections, with every quoted 68 per cent ($\Delta C = 1$) profile interval remaining above zero (Tables 4.1 and 4.2; Fig. 4.8). This behaviour contrasts with the smaller shifts of the highly ionised **pion 1** component. The **pion 3** velocity could not be constrained and was fixed at zero, so it provides no independent kinematic measurement. The recurrent 1507–1736 km s^{−1} shift of **pion 2** is specific to one zone in the adopted absorber model; it is not a common velocity of all gas along the line of sight.

In the spectral fits, the lower-ionisation **pion 3** component has a higher best-fitting column and lower best-fitting ionisation in every dip selection, although the statistical separation is not equally strong in all sectors (Fig. 4.7). These recurrent best-fitting changes are consistent with greater line-of-sight opacity during the selected dips.

The conventional interpretation associates orbital dipping with a vertically extended bulge where the accretion stream meets the outer disc (White & Swank 1982; Hu et al. 2008). The changing dip morphology and the differences among the fitted absorber components are consistent with a multiphase disc atmosphere disturbed by stream impact. Possible stream-impact structures are discussed in Section 5.3.

5.2 Orbital Dipping and a Possible Superorbital Clock

5.2.1 Evolution from regular to irregular dipping

The three sectors cover successive parts of the pn sequence, which spans approximately 91 ks. They were selected because both the recurrence and shape of the dips changed with time. The selected dip GTIs account for 10.9, 28.9, and 49.7 per cent of the classified exposure in sectors 1–3. The increase reflects more frequent and extended low-count-rate intervals across the three sectors. The exact percentages are not directly comparable because sectors 1 and 2 were selected manually, whereas sector 3 used a count-rate threshold.

Previous observations show changes on several timescales. [Smale et al. \(1988\)](#) found that the dip appearance could evolve over hours, from weak or absent dips to broad or multiple dips. [Homer et al. \(2001\)](#) reported changes within individual visits and from day to day, with the morphology repeating only approximately over 4–5 d. [Hu et al. \(2008\)](#) found dip parameters varying over 2–5 d, including a 4.87-d modulation of the dip width. The present 1.05-d pn sequence samples a shorter interval. It is consistent with changes in the covering, column density, or azimuthal extent of a non-axisymmetric absorber during a single observation.

The light curve shows a clear change in dip recurrence and morphology, while the count rate during persistent intervals shows no comparable change. The sector-resolved spectra test whether the changing dip morphology is accompanied by changes in the absorber parameters.

5.2.2 Orbital and superorbital viewing geometry

Two timescales are relevant: the orbital dip-recurrence period and the possible 4–5 d superorbital modulation introduced in Section 1.4.1. The shorter of these is the 3000.65126-s dip recurrence period used in Fig. 4.1 ([Iaria et al. 2015](#)). The recurrence of the sector 1 minima close to successive relative phase-zero markers shows that the dominant obscuration is organised by the binary orbit. This is consistent with material approximately fixed in the binary frame, as expected for the outer-disc stream-impact region ([White & Swank 1982](#); [Hu et al. 2008](#)). The additional minima and broader attenuation in sectors 2 and 3 then require either multiple structures, a wider azimuthal distribution of absorbing material, or a changing view of the same disturbed region.

The possible second clock is less precisely defined. The stable X-ray and optical periods imply a 3.9087-d beat period ([Chou et al. 2001](#)), while [Homer et al. \(2001\)](#) found a 4.74 ± 0.05 -d quasi-period in the evolving dip morphology. [Hu et al. \(2008\)](#) subsequently measured a 4.87-d modulation of the dip width and interpreted it as likely nodal precession; the detection of a negative superhump provides independent motivation for a tilted, retrogradely precessing disc ([Retter et al. 2002](#)). These measurements support a precession-related timescale of roughly 4–5 d. The reported signals are not identical,

so they are used here only to define an approximate 4–5 d timescale rather than a single ephemeris.

The pn interval analysed here spans approximately 91 ks, or 1.05 d, which is only about 0.22 of the reported 4.87-d dip-width cycle. A changing disc orientation could therefore alter both the line-of-sight elevation through the rim and the position at which the stream encounters the disc during the observation, qualitatively producing the progression from isolated to complex dips. However, the data neither cover a complete long-period cycle nor revisit the same superorbital phase. The data are compatible with a precessing disc, but do not detect this modulation. Stochastic changes in the scale height, density, or cooling of the stream-impact region remain an alternative and may operate together with precession.

5.3 Physical Structure of the Absorber

5.3.1 Evidence for a multi-zone structure

Multi-phase disc plasma is a recurring result in high-inclination systems. [Díaz Trigo et al. \(2006\)](#) explained spectral changes during dips through the column and ionisation of ionised gas; [Xiang et al. \(2009\)](#) separated a highly ionised extended component from a more variable disc-rim component; and [Psaradaki et al. \(2018\)](#) inferred two photoionised components with different spatial coverage. The persistent-spectrum model decompositions in Figs. 4.5 and 4.6 show that the three adopted `pion` zones do not produce identical absorption templates. The high-ionisation zone contributes strongly to the H- and He-like line complexes close to their laboratory energies, including the Fe xxv and Fe xxvi structure resolved by *Resolve* and the O viii, Ne x, and Mg xii features in the RGS band. The intermediate-ionisation zone produces a second set of troughs displaced towards lower energy in the Fe-K band and towards longer wavelength in the soft band, matching its large fitted redshift. The lower-ionisation zone adds further neighbouring structure and substantial soft-band opacity. Their overlapping but non-coincident signatures give the three components distinct roles within the adopted model.

The fitted parameters show the same separation. The high-ionisation zone has by far the largest column density and comparatively small velocity shifts with no uniform persistent-to-dip trend. The intermediate-ionisation zone has a much smaller column but retains a positive shift of 1507–1736 km s⁻¹ in every persistent and dip spectrum. The lower-ionisation zone has the smallest log ξ and shows a recurrent best-fit increase in column density and decrease in ionisation during dips, although the statistical separation is not equally strong in every sector (Fig. 4.7). Its velocity was fixed at zero and therefore supplies no independent evidence that this component is physically at rest; the available statistics could not distinguish zero from a small shift.

These spectral and velocity differences motivate the physical picture developed below: the fitted components are associated with an inner atmosphere, stream-impact material, and

an outer absorber. Their radial order cannot be inferred from ionisation alone because $\xi = L_{\text{ion}}/nR^2$ depends on both density and radius.

5.3.2 Stream impact, bulge, overflow, and splash material

The standard dipping picture places a vertically extended bulge where the accretion stream meets the outer disc (White & Swank 1982; Frank et al. 1987; Hu et al. 2008). This remains a natural origin for obscuration recurring at the orbital period. A single static bulge, however, does not by itself account for a modest-column component with a persistent large redshift, a much larger-column component with a smaller shift, and lower-ionisation opacity that strengthens during dips. Hydrodynamic stream–disc calculations produce a wider range of structures: depending on the cooling of the shocked gas, the collision can generate a vertically extended impact region, splash or backsplash, downstream spreading, and overflow towards smaller radii (Armitage & Livio 1998).

In the proposed geometry shown in Fig. 5.1, the intermediate-ionisation **pion 2** component is associated with gas that has only recently been deflected at, or just downstream of, the stream-impact point. This zone has a relatively low fitted column, $N_{\text{H}} \simeq 0.8\text{--}1.8 \times 10^{22} \text{ cm}^{-2}$, but the largest coherent line-of-sight shift, with a redshift of $1507\text{--}1736 \text{ km s}^{-1}$ in all six spectra. Freshly disturbed material can retain an ordered post-impact velocity while occupying only a short path through the line of sight, which could account for this combination of large shift and modest column. The spectra do not distinguish whether this motion is produced by material at the impact point, backsplash, overflow, or a downstream flow; possible contributions from projection and gravitational redshift are considered separately in Section 5.4.

The highly ionised **pion 1** component is instead associated, tentatively, with a denser disturbed atmosphere extending further across the inner disc. If post-impact gas spreads inward or downstream, differential rotation, mixing, and the superposition of material at different azimuths can reduce its coherent line-of-sight velocity centroid while increasing the accumulated path length. This could explain the much larger $N_{\text{H}} \simeq 22\text{--}73 \times 10^{22} \text{ cm}^{-2}$ of **pion 1** together with its smaller shifts across the six spectra. In this interpretation, “slower” means a smaller net line-of-sight centroid; it is not a direct measurement that the three-dimensional gas speed has decreased. The higher ionisation is compatible with greater irradiation, but does not independently locate the zone because ξ also depends on density.

The lower-ionisation **pion 3** component is a plausible tracer of outer-disc or rim material that contributes more strongly when the dip sight line crosses the stream-impact environment. Its recurrent increase in best-fitting column and decrease in ionisation during dips support this association, subject to the sector-dependent uncertainties described in Section 4.4. Its kinematics are not measured: the available statistics did not constrain the centroid, so it was fixed at zero in every fit. Material at rest and a weakly blueshifted or redshifted component therefore remain indistinguishable, and the

outer-zone location in Fig. 5.1 is not inferred from the fixed velocity.

The upper panel of Fig. 5.1 illustrates a dip sight line through a more vertically extended and structured impact region. The lower panel shows the corresponding sight line during persistent intervals. In a compact disc, rotation and shear may redistribute disturbed gas over a wider azimuth, so related absorption need not be confined to the deepest dips. This interpretation adds to the conventional bulge picture rather than excluding it. It is qualitatively similar to the clumpy, multiphase bulge inferred in 4U 1624–490 (Caruso et al. 2025), although differences in orbital period, disc size, luminosity, and data quality prevent a one-to-one identification.

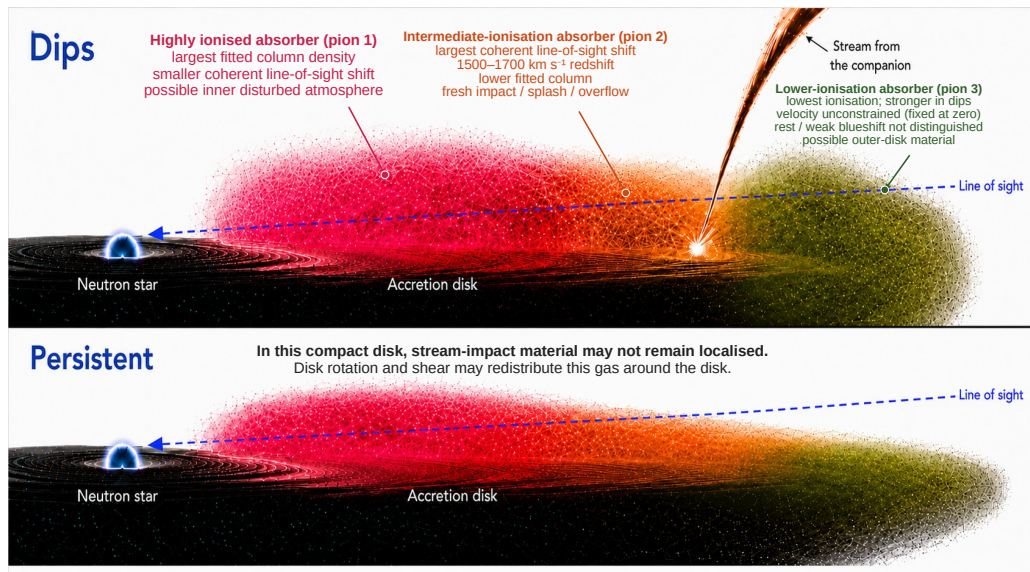


FIG. 5.1— Schematic geometry for the fitted absorber zones in dip (upper) and persistent (lower) intervals. Pink, orange, and green associate pion 1–3 with a possible inner atmosphere, impact-driven material, and a lower-ionisation outer layer. The geometry is schematic and the pion 3 velocity was fixed at zero. The interpretation and hand-drawn sketch followed discussions with the supervisors. ChatGPT Image 2.0 generated the polished illustration.

5.3.3 Disc atmosphere, wind, or failed wind?

Photoionisation establishes that the absorbing plasma is exposed to the central continuum, but it does not distinguish a bound disc atmosphere from an outflow. A disc wind requires a statistically significant blueshift, ideally repeated across several transitions, together with a physically plausible launching region (Neilsen & Degenaar 2023). No fitted zone in the present analysis meets this kinematic criterion. The pion 1 centroids are positive or boundary-limited at zero, pion 2 is redshifted in all six spectra, and the pion 3 velocity was fixed. The current data contain no secure evidence for a blueshifted disc wind.

A failed wind is gas that is lifted from the disc but does not reach the local escape speed, so it stalls or falls back. In the bright black-hole LMXB GRS 1915+105, Miller

et al. (2020) observed a low-column blueshifted phase that became redshifted as the source entered a highly obscured, Compton-thick phase. Here Compton-thick means an electron-scattering optical depth of order unity. In the high-inclination transient black-hole candidate MAXI J1305–704, density-sensitive lines observed during its 2012 outburst provided an independent absorber radius for comparison with the measured and expected velocities (Miller et al. 2014). 4U 1916–053 shows neither the transition from blueshift to redshift seen in GRS 1915+105 nor an independent density-based radius. The same failed-wind diagnosis therefore cannot be applied directly.

The redshifted **pion 2** component has $N_{\text{H}} \simeq 0.8\text{--}1.8 \times 10^{22} \text{ cm}^{-2}$. In the hydrogen-equivalent approximation, its Thomson optical depth is

$$\tau_{\text{T}} \simeq \sigma_{\text{T}} N_{\text{H}} \simeq 0.005\text{--}0.012, \quad (5.1)$$

which is roughly two orders of magnitude below the Compton-thick scale $\tau_{\text{T}} \sim 1$. The exact conversion depends on the adopted abundances, but **pion 2** is clearly not Compton-thick. The much larger column of **pion 1** is accompanied by smaller line-of-sight shifts, and none of the six spectra shows the low-column blueshift to high-column redshift transition reported for GRS 1915+105.

A second test asks whether outward-moving gas at the adopted outer-disc radius would be below the local escape speed, as expected for a failed wind. If **pion 2** is associated with the stream–disc impact region, $R_{\text{out}} \simeq (8\text{--}10) \times 10^4 r_{\text{g}}$ gives

$$v_{\text{esc}}(R_{\text{out}}) = c \left(\frac{2r_{\text{g}}}{R_{\text{out}}} \right)^{1/2} \simeq 1340\text{--}1500 \text{ km s}^{-1}. \quad (5.2)$$

This is comparable to the measured line-of-sight shift of $1507\text{--}1736 \text{ km s}^{-1}$. Since the full speed cannot be smaller than its line-of-sight component, this simple comparison does not demonstrate a flow securely below escape speed at the outer-disc radius. It also does not exclude a failed wind: the absorber radius and three-dimensional velocity are unknown, and rotation, orbital motion, and gravity may all contribute to the measured centroid. At a smaller radius, the escape speed would be higher.

The high ionisation, large column density, and comparatively small shifts of **pion 1** are consistent with a dense bound or slowly moving disc atmosphere. The recurrent redshift of **pion 2** is not the signature of a conventional outflow towards the observer, but a redshift alone does not prove inflow. Stream impact, splash, overflow, projected rotation, and gravitational redshift may all contribute. The **pion 3** velocity was fixed at zero, so the data cannot distinguish stationary gas from a low-velocity outflow in this zone. The preferred description is therefore a multiphase disc atmosphere disturbed by stream impact. A failed wind remains possible, but is not established by the present velocity and column measurements. This picture is consistent with the redshifted-atmosphere interpretation of Trueba et al. (2020).

5.4 Testing a Pure Gravitational-Redshift Interpretation

5.4.1 A conditional pure-gravity radius test

The positive absorber shifts can be tested against the gravitationally redshifted disc-atmosphere interpretation of [Trueba et al. \(2020\)](#). The test asks a conditional question: if gravity produced the full fitted shift, what radius would be required, and would it agree with the proposed outer-disc geometry? It does not measure the absorber radius.

For an emitter at rest at radius R in Schwarzschild spacetime and an observer at infinity, the gravitational-redshift relation gives ([Hamilton 2021](#), Section 7.8)

$$z_{\text{grav}} = \left(1 - \frac{2GM}{Rc^2}\right)^{-1/2} - 1, \quad (5.3)$$

where $r_g \equiv GM/c^2$. SPEX treats $\mathbf{z}_v$ as the systematic velocity of the absorber and uses the longitudinal special-relativistic Doppler relation ([SPEX Development Team 2026c](#)). For positive motion away from the observer, $\nu_{\text{obs}}/\nu_{\text{em}} = [(1 - \beta)/(1 + \beta)]^{1/2}$. Since $1 + z_{zv} \equiv \nu_{\text{em}}/\nu_{\text{obs}}$, this gives

$$1 + z_{zv} = \left(\frac{1 + \beta}{1 - \beta}\right)^{1/2}, \quad \beta \equiv \frac{v_{zv}}{c}. \quad (5.4)$$

Equating Eqs. 5.3 and 5.4 under the pure-gravity assumption gives

$$\frac{R_{\text{grav}}}{r_g} = \frac{2}{1 - (1 + z_{zv})^{-2}} = \frac{1 + \beta}{\beta} = 1 + \frac{c}{v_{zv}} \simeq \frac{c}{v_{zv}}. \quad (5.5)$$

For a $1.4 M_\odot$ neutron star, $r_g = 2.07$ km. [Trueba et al. \(2020\)](#) reported shifts of approximately 220–290 km s^{−1}, corresponding to a characteristic radius near $1200 r_g$ under this assumption. The full 1507–1736 km s^{−1} range measured for **pion 2** would instead give 174–200 r_g , or approximately 360–410 km. These values are equivalent pure-gravity radii, not measured absorber locations.

5.4.2 Consistency with the outer-disc geometry

For 4U 1916–053, the adopted outer-disc radius is approximately $(8\text{--}10) \times 10^4 r_g$ ([Iaria et al. 2015](#); [Trueba et al. 2020](#)). The gravitational shift at this radius is only

$$v_{\text{grav}}(R_{\text{out}}) \simeq \frac{c r_g}{R_{\text{out}}} \sim 3\text{--}4 \text{ km s}^{-1}, \quad (5.6)$$

far below the measured 1507–1736 km s^{−1}. Therefore, if **pion 2** is associated with the outer stream-impact region, gravity cannot dominate its fitted shift. This comparison does not by itself identify the gas motion.

5.4.3 Interpretation and modelling systematics

Based on the comparison in Section 5.4, the fitted zv parameter for **pion 2** in the three persistent and three dip spectra should be interpreted as a net line-centroid shift, not a direct radius measurement. To first order in v/c , its physical contributions can be written schematically as

$$v_{zv} \simeq v_{\text{grav}} + v_{\text{sys}} + v_{\text{orb,los}} + v_{\text{rot,los}} + v_{\text{local,los}}. \quad (5.7)$$

Here v_{grav} is the gravitational redshift expressed in velocity units, v_{sys} is the systemic radial velocity of the binary, $v_{\text{orb,los}}$ is the line-of-sight orbital velocity of the neutron star, $v_{\text{rot,los}}$ is projected disc rotation, and $v_{\text{local,los}}$ is local motion of the absorbing gas. The last term may include motion associated with stream impact, splash, overflow, or downstream material. Equation 5.7 is a schematic expression rather than a fitted dynamical model; the present spectra do not measure its terms separately. Calibration and modelling offsets may also affect the absolute centroid.

Under the preferred outer stream-impact geometry, the calculation above limits the gravitational contribution at the outer disc to only $3\text{--}4 \text{ km s}^{-1}$, so local gas motion or the other kinematic terms must provide most of the fitted shift. The redshift does not by itself prove inflow. No independent density measurement places **pion 2** at the equivalent $174\text{--}200 r_g$ pure-gravity radius, although a smaller gravitational contribution remains possible.

5.5 Type-I Bursts and the Short-Timescale Response

The three EPIC-pn bursts test the short-timescale response of the persistent continuum. Because bursts 2 and 3 occur near morphology-interval boundaries, their timing is also compared with the later dip evolution. The oscillation search tests whether the previously reported $\sim 270\text{-Hz}$ signal recurs.

5.5.1 Burst energetics, cooling, and PRE classification

The $1\text{--}2 \text{ s}$ rises and $D_5 = 32.6\text{--}54.1 \text{ s}$ durations are typical of short type-I bursts (Galloway & Keek 2017; Galloway et al. 2020). The 2-s peak-average burst-blackbody fluxes are $0.73\text{--}0.80$ of the historical mean PRE-burst flux. None of the three bursts shows radius expansion and contraction at nearly constant flux, so none meets the criteria for a secure PRE classification (Lewin et al. 1993; Galloway & Keek 2017). Burst 1 reaches a one-second fitted blackbody flux above the historical reference, but this value depends on the division between the burst blackbody and the f_a -scaled pre-burst continuum and is not accompanied by the expected radius evolution.

5.5.2 Burst-oscillation search

The earlier *RXTE* signal near 270 Hz occurred in one of the ten bursts examined by [Galloway et al. \(2001\)](#). None of the three EPIC-pn bursts passes the adopted trial-corrected threshold. The 95 per cent fractional-rms upper limits are 19.1–20.8 per cent for the 0.5-s search and 10.7–19.1 per cent for the 2-s search. These limits do not exclude the historical 6–12 per cent signal, which was measured over 2–60 keV and increased with energy, whereas the present search uses 0.3–10 keV events. The non-detection provides no new constraint on ignition latitude or flame spreading.

5.5.3 Persistent-emission response and the meaning of f_a

The fitted f_a values rise at the burst peaks: the 2-s values are 6.28, 5.47, and 6.53 for bursts 1–3, compared with full-tail averages of 1.566, 1.490, and 1.336. When an additional power law is included, it is degenerate with the scaled persistent continuum and drives f_a to approximately 0.15 or zero for bursts 1 and 2; for burst 3, the power-law normalisation converges to zero. This test shows that f_a depends on the continuum decomposition. It is therefore not used as a direct measurement of the instantaneous accretion rate ([Wörpel et al. 2013, 2015](#); [Degenaar et al. 2018](#)).

The burst fits contain no local *pion* zones or burst-resolved high-resolution line measurements, so they do not measure a change in the ionised absorber.

5.5.4 Relation to the sector evolution

Bursts 2 and 3 occur close to the ends of sectors 1 and 2, whereas burst 1 does not occur near a sector boundary. The timing pattern is not repeated across all three events. The persistent and dip spectra exclude the bursts and average several kiloseconds, so they cannot test an immediate absorber response. These observations do not support a causal link between the bursts and the hours-long change in dip morphology.

5.6 Comparison with Previous High-Resolution Studies

5.6.1 Relation to the previous *Chandra* result

[Trueba et al. \(2020\)](#) provide the closest source-specific benchmark for the present velocity measurements. They associated a modest redshift of highly ionised persistent absorption with an inner disc atmosphere and used absorption measured during dips as an outer-disc reference. Section 5.4 tests the same pure-gravity interpretation quantitatively for the present fits and shows why it cannot account for the large shift of the intermediate-ionisation component under the adopted stream-impact geometry.

The fitted components should not, however, be matched one-to-one between the two

analyses. In the present three-zone model, the highly ionised `pion 1` component occupies the more comparable low-velocity regime, whereas the much larger and persistent redshift belongs to the lower-column intermediate-ionisation `pion 2` component. The latter should therefore not be described simply as acceleration of the absorber detected by [Trueba et al. \(2020\)](#). Differences in the transitions fitted, spectral resolution, observing epoch, selection criteria, velocity convention, and photoionisation model can change how blended absorption is assigned among components.

The present study fits the RGS soft band and the *Resolve* Fe-K band with one absorber model, using separate persistent and dip spectra in three timing sectors. The large positive shift belongs to one zone and persists across all six spectra; the other zones show different column, ionisation, and velocity behaviour. Any claim of an epoch-to-epoch velocity change requires a matched reanalysis of the *Chandra* and current data with the same transitions and model hierarchy.

5.6.2 Other dipping and ultracompact binaries

The earlier XMM-*Newton* spectrum of 4U 1916–053 already established that highly ionised Fe absorption is present outside dips and that the spectrum becomes less ionised during dipping ([Boirin et al. 2004](#)). Across a wider sample of dipping LMXBs, [Díaz Trigo et al. \(2006\)](#) likewise found that much of the persistent-to-dip spectral evolution can be described by an increasing ionised column and decreasing ionisation, rather than by a single additional screen of cold gas. The recurrent increase in the best-fitting `pion 3` column and decrease in its ionisation therefore place 4U 1916–053 within an established class of dip-dependent plasma changes. The non-uniform statistical separation across the three sectors, however, prevents this correspondence from being used as evidence that the same number of physical zones is present in every system.

High-resolution studies also show that a multiphase absorber need not have one universal geometry. In the 20.89-h neutron-star dipper 4U 1624–490, [Xiang et al. \(2009\)](#) associated a highly ionised component with an extended disc atmosphere and a more variable, less-ionised component with the disc rim. Time- and intensity-resolved XMM-*Newton* spectroscopy subsequently showed that its impact-region absorber is clumpy and multiphase ([Caruso et al. 2025](#)). In the 3.82-h eclipsing neutron-star LMXB EXO 0748–676, two photoionised atmospheric components were inferred, with the lower-ionisation component tentatively connected to stream-impact material ([Psaradaki et al. 2018](#)). A layered, impact-perturbed atmosphere is therefore plausible, but these comparisons do not locate the three `pion` components fitted here.

Kinematics provide the more discriminating comparison. Full-orbit XRISM/*Resolve* spectroscopy of 4U 1624–49 recently recovered a phase-dependent velocity curve and a net outflow of approximately 200–320 km s^{−1} ([Díaz Trigo et al. 2026](#)). By contrast, the present fits contain a much larger positive shift confined to the intermediate-ionisation zone. This difference illustrates why high inclination and ionised absorption alone cannot

identify an atmosphere, wind, or stream-impact flow: the centroid must be followed with orbital phase and tied to specific line families. The simultaneous soft-band and Fe-K coverage identifies a candidate impact-driven component in every persistent and dip selection. A matched analysis of other short-period neutron-star dippers, such as 4U 1323–62 (2.94 h) and EXO 0748–676 (3.82 h), could test whether a similar component is also present in systems with larger discs (Díaz Trigo et al. 2006; Psaradaki et al. 2018). Such a comparison would not by itself show that the component is characteristic of ultracompact discs.

5.7 Limitations and Systematic Uncertainties

The first limitation is temporal sampling. The coordinated observation provides only one approximately 91-ks EPIC-pn sequence, corresponding to about 0.22 of the reported 4.87-d dip-width cycle. The three sectors are operational descriptions of this observation and are not known to recur. Moreover, the sector 1 and sector 2 dip intervals were selected manually, while sector 3 used a count-rate threshold. Their dip fractions and spectral differences therefore cannot be compared as though they came from one uniform selection function. No systematic grid of alternative sector boundaries or dip thresholds was propagated through the full spectral analysis.

The second limitation is statistical and model dependence. The quoted uncertainties are 68 per cent ($\Delta C = 1$) one-parameter profile intervals conditional on the adopted continuum and three-zone absorber. Several velocity widths reached imposed boundaries, and the `pion` 3 centroid and width were fixed because the available statistics did not constrain them. The sector 3 persistent absorber intervals require additional caution because the corresponding scans did not all share a verified common minimum. For the sector 2 dip spectrum, some initial searches were started from a solution approximately $\Delta C = 14$ above a subsequently identified lower minimum and produced spuriously large intervals. Repeating the affected searches around the lower- C solution, with neighbouring absorber columns fixed where required, yielded finite profile intervals for the reported parameters. This resolves the earlier numerical problem but does not remove the model dependence or possible covariance between absorber zones. Boundary-limited, fixed, and upper-limit cases remain explicitly identified.

The component count is also not unique. No complete sequence of one-, two-, and three-zone model comparisons was retained, and correlations among N_{H} , $\log \xi$, broadening, and centroid shift can redistribute opacity between neighbouring components. In the dip fits, the continuum shape was fixed to the sector-matched persistent solution and the scattered fraction was fixed at $f_{\text{sc}} = 0.09$. The dip-local `hot` columns were upper limits rather than detections. The figures show how the adopted decomposition behaves; they do not provide an independent detection significance or a unique physical identity for every zone in every spectrum.

Instrumental and dynamical systematics are especially relevant to the velocity

interpretation. The RGS and *Resolve* spectra have instrument-specific screening, effective GTIs, responses, and energy-scale calibration, even though their persistent and dip selections were transferred from the same pn timing reference. The reported profile intervals quantify statistical variation within the fit; they do not by themselves encompass every possible cross-calibration or absolute energy-scale offset. In addition, the fitted centroid combines systemic, orbital, rotational, gravitational, and local-flow contributions. Because these terms were not separated with an orbital radial-velocity model, the measured positive shift cannot be converted directly into an inflow speed or absorber radius.

The geometrical mapping in Fig. 5.1 is the most interpretation-dependent part of the thesis. No independent density measurement fixes the radii of the zones, and $\xi = L_{\text{ion}}/nR^2$ does not yield a radius without n . The association of **pion 2** with fresh impact, splash, or overflow material is motivated by its stable large shift and modest column, while the assignments of **pion 1** and **pion 3** to an inner disturbed atmosphere and an outer layer remain plausible but non-unique. The failed-wind and pure-gravitational alternatives are disfavoured only under the adopted outer-disc geometry; neither is excluded for an independently established smaller radius.

The burst analysis has limited scope: the f_a values depend on a fixed pre-burst continuum template, and the fits do not measure an absorber response. The apparent radii and oscillation upper limits remain conditional on the adopted spectral model and search setup. These caveats do not change the conclusions that no secure PRE sequence or burst oscillation was detected and that the bursts show no repeatable relation to the sector evolution.

These limitations set the strength of the conclusions. The changing dip morphology is directly observed. The repeated positive **pion 2** shift is the most significant spectroscopic result within the adopted model, although the systematics above affect its absolute physical velocity. The recurrent **pion 3** dip trend is suggestive but not statistically separated in every sector. The spatial ordering of the zones and the detailed stream-impact picture remain testable interpretations rather than direct measurements.

Chapter 6

Conclusions and Future Prospects

6.1 Main Conclusions

This thesis examined how the ionisation structure and line-of-sight kinematics of disc-associated plasma change with dip morphology in 4U 1916–053. The main conclusions are:

1. The 100-s EPIC-pn light curve evolves from relatively isolated dips to broader attenuation and closely separated multiple dips (Section 4.1). The three sectors capture this progression for persistent and dip spectroscopy, but do not by themselves establish a periodic long-term modulation.
2. Within the adopted three-zone model, the RGS soft band and *Resolve* Fe-K band contain overlapping but non-identical absorption signatures from high-, intermediate-, and lower-ionisation plasma (Section 4.3). The decomposition gives a common description of all sectors and both source selections, although the analysis does not establish that this is the only viable absorber decomposition.
3. The lower-ionisation **pion 3** component has a higher best-fitting column and lower best-fitting ionisation in every dip spectrum (Section 4.4). This recurrent tendency is consistent with a denser, cooler sight line through outer or impact-region material, but the persistent and dip intervals are not statistically separated for every parameter in every sector.
4. The most significant spectroscopic result is the positive line-centroid shift of 1507–1736 km s^{−1} in the intermediate-ionisation **pion 2** component in all six persistent and dip fits (Fig. 4.8). If this zone is associated with the outer stream–disc interaction, the shift is much larger than expected from gravitational redshift alone. Its moderate column and recurrent redshift also differ from the high-column failed-wind comparison considered in the discussion. We interpret the shift as coherent line-of-sight motion generated by stream impact, splash, overflow, or downstream gas. Orbital projection and a smaller gravitational contribution may also be present.
5. None of the three bursts shows a secure PRE sequence (Section 4.6). The fitted f_a enhancement is model dependent. No significant burst oscillation was detected (Section 4.6.2), and the burst timing provides no evidence for a causal connection to the subsequent dip-morphology evolution.

6.2 Future Prospects

The working picture is a multiphase disc atmosphere disturbed by stream impact. In this toy geometry, the redshifted **pion 2** component may be associated with gas motion near the impact site or in overflow and downstream material. The lower-ionisation **pion 3**

component has a larger fitted column during the dips and may trace cooler outer material. A failed wind, projected orbital or disc motion, and gravitational redshift at a smaller radius remain possible alternatives for the `pion 2` shift. The measurements below would test these interpretations.

6.2.1 Orbital-phase-resolved kinematics

The immediate priority is to measure the strongest transitions from `pion 1` and `pion 2` in common orbital-phase bins. The velocity model should include the systemic velocity, neutron-star orbital motion, projected disc rotation, impact azimuth, and the absolute energy-scale uncertainty of each instrument. Persistent and dip intervals should be retained as separate selections within this phase grid so that changes in their relative coverage are not mistaken for a velocity curve. The full-orbit *Resolve* analysis of 4U 1624–49 provides a direct methodological precedent (Díaz Trigo et al. 2026).

The main test is whether the `pion 2` velocity changes with orbital phase after accounting for the systemic velocity, neutron-star orbital motion, and instrument energy-scale uncertainties. This would show whether the measured redshift contains a phase-dependent projected-motion component or remains as a similar offset throughout the orbit.

6.2.2 Long-term dipping and the 4–5-d modulation

The aim of a longer study is to test whether the dip morphology and absorber parameters repeat on the reported 4–5-d timescale. The present 1.05-d pn sequence covers only part of such a cycle. Existing XMM-*Newton* and *Chandra* observations, including those with weak or absent dips, should be reduced with the same energy bands and divided into the same orbital-phase bins. The number, width, and hardness of the dips and the fitted absorber columns can then be compared at the same long-period phase. The published 3.9087-d beat period, 4.74 ± 0.05 -d quasi-period, and 4.87-d dip-width modulation should be tested separately because they are not one exact ephemeris (Chou et al. 2001; Homer et al. 2001; Hu et al. 2008). Repeated dip properties at the same phase would support a tilted or warped precessing disc. A lack of repetition would be more consistent with irregular changes in the stream-impact region.

6.2.3 Selection Criteria and Spectral-Model Robustness

To test whether the absorber trends depend on how the persistent and dip intervals were selected, the count-rate classification should be repeated with a hardness-ratio criterion and with a grid of sector boundaries and dip thresholds. The same selection method should be applied to every sector. This is particularly important for sector 2, where the broad attenuation is not cleanly separated from the surrounding variability. The RGS and *Resolve* spectra should also be reconstructed from strictly intersected GTIs to

measure any dilution caused by unequal coverage. These checks would show whether the fitted persistent–dip trends remain when the selections are changed.

The three-zone absorber decomposition should also be tested against one- and two-zone models, with simulations used to determine when neighbouring zones can be distinguished. The continuum should be refitted with alternatives that still reproduce the broad-band curvature, for example by replacing `comt` with a cut-off power law while retaining `bb` and `dbb`. The thermal-component parameters and hard-band curvature should remain within the ranges allowed by the broad-band fit that includes NuSTAR. The scattered fraction, velocity widths, abundances, and instrument energy-scale offsets should also be varied.

Separate profile-likelihood scans should test the `pion 2` velocity in the RGS soft-X-ray absorption lines and in the *Resolve* Fe XXV and Fe XXVI lines. This would show whether the two instruments independently require a positive velocity shift, without relying on the complete broad-band model.

6.2.4 Density, radius, and transient response

These measurements can test whether the `pion 2` redshift could come from a failed wind: gas lifted from the disc that does not reach the local escape speed and then stalls or falls back. This requires an independent estimate of the absorber density and radius. A density-sensitive transition or an ionisation-response timescale could constrain n ; combining this with $\xi = L_{\text{ion}}/nR^2$ would then give an estimate of R . The measured velocity could be compared with the escape speed at that radius. A line-of-sight speed above the local escape speed would disfavour a bound failed wind there. A lower speed would leave the failed-wind interpretation possible, but would not establish it because the full three-dimensional velocity is unknown. The same density-based radius would also help distinguish an outer stream-impact location from the much smaller radius required for a pure gravitational redshift. Hydrodynamic calculations tailored to the short-period binary could then predict the projected velocities and columns of the impact shock, splash, overflow, and downstream material for comparison with the fitted zones.

The bursts offer a secondary route to a response timescale. A future observation with the planned NewAthena/X-IFU could search for changes in the same absorption lines during the burst rise and cooling tail (European Space Agency 2026). A detection would constrain recombination and density; a non-detection with useful limits would still restrict the distance or covering of the gas. The continuum analysis should in parallel test reprocessing and Comptonisation components instead of requiring one f_a factor to describe all non-blackbody evolution. Comparing these alternatives would test whether the large f_a values trace a change in the persistent emission or the reprocessing of burst photons, and would make the inferred burst-blackbody flux less dependent on the adopted continuum decomposition.

6.2.5 Comparison with Other Short-Period Dippers

Applying the same timing selection, transmission decomposition, and phase-resolved velocity model to other short-period dippers would test whether a redshifted impact component is common in compact systems or specific to this epoch of 4U 1916–053. Such comparisons would test the stream-impact picture through the predicted velocity and opacity as functions of viewing phase, rather than through a time-averaged schematic alone.

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Appendix A

Analysis Provenance and Supplementary Diagnostics

A.1 Data reduction provenance

Chapter 2 describes the screening choices that affect the scientific products. This appendix records the exact software and calibration context, together with the shortest command chain needed to reproduce the main pn, RGS, and *Resolve* products. Directory paths and long archive filenames are shortened for readability. In the listings, `SELECTION.gti` denotes one of the six GTI files in Table A.1; the same extraction was repeated for each selection and detector. File-transfer, exploratory plotting, and discarded test commands are not included.

A.1.1 Software and calibration record

The XMM-*Newton* products were reduced with SAS v22.1.0. The CALINDEX metadata are stored in the adopted screened pn event product. Its ANALDATE is 2025-09-28T20:10:27; this product-level value defines the CCF provenance used in the thesis. The RGS products were produced in the same SAS environment.

The *Resolve* products were reduced with HEASoft 6.34, XRISM software release 21Aug2024 V001, and calibration set `gen20240315_xtd20240815_rs120240815`. The pipeline was run through stages 1–3 before the additional screening described in Chapter 2.

A.1.2 Compact command record

EPIC-pn screening and products. Listing A.1 gives the decisive screening, light-curve, and persistent and dip spectrum-extraction steps. The response commands were repeated for all six pn spectra.

```
evselect table=pn.fits filteredset=pn_filt.fits \
  expression="(PATTERN<=4)&&(PI in [300:12000])&&#XMMEA_EP"

evselect table=pn_filt.fits withrateset=yes \
  rateset=pn_hi_bkg_lc.fits maketimecolumn=yes \
  makeratecolumn=yes timebinsize=100 \
  expression="(PATTERN==0)&&(PI in [10000:12000])&&(RAWX in [5:10])"

tabgtigen table=pn_hi_bkg_lc.fits gtiset=GTI_timecut.fits \
  expression="TIME<=8.29727e8"

evselect table=pn_filt.fits withfilteredset=yes \
  filteredset=pn_filt_gti.fits keepfilteroutput=yes \
  expression="gti(GTI_timecut.fits,TIME)"

evselect table=pn_filt_gti.fits withrateset=yes \
  rateset=pn_total_lc_raw_20.fits maketimecolumn=yes \
  makeratecolumn=yes timebinsize=20 \
  expression="(PI in [300:12000])&&(RAWX in [28:48])&&#XMMEA_EP&&(PATTERN<=4)"

epiclccorr srctslst=pn_total_lc_raw_20.fits \
  eventlist=pn_filt_gti.fits outset=pn_total_lc_20.fits \
```

```
applyabsolute corrections=yes

SELECTION=SELECTION.gti
PN_FILTER="(FLAG==0)&&(PATTERN<=4)&&(PI in [300:10000])"
PN_REGION="(RAWX in [28:48])"
evselect table=pn_filt_gti.fits withspectrumset=yes \
  spectrumset=pn_SELECTION.pha energycolumn=PI spectralbinsize=5 \
  expression="GTI({SELECTION},TIME)&&${PN_FILTER}&&${PN_REGION}"
backscale spectrumset=pn_SELECTION.pha \
  badpixlocation=pn_filt_gti.fits
rmfgen spectrumset=pn_SELECTION.pha rmfset=pn_SELECTION.rmf
arfgen spectrumset=pn_SELECTION.pha arfset=pn_SELECTION.arf \
  withrmfset=yes rmfset=pn_SELECTION.rmf \
  badpixlocation=pn_filt_gti.fits detmaptype=psf
```

LISTING A.1— Key EPIC-pn screening, light-curve, and persistent and dip spectrum-extraction commands.

Sector-3 threshold GTIs. The final threshold was applied only within the adopted sector-3 time range. Burst intervals were excluded when the persistent and dip GTIs were applied to the spectral products.

```
TMIN=829697275
TMAX=829727035
tabgtigen table=pn_total_lc_20.fits \
  gtiset=pn_s3_pers.gti \
  expression="(RATE>65)&&((TIME>=$TMIN)&&(TIME<=$TMAX))"
tabgtigen table=pn_total_lc_20.fits \
  gtiset=pn_sector3_dips.gti \
  expression="(RATE<65)&&((TIME>=$TMIN)&&(TIME<=$TMAX))"
```

LISTING A.2— Threshold construction of the sector-3 pn GTIs.

RGS persistent and dip products. The variable names below stand for the pipeline event and source-list products. The same sequence was run for RGS1 and RGS2.

```
RGS_EVENTS=rgs_event_list.fits
RGS_SRCLIST=rgs_source_list.fits
SELECTION=SELECTION.gti

evselect table=$RGS_EVENTS \
  filteredset=rgs_SELECTION_events.fits keepfilteroutput=yes \
  expression="gti({SELECTION},TIME)"
rgsspectrum evlist=rgs_SELECTION_events.fits \
  srclist=$RGS_SRCLIST bkgcorrect=yes order=1 \
  spectrumset=rgs_SELECTION_src.fits withbkgset=yes \
  bkgset=rgs_SELECTION_bkg.fits
rgsrmfgen evlist=rgs_SELECTION_events.fits \
  srclist=$RGS_SRCLIST spectrumset=rgs_SELECTION_src.fits \
  rmfset=rgs_SELECTION.rmf order=1
```

LISTING A.3— Transfer of a pn selection GTI to one RGS detector.

Resolve screening and responses. Listing A.4 retains the observation-specific filters and response settings. The long CALDB filenames supplied to `xaarfgn` are represented by CALDB; their release is recorded above.

```
xapipeline indir=. outdir=./xpipe_resolve \
  steminputs=xa300039010 stemoutputs=DEFAULT \
  entry_stage=1 exit_stage=3 instrument=resolve \
  verify_input=no

ftcopy \
  infile="xa300039010rsl_p0px1000_cl.evt[EVENTS] [(PI>=600)&&((((RISE_TIME+0.00075*DERIV_MAX)>46)
    &&((RISE_TIME+0.00075*DERIV_MAX)<58))&&ITYPE<4)|| (ITYPE==4))&&(STATUS[4]==b0)]" \
  outfile=xa300039010rsl_p0px1000_cl2.evt \
  copyall=yes clobber=yes history=yes
ftcopy \
  infile="xa300039010rsl_p0px1000_cl2.evt[EVENTS] [ITYPE<4]" \
  outfile=xa300039010rsl_p0px1000_wols_cl2.evt \
  copyall=yes clobber=yes history=yes

barycen xa300039010rsl_p0px1000_wols_cl2.evt \
  xa300039010rsl_p0px1000_wols_bc_cl2.evt \
  xa300039010.orb 289.69937 -5.23806 orbext=ORBIT

# Applied in XSELECT before extracting each selection product:
# CORTIME>8, ANG_DIST<0.1, PIXEL=0:11,13:26,28:35,
# PHA_CUTOFF=4000:19999, and the corresponding SELECTION time file.

rslmkrmf infile=resolve_SELECTION.evt \
  outfileroot=resolve_SELECTION regmode=DET whichrmf=L \
  resolist=0 regionfile=NONE \
  pixlist=0-11,13-26,28-35
xaexpmap ehkfile=xa300039010.ehk \
  gtifile=resolve_SELECTION.evt instrume=RESOLVE \
  badingfile=NONE pixgtifile=resolve_exposure.gti \
  outfile=resolve_SELECTION.expo outmatype=EXPOSURE \
  delta=20.0 numphi=1
xaarfgn xrtevtfile=resolve_SELECTION_raytrace.fits \
  source_ra=289.69946 source_dec=-5.23808 \
  telescop=XRISM instrume=RESOLVE \
  emapfile=resolve_SELECTION.expo regmode=DET \
  regionfile=rsl_pix_no27.reg sourcetype=POINT \
  rmffile=resolve_SELECTION.rmff erange="2 15.0 0 0" \
  outfile=resolve_SELECTION.arf numphoton=300000 \
  qefile=CALDB contamifile=CALDB gatevalvefile=CALDB \
  fwfile=CALDB onaxisffile=CALDB onaxiscfile=CALDB \
  mirrorfile=CALDB obstructionfile=CALDB scatterfile=CALDB
```

LISTING A.4— Key *Resolve* reprocessing, screening, and response commands.

A.2 Persistent and Dip GTIs

The persistent and dip intervals in sectors 1 and 2 were chosen by inspecting the 20-s pn light curve. Clear, continuous decreases in count rate were marked as dips. The remaining valid non-burst intervals were assigned to the persistent intervals.

Sector 3 contained dips during much of the observation, so a simple count-rate threshold was used instead. Each 20-s bin above 65 counts s⁻¹ was classified as a persistent interval, while each bin below this level was classified as a dip interval.

Table A.1 lists the six resulting GTI files and their total durations. These durations are the summed pn time intervals and have not been corrected for detector dead time. The usable RGS and *Resolve* exposures can be shorter because each instrument also has its own valid-time selection.

TABLE A.1— Summary of the six pn GTI files used for the persistent and dip RGS and *Resolve* products.

Sector	Selection	File	Intervals	Duration (ks)	Construction
1	Persistent	pn_s1_pers.gti	31	29.848	Manual non-dip windows
1	Dip	dips_sector1.gti	29	3.645	Manual dip windows
2	Persistent	pn_s2_pers.gti	24	19.401	Manual non-dip windows
2	Dip	pn_sector2_dips.gti	23	7.880	Manual dip windows
3	Persistent	pn_s3_pers.gti	184	14.980	$R_{\text{pn},20\text{s}} > 65 \text{ counts s}^{-1}$
3	Dip	pn_sector3_dips.gti	183	14.780	$R_{\text{pn},20\text{s}} < 65 \text{ counts s}^{-1}$

A.3 Supplementary fit diagnostics

Table A.2 gives the complete SPEX fit-statistic summary for the six persistent and dip fits. The expected statistic is calculated by SPEX for the fitted model and binning. The final column reports $C_{\text{obs}}/C_{\text{exp}}$. These values complement the best-fitting parameters in Tables 4.1 and 4.2; the parameter tables are not repeated here.

TABLE A.2— Observed and expected C-statistics for the six persistent and dip fits.

Sector	Selection	C_{obs}	dof	C_{exp}	$C_{\text{obs}}/C_{\text{exp}}$
1	Persistent	2864.98	2624	2682.38	1.068
1	Dip	1866.11	1560	1665.53	1.120
2	Persistent	3005.18	2924	2980.50	1.008
2	Dip	2850.88	2586	2702.97	1.055
3	Persistent	1797.50	1642	1683.63	1.068
3	Dip	2836.90	2700	2787.98	1.018

The sector-1 dip spectrum has the largest ratio, $C_{\text{obs}}/C_{\text{exp}} = 1866.11/1665.53 = 1.12$. This shows that its observed statistic lies above the SPEX expectation, but it does not identify the responsible energy range or measure the significance of an individual absorber parameter.

The detailed fixed, tied, and boundary markers remain attached to the relevant entries in Tables 4.1 and 4.2. In all six fits, the `pion` 3 velocity width and shift were fixed at 100 km s⁻¹ and zero, respectively. The `comt` electron temperature, optical depth, and normalisation were fixed, its seed temperature was tied to the `bb` temperature, and the scattered continuum fraction was fixed at 0.09. For each dip fit, the intrinsic continuum was fixed to the corresponding persistent solution. Physical model parameters were

coupled across the simultaneously fitted *Resolve*, RGS1, and RGS2 regions, while each region retained its own response.

Several velocity widths reached the imposed 100–200 km s^{−1} search range. These comprise `pion 1` and `pion 2` in sector-1 persistent, `pion 1` and `pion 2` in sector-1 dip, `pion 2` in both sector-2 fits, and both zones in sector-3 persistent. In the sector-3 dip fit, the `pion 1` velocity shift reached zero and the `pion 2` width reached 100 km s^{−1}. The dip-local `hot` columns are one-sided upper limits from best-fitting values of zero. The sector-3 persistent profile searches did not share a verified common minimum; the caution below Table 4.1 therefore applies.

Appendix B

Statement on the Use of Generative AI

B.1 Scope and frequency

I used OpenAI Codex occasionally during the analysis and preparation of this thesis, mainly to help write and check Python code, prepare Matplotlib figures, check LaTeX, suggest sentence-level wording, and help reorder some paragraphs. I later used UvAChat for the final language revision to simplify wording in response to supervisor comments.

B.2 Examples and checks

TABLE B.1— Examples of generative-AI use and the checks applied to the output.

Task	Example request	Tool and frequency	What I checked
Burst code	<i>Implement my selected D_5 procedure and mark the adopted intervals.</i>	Codex; used several times during the burst analysis	I ran the code, compared the intervals with the EPIC-pn light curves, and inspected the fits before choosing the models and intervals.
Spectral figures	<i>Use the QDP files exported by SPEX to plot the data, model, and residuals with Python and Matplotlib, using the chosen colours and energy ranges.</i>	Codex; repeated during figure preparation	I ran the scripts and checked the input files, axes, units, data, model, and residuals against the corresponding SPEX plots.
Scientific writing	<i>Revise this paragraph into concise scientific English. Preserve the values, citations, and level of uncertainty, separate the measured result from its interpretation, and do not add new claims.</i>	Codex during drafting and UvAChat during the final language revision	I checked each passage against the fit results and citations, and made sure that it addressed the relevant supervisor comments.
LaTeX checks	<i>Check references, captions, and instrument and model names.</i>	Codex; occasionally during writing and final revision	I compiled the thesis and checked the revised pages, references, and captions.

Note. A QDP file is a plain-text plotting table exported by SPEX. It contains the numerical arrays used for the displayed spectrum, model, and residuals and can be imported into Python for customised plotting.

B.3 Scientific choices and reflection

I chose the burst-duration method after reading the MINBAR catalogue paper by [Galloway et al. \(2020\)](#), because its D_5 definition provides a reproducible way to measure burst duration and compare it with previous work. I used the MINBAR decay-model comparison as a reference. The additional two-exponential model was an empirical description of these EPIC-pn light curves, and I used it to define the extraction windows. It is not part of the MINBAR method, and I did not interpret its two terms as separate physical decay components. Codex helped me implement the procedure I had chosen and prepare the figures. I inspected the light curves, ran and assessed the fits, chose the intervals and models, and drew the conclusions from the results. In an early draft written with AI assistance, I did not clearly separate the MINBAR definition from the choices I made for these data, so I rewrote this section.

The physical interpretation and initial hand-drawn sketch for Fig. 5.1 came from discussions with my supervisors. I used ChatGPT Image 2.0 to produce the polished illustration, then added and checked the scientific labels myself.

I first drew the nodal- and apsidal-precession schematic in Fig. 1.9 by hand, then used ChatGPT Image 2.0 to polish the illustration.

Codex suggested the four-step outline used to present the pure-gravity test in Section 5.4: state the pure-gravity assumption, express it using the Schwarzschild gravitational-redshift relation and the longitudinal special-relativistic Doppler relation used for the SPEX *zv* parameter, substitute the parameters adopted for 4U 1916–053, and compare the implied radius with the expected outer-disc radius. I carried out the algebra and numerical calculation myself. Codex then helped me order the equations and explanation in the thesis. I derived the radius expression by equating the two relations; it is not taken from [Trueba et al. \(2020\)](#). The shifts and characteristic radius reported by [Trueba et al. \(2020\)](#) are used only for comparison. Under the adopted outer-disc interpretation, the calculation shows that pure gravitational redshift cannot account for the fitted shift.

I initially considered only gravitational redshift and local gas motion when interpreting the fitted shift. Codex suggested that I should also check the possible contributions from the systemic velocity, neutron-star orbital motion, and projected disc rotation. I then checked these effects and included them in the schematic expression in Section 5.4. The spectra do not separate the individual contributions. I checked the equations and numerical inputs against the cited sources and the SPEX documentation. I am responsible for the final scientific content and any remaining errors.